REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	OT	JO	SK	07-25-2025
B	90% CD	OT	JO	SK	08-20-2025
C	100% CD	OT	JO	SK	11-05-2025
D	FAO REVIEW	OT	JO	SK	01-05-2026
E	BIO REVIEW	OT	JO	SK	01-26-2026

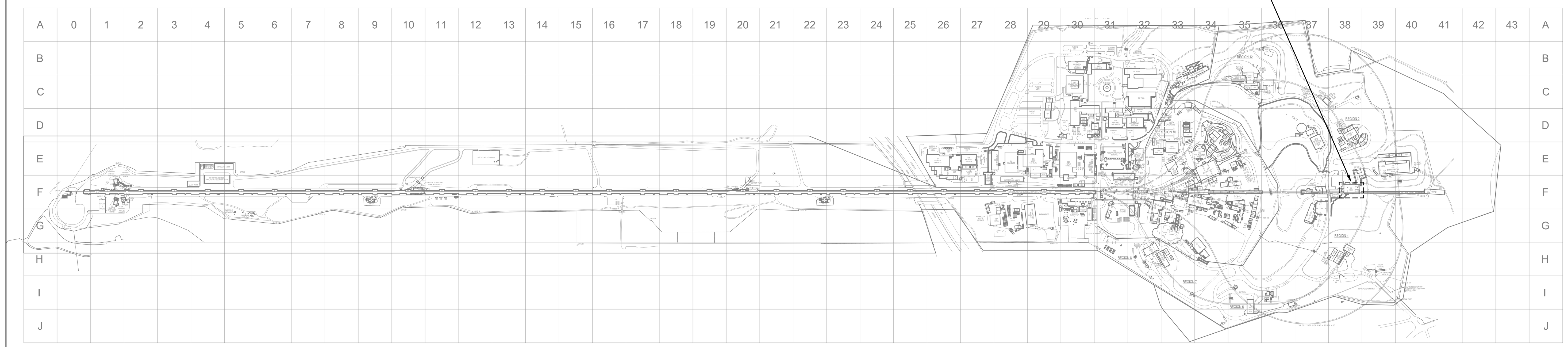
SLAC NATIONAL ACCELERATOR LABORATORY

SLAC NATIONAL ACCELERATOR LABORATORY
2575 SAND HILL ROAD, MENLO PARK,
CALIFORNIA 94025-7015

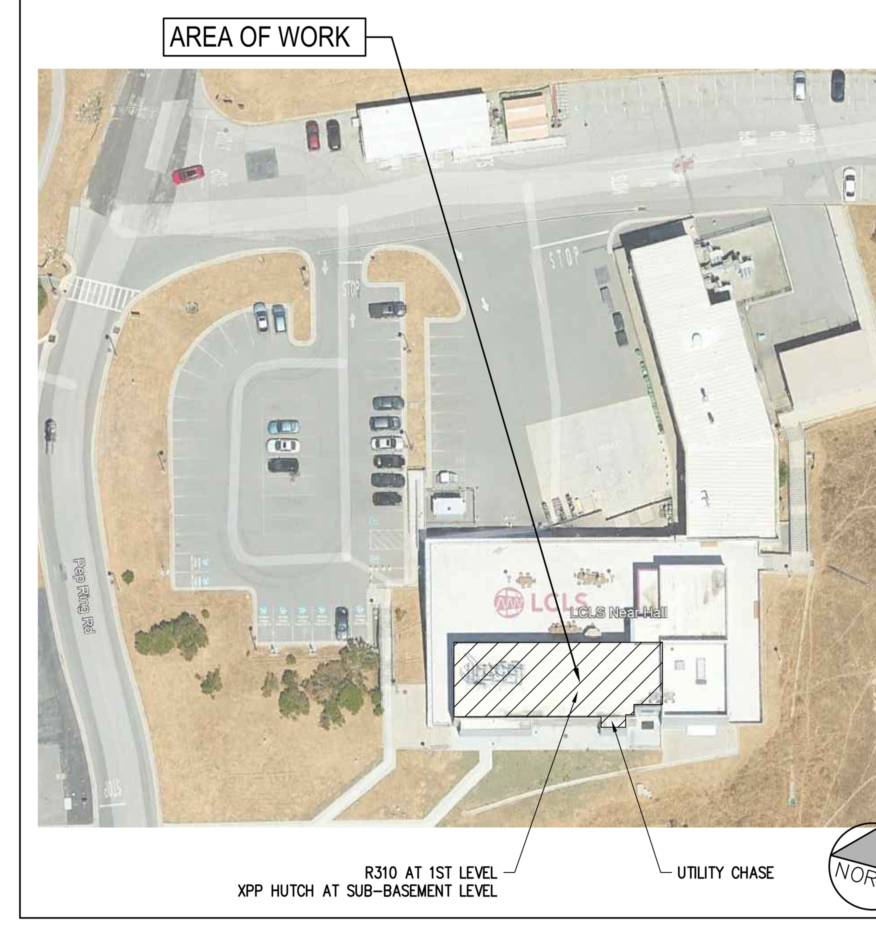
B950 MECH ROOM CO2 CHILLERS

PROJECT VICINITY

PROJECT SITE



AREA OF WORK



PROJECT TEAM

OWNER REPRESENTATIVE

LINAC COHERENT LIGHT SOURCE (LCLS)
SLAC NATIONAL ACCELERATOR LABORATORY (SLAC)
2575 SAND HILL ROAD, MENLO PARK, CA 94025-7015
P: 740.816.6210
CONTACT: CIBIL ZACHARIAH

MECHANICAL ENGINEER

MHC ENGINEERS
150 8TH STREET
SAN FRANCISCO, CA 94103
P: 415.512.7141
CONTACT: JIMMY LI, P.E.

ELECTRICAL ENGINEER

VEKTOR ENGINEERING AND CONSULTING SERVICES, INC.
2603 CAMINO RAMON, SUITE 417
SAN RAMON, CA 94583
P: 925.835.8671
CONTACT: SERGEY KOROLEV, P.E.

STRUCTURAL ENGINEER

ZFA STRUCTURAL ENGINEERS
601 MONTGOMERY STREET, STE 1450
SAN FRANCISCO, CA 94111
P: 415.243.4091
CONTACT: RYAN BOGART, S.E.

APPLICABLE CODES AND DESIGN CRITERIA

E

CONSTRUCTION SHALL MEET OR EXCEED REQUIREMENTS OF THE FOLLOWING CODES, STANDARDS AND REGULATIONS LISTED. IN THE EVENT OF CONFLICTING REQUIREMENTS, THE LATEST REQUIREMENT PROVIDING THE GREATEST PROTECTION TO SLAC SHALL GOVERN. ALL CODES, STANDARDS AND REFERENCED SPECIFICATIONS INDICATED SHALL MEAN THE LATEST EDITION, INCLUDING SUPPLEMENT(S) WHEN SUCH EXISTS, UNLESS OTHERWISE STATED; THE MOST RESTRICTIVE AND STRINGENT CODE SHALL BE FOLLOWED.

NON-GOVERNMENTAL STANDARDS DEVELOPING ORGANIZATIONS:

A. AAMA - AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION:
   1. AAMA 501.2 - FIELD CHECK OF METAL STOREFRONTS, CURTAIN WALLS, AND SLOPED GLAZING SYSTEMS FOR WATER LEAKAGE AAMA 1503 - VOLUNTARY TEST METHOD FOR THERMAL TRANSMISSION AND CONDENSATION RESISTANCE OF WINDOWS, DOORS, AND GLAZED WALL SECTIONS.
   2. AAMA 2604 - VOLUNTARY SPECIFICATION, PERFORMANCE REQUIREMENTS AND TEST PROCEDURES FOR HIGH PERFORMANCE ORGANIC COATINGS ON ALUMINUM EXTRUSIONS AND PANELS.
   3. AAMA GDSG-1 - GLASS DESIGN FOR SLOPED GLAZING.
B. AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC).
C. AMERICAN IRON & STEEL INSTITUTE (AISI).
D. AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI):
   1. ANSI A10 SERIES – SAFETY REQUIREMENTS FOR CONSTRUCTION.
E. AMERICAN SOCIETY OF TESTING AND MATERIALS (ASTM).
F. AMERICAN WELDING SOCIETY, (AWS) STANDARDS.
G. NFPA - NATIONAL FIRE PROTECTION ASSOCIATION:
   1. NFPA 101 - LIFE SAFETY CODE 2024.
   2. NFPA 10 - PORTABLE FIRE EXTINGUISHERS 2022.
   3. NFPA 13 - STANDARD FOR INSTALLATION OF SPRINKLER SYSTEM 2022.
   4. NFPA 72 – NATIONAL FIRE ALARM AND SIGNALING CODE 2022.
   5. NESC C2 - NATIONAL ELECTRICAL SAFETY CODE.

SLAC NATIONAL ACCELERATOR LABORATORY:

1. DOE ORDER 420.1C CHANGE 3 - FACILITIES SAFETY.
2. SLAC CURRENT ESH MANUAL STANDARDS 3, 4
3. SLAC- I-720-0A24E-001 SLAC CURRENT SEISMIC DESIGN SPECIFICATION FOR BUILDING STRUCTURES, EQUIPMENT AND SYSTEMS.
4. SLAC-I-708-104-007 R00 SLAC CURRENT ELECTRICAL DESIGN STANDARD.
5. SLAC-I-708-404-096 OPERATIONS ON SYSTEMS WITH SUSPECTED SHARED NEUTRALS.
6. SLAC-I-708-406-004 MECHANICAL EQUIPMENT ID STANDARD.
7. SLAC-I-708-406-006 PIPING IDENTIFICATION STANDARD.
8. SLAC-I-708-406-008 VALVE & INSTRUMENT ID STANDARD.

PRIVATE EVALUATION ORGANIZATIONS:

A. FM - FACTORY MUTUAL SYSTEM:
   1. FM P7825 - APPROVAL GUIDE; CURRENT EDITION.
B. UL - UNDERWRITERS LABORATORIES INC.:
   1. UL (FPED) - FIRE PROTECTION EQUIPMENT DIRECTORY; CURRENT EDITION UL (FRD) - FIRE RESISTANCE DIRECTORY; CURRENT EDITION.

APPLICABLE CODES AND DESIGN CRITERIA (CONT.)

E

GOVERNMENTAL REGULATIONS AND PUBLICATIONS:

A. CFR - CODE OF FEDERAL REGULATIONS, UNITED STATES GOVERNMENT:
   1. CFR 29 PART 1910, OCCUPATIONAL SAFETY AND HEALTH STANDARDS FOR THE INDUSTRY, AND 1926, OCCUPATIONAL SAFETY AND HEALTH STANDARDS FOR THE CONSTRUCTION INDUSTRY BY OSHA, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION.
   2. CFR 36 - NONDISCRIMINATION BY PUBLIC ACCOMMODATIONS AND IN COMMERCIAL FACILITIES; FINAL RULE; DEPARTMENT OF JUSTICE; CURRENT EDITION.
   3. CFR 435 – MANDATORY FEDERAL ENERGY EFFICIENCY REQUIREMENTS.
   4. CFR 436 – FEDERAL ENERGY MANAGEMENT & PLANNING PROGRAMS.
   5. CFR 851 - WORKER SAFETY AND HEALTH.
   6. CFR 1191 - AMERICANS WITH DISABILITIES ACT ACCESSIBILITY GUIDELINES FOR BUILDINGS AND FACILITIES; FINAL GUIDELINES AND AMENDMENT TO FINAL GUIDELINES (ADAAG); ARCHITECTURAL AND TRANSPORTATION BARRIERS COMPLIANCE BOARD; CURRENT EDITION; REPRINTED COMPILING ALL REVISIONS.
   7. AMERICANS WITH DISABILITIES ACT (UFAS, ADA/ABA, SECTION 504 OF REHABILITATION ACT).
   8. CFR 1201 - SAFETY STANDARD FOR ARCHITECTURAL GLAZING MATERIALS; CONSUMER PRODUCT SAFETY COMMISSION; CURRENT EDITION.
   9. CFR 1910 - OCCUPATIONAL SAFETY AND HEALTH STANDARDS; OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION; CURRENT EDITION.
   10. CFR 1926 - SAFETY AND HEALTH REGULATIONS FOR CONSTRUCTION; OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION; CURRENT EDITION.
   11. DOE ORDER 420.1.C, FACILITY SAFETY, ATTACHMENT II, FACILITY SAFETY REQUIREMENTS, CHAPTER 2, FIRE PROTECTION SAFETY, CHG. 3, 11/14/2019
   12. DOE STANDARD 1066-2016, FIRE PROTECTION, 2016
B. STATE OF CALIFORNIA GOVERNMENT STANDARDS:
   1. CCR, TITLE 8, CAL OSHA.
   2. CCR, TITLE 19, PUBLIC SAFETY.
   3. DIVISION 1, STATE FIRE MARSHALL REGULATIONS.
   4. CCR, TITLE 24 CALIFORNIA BUILDING STANDARDS CODE.
   5. PART 2, CALIFORNIA BUILDING CODE (CBC).
   6. PART 9, CALIFORNIA FIRE CODE INCLUDING BUT NOT LIMITED TO THE IFC.
   7. PART 12, CALIFORNIA REFERENCED STANDARDS CODE.
   8. APPENDIX J, CALIFORNIA BUILDING CODE (CBC).
   9. CHAPTER 11, CALIFORNIA BUILDING CODE.
   10. DIR, CALIFORNIA DEPARTMENT OF INDUSTRIAL RELATIONS.
C. MODEL CODE ORGANIZATIONS:
   1. CBC – CALIFORNIA BUILDING CODE, 2025 EDITION.
   2. CEBC - CALIFORNIA EXISTING BUILDING CODE, 2025 EDITION.
   3. CEC - CALIFORNIA ELECTRICAL CODE, 2025 EDITION.
   4. CFC - CALIFORNIA FIRE CODE, 2025 EDITION.
   5. CGBSC - CALIFORNIA GREEN BUILDING STANDARDS CODE, 2025 EDITION
   6. CMC – CALIFORNIA MECHANICAL CODE, 2025 EDITION.
   7. CPC – CALIFORNIA PLUMBING CODE, 2025 EDITION.
   8. CWUIC - CALIFORNIA WILDLAND-URBAN INTERFACE CODE, 2025 EDITION
   9. NFPA 70 – NATIONAL ELECTRICAL CODE (NEC), 2026 EDITION.
   10. NFPA 70E - STANDARD FOR ELECTRICAL SAFETY IN THE WORKPLACE, 2024 EDITION.

NOTE: THERE MAY BE ADDITIONAL CODES AND REQUIREMENTS LISTED UNDER SPECIFIC DESIGN DISCIPLINES, AND THEY SHALL BE APPLICABLE UNLESS OTHERWISE DETERMINED BY THE OWNER REPRESENTATIVE.

GENERAL SCOPE OF WORK

1. MOBILIZE, PROVIDE TEMPORARY FACILITIES, AND DUST AND OTHER ENVIRONMENTAL CONTROLS.
2. COORDINATE W/ SLAC OWNER REPRESENTATIVE TO IDENTIFY AND LOCATE (E) DECOMMISSIONED GENERATOR AND ATS SYSTEM COMPONENTS, PARTS OF THIS SYSTEM SHALL BE RE-USED AS PART OF THIS PROJECT, SEE RESPECTIVE DRAWINGS. ALL ADDITIONAL DECOMMISSIONING ACTIVITIES AND INSTALLATION OF "DO NOT USE" TAGS AND LOCKS BY SLAC.
3. PROVIDE (N) SUPPORTING ELECTRICAL INFRASTRUCTURE FOR THE INSTALLATION OF (2) (N) CO2 CHILLERS W/ PROVISIONS FOR FUTURE (4) (N) CO2 CHILLERS AS SHOWN ON RESPECTIVE DRAWINGS.
4. PROVIDE (N) SUPPORTING MECHANICAL INFRASTRUCTURE FOR THE INSTALLATION OF (2) (N) CO2 CHILLERS W/ PROVISIONS FOR FUTURE (4) (N) CO2 CHILLERS AS SHOWN ON RESPECTIVE DRAWINGS.
5. PROVIDE (N) SUPPORTS FOR (N) CO2 CHILLER TRANSFER HOSE PIPING. (N) CO2 CHILLER TRANSFER HOSE PIPING PROVIDED BY SLAC, ROUTE SHOWN FOR REFERENCE.
6. (2) (N) CO2 CHILLER SKID ASSEMBLIES PROVIDED BY SLAC, APPROXIMATE LOCATIONS SHOWN FOR REFERENCE.
7. FOR THE SPECIFIC SCOPE OF WORK FOR THE MECHANICAL AND ELECTRICAL TRADES SEE E0.01 AND M0.01.
8. REMOVE TEMPORARY FACILITIES, DUST AND OTHER ENVIRONMENTAL CONTROLS, PASS ALL INSPECTIONS, PROVIDE PROJECT CLOSE OUT MATERIALS, AND DEMOBILIZE.

GENERAL PROJECT NOTES

1. THE CONTRACTOR SHALL VISIT THE JOB SITE AND VERIFY ALL DIMENSIONS AND CONDITIONS OF THE WORK PRIOR TO SUBMITTING BID. IMMEDIATELY NOTIFY THE OWNER REPRESENTATIVE OF ALL DISCREPANCIES BETWEEN DRAWINGS AND THE PROJECT SITE CONDITIONS. THE PRIME CONTRACTOR SHALL INCLUDE THE COST TO CORRECT THE DISCREPANCIES IN THEIR BID. NO EXTRA COSTS SHALL BE CHARGED TO SLAC AFTER BID AWARD.
2. THE CONTRACTOR SHALL NOT PROCEED WITH ANY CHANGES WITHOUT THE APPROVAL OF THE OWNER REPRESENTATIVE.
3. THE CONTRACTOR SHALL BE SOLELY AND COMPLETELY RESPONSIBLE FOR JOB SITE CONDITIONS INCLUDING SAFETY OF PUBLIC, WORKERS, PROPERTY, AND ENSURE COMPLIANCE WITH STATE OSHA, AND SLAC GUIDELINES AND SAFETY REQUIREMENTS.
4. THE CONTRACTOR SHALL ENSURE THAT ALL WORK PERFORMED MEETS OR EXCEEDS THE REQUIREMENTS OF THE LATEST ADOPTED EDITIONS OF THE APPLICABLE CODES MENTIONED ON THIS SHEET.
5. THE CONTRACTOR SHALL PROTECT THE EXISTING BUILDINGS ADJACENT TO THE CONSTRUCTION SITE, THEIR EQUIPMENT, SYSTEM, FINISHES, AND FURNISHINGS FROM ANY DAMAGE DURING THE COURSE OF CONSTRUCTION. ACCESS TO ALL BUILDING AREAS MUST BE MAINTAINED AT ALL TIMES. THE CONTRACTOR SHALL REPAIR ALL DAMAGES TO THE ORIGINAL CONDITIONS AT NO COST TO SLAC. THIS COST SHALL BE PAID BY THE PRIME CONTRACTOR.
6. THE CONTRACTOR SHALL COORDINATE WITH THEIR SUBCONTRACTORS TO IDENTIFY ALL LONG LEAD MATERIALS IMMEDIATELY UPON BEING AWARDED THE CONTRACT. THE CONTRACTOR AND THEIR SUBCONTRACTORS SHALL SUBMIT SHOP DRAWINGS AND SUBMITTALS FOR APPROVAL TO THE RESPECTIVE DESIGN CONSULTANT WITHIN 2 WEEKS TIME AFTER BID AWARD OTHERWISE CONTRACTOR SHALL BARE THE COSTS OF DELAY.
7. THE CONTRACTOR AND THEIR RESPECTIVE SUBCONTRACTORS SHALL BE COMPLETELY RESPONSIBLE FOR ALL GENERAL NOTES, SPECIFICATIONS, AND OTHER PERTINENT INFORMATION AS INDICATED WITHIN THE RESPECTIVE CONSTRUCTION DRAWINGS FOR THEIR TRADE. DO NOT DELAY IF CLARIFICATION IS REQUIRED, SUBMIT REQUESTS FOR INFORMATION TO THE DESIGN CONSULTANT FOR THE RESPECTIVE DISCIPLINE AND FOLLOW FORMAL CONSTRUCTION INFORMATION EXCHANGE PRACTICES.
8. ACCESS TO THE JOB SITE ON ROADS AND PARKING LOTS SHALL BE ARRANGED BETWEEN CONTRACTORS AND OWNER REPRESENTATIVE. CONTRACTORS' ENTRANCE TO THE PROJECT LOCATION AND AREA OF WORK WILL BE LIMITED TO LOCATIONS AS DESIGNATED BY OWNER REPRESENTATIVE.
9. THE CONTRACTOR SHALL SCHEDULE WORK WITH MINIMUM INTERFERENCE WITH PROJECT SITE AS WELL AS SLAC ACTIVITIES AND OPERATIONS. LEGALLY DISPOSE OF DEBRIS AFTER EACH WORKING DAY SO AS NOT TO DISTURB PEDESTRIAN AND SLAC ACTIVITIES, DO NOT OBSTRUCT FIRE LANES AND EXITS.
10. AFTER COMPLETION OF THE PROJECT THE CONTRACTOR SHALL CLEAN THE JOB SITE BEFORE ACCEPTANCE BY OWNER REPRESENTATIVE.

SHEET INDEX

SHEET NUMBER	SHEET TITLE/DESCRIPTION	SLAC-ISSUED DRAWING NUMBER
GENERAL		
G0.01	COVER SHEET	B0950-G00-0415
MECHANICAL		
M0.01	COVER SHEET	B0950-M00-0416
M2.00	3RD FL CHILLED WATER PIPING PLAN	B0950-M01-0417
M2.01	3RD FL CO2 PIPING PLAN	B0950-M01-0418
M2.02	BASEMENT CO2 PIPING PLAN	B0950-M01-0419
M2.03	SUB-BASEMENT CO2 PIPING PLAN	B0950-M01-0420
ELECTRICAL		
E0.01	COVER SHEET	B0950-E00-0421
E0.02	EQUIPMENT SCHEDULES	B0950-E06-0422
E3.01D	LCLS B950 R310 DEMO POWER PLAN	B0950-ED1-0423
E3.01	LCLS B950 R310 POWER PLAN	B0950-EP1-0424
E5.01D	SINGLE LINE DIAGRAM DEMO	B0950-ED7-0425
E5.02	SINGLE LINE DIAGRAM	B0950-E07-0426
E5.03	SINGLE LINE DIAGRAM NEW	B0950-E07-0427
E5.04	PANEL SCHEDULES	B0950-E06-0428
E5.05	PANEL SCHEDULES	B0950-E06-0429
E6.01	DETAILS	B0950-E05-0438
STRUCTURAL		
S0.01	COVER SHEET AND DETAILS	B0950-S00-0439

DEFERRED SUBMITTALS

1. CO2 CHILLER SKID SHOP DRAWINGS (BY SLAC).
2. CO2 CHILLER SKID ANCHORAGE AND STRUCTURAL ANALYSIS (BY SLAC).

B950 INFORMATION

BUILDING INFORMATION HAS BEEN GATHERED FROM AVAILABLE AS-BUILT DRAWINGS OF THE BUILDING AND THE MOST RECENT MAJOR CONSTRUCTION PROJECTS, REF. SLAC PROJECT # B0950.2021.02. OVERALL BUILDING OCCUPANCY CLASSIFICATION PER LCLS-II FIRE HAZARD ANALYSIS REPORT.

FOR THE PURPOSES OF INTERPRETING CBC REQUIREMENTS, THE GENERAL OCCUPANCY CLASSIFICATION FOR THE B950 NEH IS F1 - FACTORY INDUSTRIAL MODERATE HAZARD (MACHINERY) OCCUPANCY PER CBC 306.2.

CALCULATED GROSS SQUARE FOOTAGE ARE AS FOLLOWS:

SUB-BASEMENT	13,947 SF	
BASEMENT	13,947 SF	
FIRST FLOOR	6,000 SF	(MECHANICAL SUITE R310 2,845 SF)
TOTAL	33,894 SF	

CONSTRUCTION TYPE II-A

AUTOMATIC FIRE SUPPRESSION SYSTEM: YES

AUTOMATIC FIRE ALARM SYSTEM: YES

E

SLAC BUILDING INSPECTION OFFICE
Construction Authorization (IFC)
BY: cnadler
DATE: 2/18/2026 PRS#: 26-009
Revised Authorization ☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

Vektor Engineering & Consulting Services, Inc.
"Where engineering and technology drive innovation"
2603 Camino Ramon, Suite 417
San Ramon, CA 94583
+1 (866) VEKTOR1 (835-8671)

REGISTERED PROFESSIONAL ENGINEER SERGEY V. KOROLEV E20491 EXP: 09/30/27 ELECTRICAL STATE OF CALIFORNIA

SCALE: AS NOTED	DO NOT SCALE DRAWING	B0950.2025.05		G0.01

SLAC NATIONAL ACCELERATOR LABORATORY — U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
GENERAL
COVER SHEET

		DATE	APPROVALS	DATE
ENGR	KOROLEV, SERGEY	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TURCHINA, OLENA	01-26-2026		
CHKR	ORTIZ, JUSTIN	01-26-2026		

DRAWING NUMBER: B0950-G00-0415 REVISION: E E

B0950-G00-0415 - E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	KT	JL	MHC	07-23-2025
B	90% CD	KT	JL	MHC	08-20-2025
C	100% CD	KT	JL	MHC	11-05-2025
D	FAO REVIEW	KT	JL	MHC	11-21-2025
E	BIO REVIEW	KT	JL	MHC	01-26-2026

GENERAL NOTES

1. ALL WORK SHALL CONFORM TO THE 2022 CALIFORNIA MECHANICAL/PLUMBING CODE.
2. THE SUBMITTAL OF A BID PROPOSAL SHALL BE CONSIDERED CONCLUSIVE EVIDENCE THAT THE CONTRACTOR IS THOROUGHLY FAMILIAR WITH THE INTENT OF THE CONTRACT DOCUMENTS. THE CONTRACTOR, PRIOR TO BIDDING, SHALL CHECK EXISTING INSTALLATIONS AND SYSTEMS RELATED TO HIS WORK AND SHALL IN THE BID PROPOSAL INCLUDE ALL LABOR AND MATERIAL REQUIRED FOR COMPLETION OF THE SYSTEM.
3. CONTRACTOR SHALL BE RESPONSIBLE FOR HAVING VISITED THE SITE AND SATISFIED HIMSELF AS TO THE CONDITIONS UNDER WHICH THE WORK WILL BE PERFORMED. THE CONTRACTOR SHALL CHECK ALL EXISTING CONDITIONS WHICH MAY AFFECT HIS WORK. NO ALLOWANCE WILL BE MADE FOR ANY EXTRA EXPENSE WHICH MAY BE INCURRED DUE TO NEGLECT OR FAILURE TO EXAMINE THE SITE, EXISTING CONDITIONS, UTILITIES, ETC. THE SITE VISIT SHALL BE DURING THE BID WALK AND/OR PREARRANGED WITH THE OWNER.
4. CONTRACTOR SHALL PROVIDE QUALIFIED TRADE PROFESSIONALS WITH ADEQUATE SAFETY TRAINING IN CONSTRUCTION OF THESE TYPES OF FACILITIES. LACK OF SAFETY PRACTICE SHALL DISQUALIFY CONTRACTOR AT OWNER DISCRETION.
5. CONTRACTOR SHALL SCHEDULE AND COORDINATE THE FOLLOWING WITH OWNER OR OWNER'S PROJECT MANAGER.
   A. ANY AND ALL UTILITIES SHUT DOWN. CONTRACTOR SHALL GIVE A MINIMUM OF TWO WEEKS PRIOR NOTICE FOR EACH SHUT DOWN.
   B. ANY AND ALL WORK BEING PERFORMED IN AREAS WHERE THE FACILITY WILL REMAIN IN OPERATION.
   C. ANY WORK WITHIN THE PARAMETER OF ITEMS A AND B ABOVE SHALL BE COORDINATED WITH THE GENERAL CONTRACTOR.
6. ALL MATERIALS AND WORKMANSHIP ARE SUBJECT TO APPROVAL BY OWNER. ANY DEFECTIVE PORTION OF THE WORK SHALL BE REPLACED BY THE CONTRACTOR AS PART OF THIS CONTRACT AT NO ADDITIONAL COST TO THE OWNER.
7. ABSOLUTE ACCURACY OF DRAWINGS CANNOT BE GUARANTEED, WHILE EVERY EFFORT HAS BEEN MADE TO COORDINATE THE LOCATION OF THE EXISTING EQUIPMENT, DUCTWORK, PIPING ETC. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO COORDINATE THE EXACT REQUIREMENTS GOVERNED BY ACTUAL JOB CONDITIONS.
8. IN THE AREA OF NEW CONSTRUCTION WILL BE FOUND A NUMBER OF EXISTING SERVICES. CONTRACTOR SHALL PROTECT AND MAINTAIN ALL ACTIVE LINES IN GOOD OPERATING CONDITION.
9. PROTECT EXISTING BUILDING STRUCTURES AND ADJACENT FINISHED SURFACES DURING CONSTRUCTION. PATCH, REPAIR AND REFINISH EXISTING WORK DAMAGED BY WORK UNDER THIS DIVISION TO MATCH ADJACENT UNDISTURBED AREAS. PATCHING, REPAIRING AND REFINISHING SHALL BE PERFORMED BY WORKMEN SKILLED IN THE TRADES INVOLVED.
10. CONTRACTOR SHALL SUBMIT SHOP DRAWINGS (1/4"=1'-0" SCALE MINIMUM) FOR REVIEW BY OWNER AND/OR HIS REPRESENTATIVE PRIOR TO COMMENCING WORK.
11. ALL MOVING AND RIGGING OF PRODUCTION EQUIPMENT SHALL BE PROVIDED BY OWNER.
12. ALL MECHANICS INVOLVED IN PIPE AND DUCT DEMOLITION SHALL WEAR PROPER PERSONAL PROTECTIVE GEAR AND SHALL MAKE THE NECESSARY PROVISIONS TO PREVENT EXPOSURE TO HAZARDOUS MATERIALS.
13. WELDING OR OPEN FLAME ARE NOT PERMITTED DURING THE DEMOLITION PROCESS OF ANY FLAMMABLE OR COMBUSTIBLE PIPING OR EXHAUST DUCT SYSTEMS.
14. CONTRACTOR SHALL SUBMIT EQUIPMENT, PIPING AND DUCT MATERIAL SUBMITTALS TO OWNER AND TO THE DESIGN ENGINEER FOR REVIEW AND APPROVAL, PRIOR TO PURCHASING AND INSTALLATION.
15. CONTRACTOR SHALL COORDINATE FINAL INSPECTION OF COMPLETED WORK WITH DESIGN ENGINEER AND OWNER OR OWNER'S REPRESENTATIVE, FOR FINAL APPROVAL AND ACCEPTANCE.
16. ANY NEW OR EXISTING DUCTWORK OR PIPING OFFSETS REQUIRED AS A RESULT OF EXISTING JOB CONDITIONS OR LACK OF COORDINATION WITH OTHER TRADES, SHALL BE PROVIDED AT NO ADDITIONAL COST TO OWNER AND SUBJECT TO OWNER'S REVIEW.
17. COORDINATE LOCATION OF EXISTING DUCTWORK OR PIPE RACKS AND NEW DUCTWORK OR PIPE RACKS WITH OTHER TRADES.
18. ALL DUCTWORK AND PIPING SHALL BE SUPPORTED ACCORDING TO SMACNA AND SEISMICALLY BRACED PER SMACNA'S SEISMIC RESTRAINT MANUAL TO MEET BUILDING CODE.
19. MAINTAIN A MINIMUM HEAD CLEARANCE OF 84" ABOVE FINISHED FLOOR UNDER SUSPENDED DUCTWORK AND PIPING.

SAFETY NOTES

1. OPERATION OF ENERGY ISOLATING DEVICES SHALL BE PERFORMED BY SLAC WORKERS. CONTACT THE FIELD CONSTRUCTION MANAGER FOR ASSISTANCE WITH EQUIPMENT OPERATION.
2. SYSTEMS POTENTIALLY CONTAINING HAZARDOUS ENERGY SHALL BE LOCKED OUT IN ACCORDANCE WITH SLAC ESH MANUAL CHAPTER 51.

MECHANICAL SCOPE SUMMARY

1. CONNECT NEW 2" CHILLED WATER SUPPLY AND RETURN PIPING WITH INSULATION TO EXISTING 4" CHILLED WATER U-JOINT AT CEILING.
2. PROVIDE NEW 1-1/4" CHILLED WATER SUPPLY AND RETURN BRANCH WITH SHUT-OFF VALVES TO SERVE AS A STUB-OUT FOR FUTURE CO2 CHILLERS.
3. EXTEND NEW 1" CHILLED WATER SUPPLY AND RETURN PIPING FROM MAIN AFTER 1-1/4" CHILLED WATER SUPPLY AND RETURN STUB-OUTS WITH INSULATION TO (2) CO2 CHILLERS.
4. PROVIDE NEW 3/4" CHILLED WATER SUPPLY BRANCH WITH SHUTOFF VALVE AT CEILING. SLAC SHALL BE RESPONSIBLE FOR DIRECT CHILLED WATER PIPE CONNECTIONS FROM SHUTOFF VALVES TO CO2 CHILLERS.
5. PROVIDE NEW 3/4" CHILLED WATER RETURN BRANCH WITH CIRCUIT SETTER (SET TO 3 GPM) AND SHUTOFF VALVE AT CEILING. SLAC SHALL BE RESPONSIBLE FOR DIRECT CHILLED WATER PIPE CONNECTIONS FROM SHUTOFF VALVES TO CO2 CHILLERS.
6. PROVIDE NEW UNISTRUT SUPPORT FOR CO2 SUPPLY AND RETURN PIPING FROM NEW CHILLERS TO SHAFT.
7. PROVIDE NEW UNISTRUT SUPPORT FOR CO2 SUPPLY AND RETURN PIPING STUB-OUT 1' FROM SHAFT OPENING.

ABBREVIATIONS

ABBREV	DESCRIPTION
AG	ABOVE GRADE
BG	BELOW GRADE
BOP	BOTTOM OF PIPE
CBC	CALIFORNIA BUILDING CODE
CDL	CONDENSATE DRAIN LINE
CFC	CALIFORNIA FIRE CODE
CHWR	CHILLED WATER RETURN
CHWS	CHILLED WATER RETURN
CMC	CALIFORNIA MECHANICAL CODE
CO2R	CO2 RETURN
CO2S	CO2 SUPPLY
CONN	CONNECTION
COTG	CLEANOUT TO GRADE
CPC	CALIFORNIA PLUMBING CODE
CTE	CONNECT TO EXISTING
CU	COPPER
DN	DOWN
DR	DRAIN
DWG	DRAWING
EXIST. OR (E)	EXISTING
(F)	FUTURE
FNPT	FEMALE NATIONAL PIPE THREAD
GAL	GALLON
ID	INNER DIAMETER
IE	INVERT ELEVATION
IW	INDUSTRIAL WASTE
MAX	MAXIMUM
MOD	MODEL
MECH	MECHANICAL
MFG	MANUFACTURER
MIN	MINIMUM
MNPT	MALE NATIONAL PIPE THREAD
(N)	NEW
NIC	NOT IN CONTRACT
NTS	NOT TO SCALE
POC	POINT OF CONNECTION
PPC	PRIMARY POLY TANK
PW	PUMPED WASTE
(R)	RELOCATED
REQ	REQUIRED
SCH	SCHEDULE
SPECS	SPECIFICATIONS
SPT	SECONDARY POLY TANK
SS	SANITARY SEWER
TYP	TYPICAL
UON	UNLESS OTHERWISE NOTED
W/	WITH
W/O	WITHOUT

FLUID PIPING SYSTEMS COMPLIANCE NOTES

LEAK TESTING AND CERTIFICATION

PIPING SYSTEMS SHALL BE TESTED IN COMPLIANCE WITH:

1. THE 2022 CALIFORNIA MECHANICAL CODE.
2. THE 2022 CALIFORNIA PLUMBING CODE.
3. THE 2022 CALIFORNIA FIRE CODE.
4. AND AS LISTED PER TABLE BELOW, WHICHEVER IS MORE STRINGENT.

PIPING SYSTEMS PRESSURE TEST TABLE

SYSTEM	DESIGN PRESSURE	TEST PRESSURE	DURATION	TEST MEDIUM
CHILLED WATER: (PRESSURIZED)	MAX. 40 PSI.	150% THE MAXIMUM SYSTEM DESIGN PRESSURE BUT NOT LESS THAN 100 PSI, PER CHAPTER 14 SECTION 2.4 SLAC ESH TEST PRESSURE	1HR	WATER

NOTE:
THIS PRESSURE TESTING TABLE IS GENERALIZED FOR PIPING SYSTEMS.

LABELING REQUIREMENTS

1. LABEL ALL PIPING WITH CONTENTS AND DIRECTION OF FLOW EVERY 20' ON CENTER BEFORE AND AFTER EVERY VALVE, BEFORE AND AFTER EVERY WALL PENETRATION AND AT EACH CHANGE IN DIRECTION.
2. LABEL ALL PIPING, DUCTING AND EQUIPMENT PER ASME A13.1.
3. ALL CONTROLS VALVES SHALL BE LABELED AND LOCATED SO AS TO BE VISIBLE AND READILY ACCESSIBLE.

REFERENCE SYMBOLS

SYMBOL	DESCRIPTION
	DETAIL # / SHEET #
	SECTION # / SHEET #
	EQUIPMENT TAG # / SCHEDULE
	REVISION #
	KEYED NOTE #
	POINT OF CONNECTION TO EXISTING (POC)
	POINT OF DEMOLITION TO EXISTING (POD)
——x——	EXISTING TO BE REMOVED (DEMOLISHED)
	EXISTING TO REMAIN
	NEW
	CIRCUIT SETTER
	SHUT-OFF VALVE

PIPE MATERIAL SCHEDULE

CODE	ITEM	MATERIAL DESCRIPTION
CHWS CHWR	ABOVE GROUND CHILLED WATER PIPING	PIPE CLASS C1, TYPE L WITH 95-5 TIN ANTIMONY SOLDER (COPPER) OR STAINLESS STEEL PIPE. FIELD VERIFY TO MATCH EXISTING CONDITIONS.

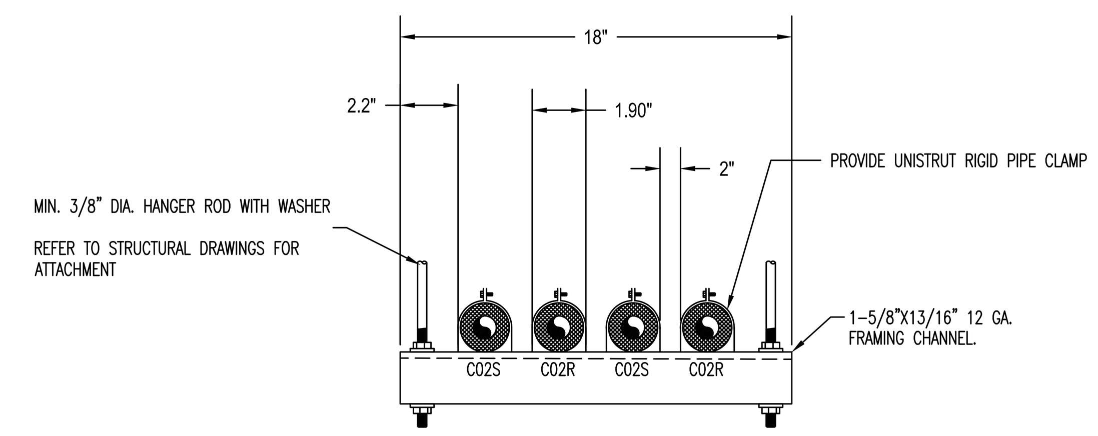
VALVE SCHEDULE

ITEM	MATERIAL DESCRIPTION
HV-950-50 HV-950-51	1-1/4" APOLLO 9A-105-01 BALL VALVE STANDARD CONFIGURATION. COMPLETE WITH INSULATION AND JACKETING
HV-950-52 HV-950-53 HV-950-54 HV-950-55	3/4" APOLLO 9A-105-01 BALL VALVE STANDARD CONFIGURATION. COMPLETE WITH INSULATION AND JACKETING
CBV-950-56 CBV-950-57	3/4" NPT BELL & GOSSETT CB-3/4 LEAD FREE CIRCUIT SETTER BALANCE VALVE

E

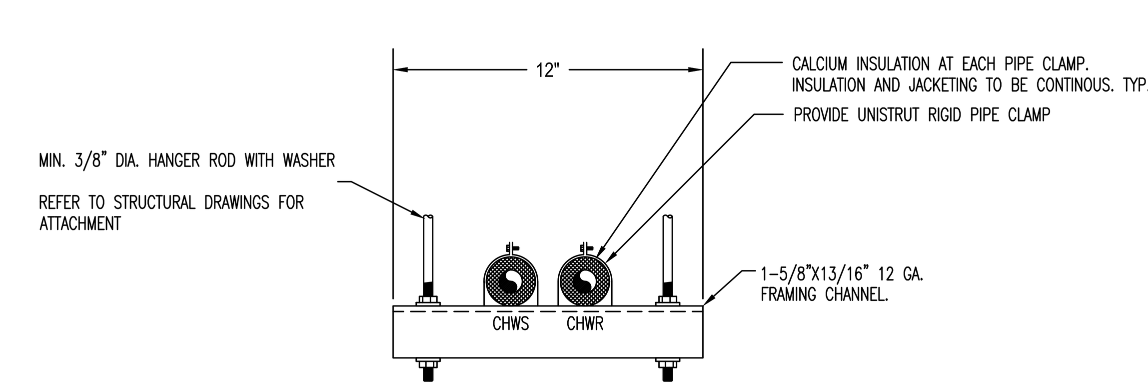
CHILLED WATER PIPE INSULATION THICKNESS TABLE PER CALIFORNIA TITLE 24 SECTION 120.3, TABLE 120.3

CODE	NOMINAL PIPE DIAMETER	MINIMUM PIPE INSULATION REQUIRED (THICKNESS IN INCHES)
CHWS/R	<1"	0.5"
CHWS/R	1" TO <1.5"	0.5"
CHWS/R	1.5" TO <4"	1"



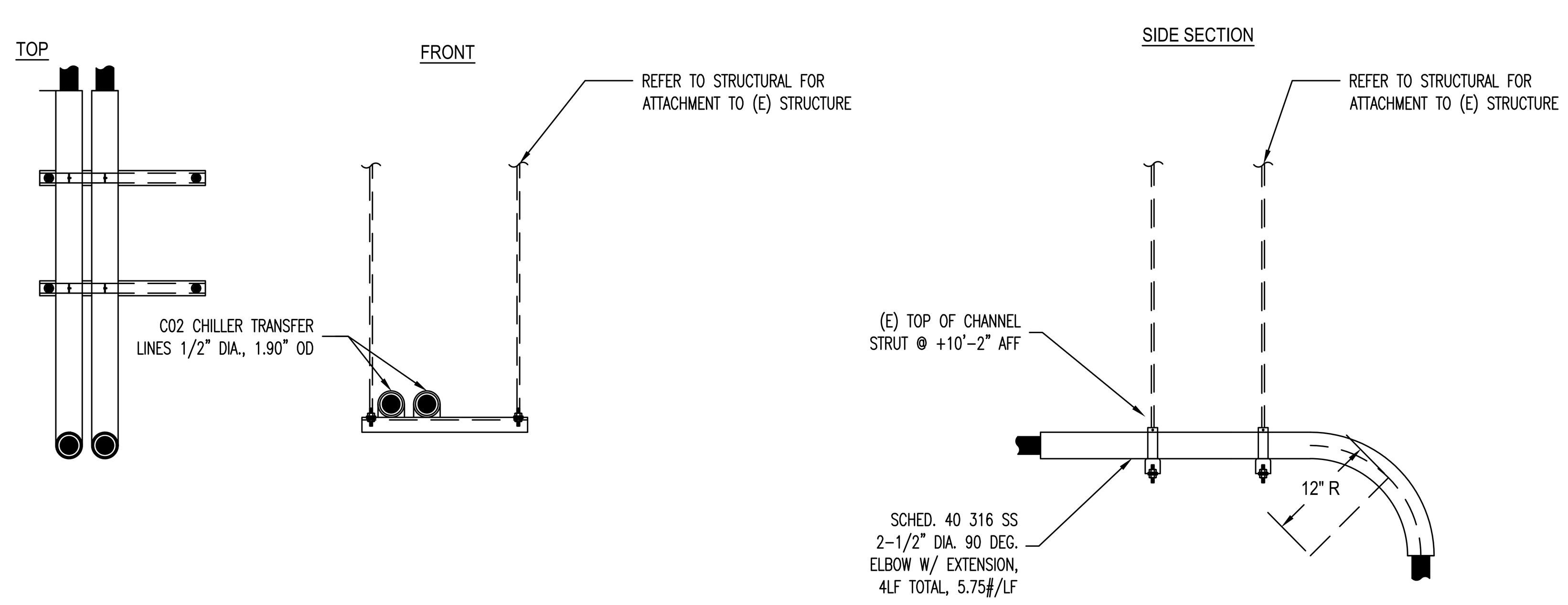
NOTES:
1. SPACING & MATERIALS OF SUPPORTS SHALL COMPLY WITH CODES.
2. SERVICES AND ARRANGEMENT MAY VARY.
3. WRAP PIPING WITH FLEXIBLE ELASTOMER MATERIAL AT SUPPORTS TO AVOID METAL TO METAL CONTACT.

1 CO2 CEILING PIPE RACK SUPPORT DETAIL
SCALE: N.T.S.

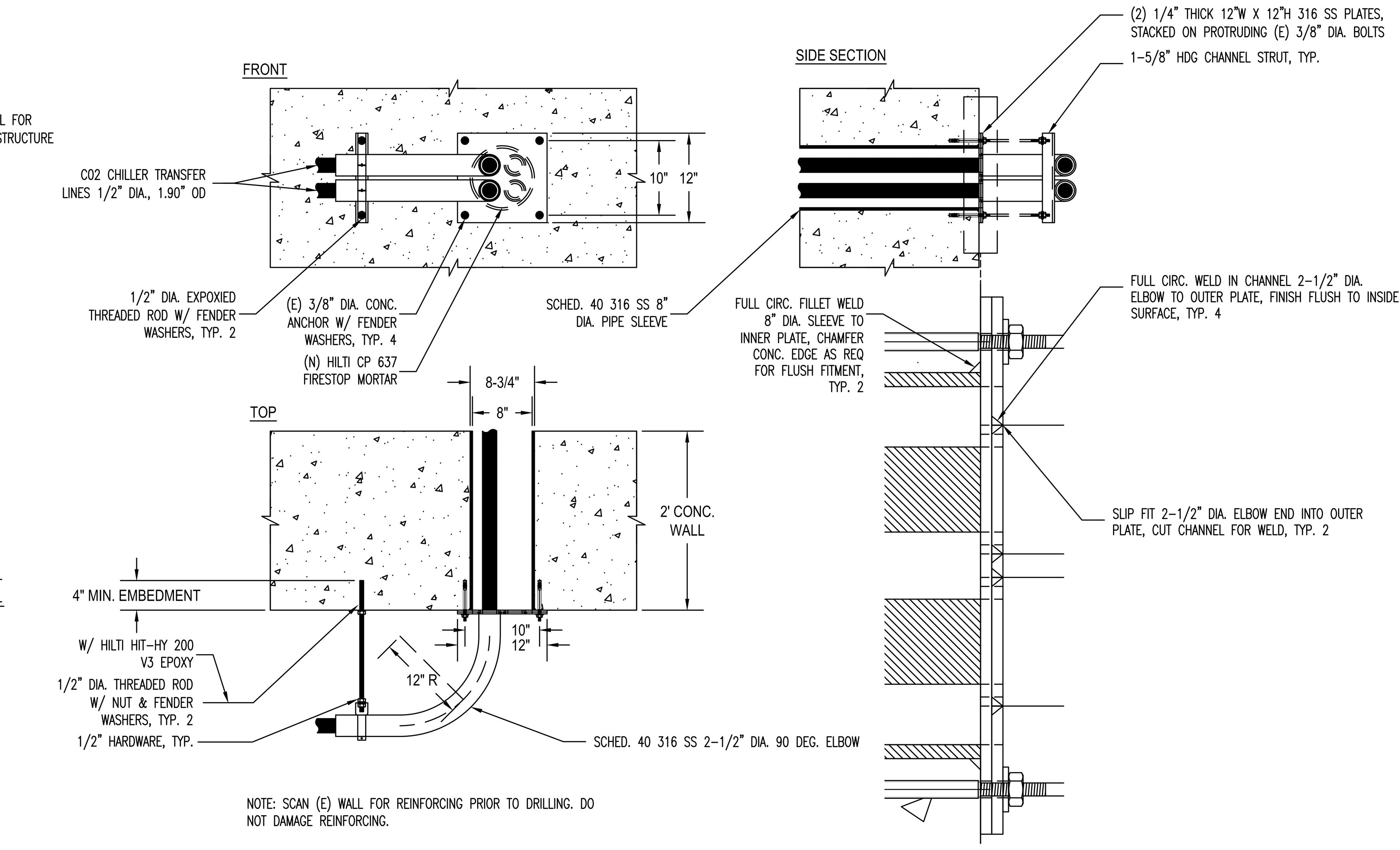


NOTES:
1. SPACING & MATERIALS OF SUPPORTS SHALL COMPLY WITH CODES.
2. SERVICES AND ARRANGEMENT MAY VARY.
3. WRAP PIPING WITH FLEXIBLE ELASTOMER MATERIAL AT SUPPORTS TO AVOID METAL TO METAL CONTACT.

2 CHILLED WATER CEILING PIPE RACK SUPPORT DETAIL
SCALE: N.T.S.



3 (N) PIPE SUPPORT AT TRANSITION FROM R302 TO UTILITY CHASE
SCALE: N.T.S.



NOTE: SCAN (E) WALL FOR REINFORCING PRIOR TO DRILLING. DO NOT DAMAGE REINFORCING.

4 (N) PEN. SLEEVE & PIPE SUPPORT AT XPP
SCALE: N.T.S.

SLAC BUILDING INSPECTION OFFICE

Construction Authorization (IFC)

BY: cnadler

DATE: 2/18/2026 **PRS#:** 26-009

Revised Authorization ☑ No ☐ Yes

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MHC ENGINEERS
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FAX (415) 512-7120

M19582
EXP. 9/30/27

SCALE: AS NOTED	DO NOT SCALE DRAWING	B0950.2025.05		M0.01

SLAC NATIONAL ACCELERATOR LABORATORY — U.S. DEPARTMENT OF ENERGY

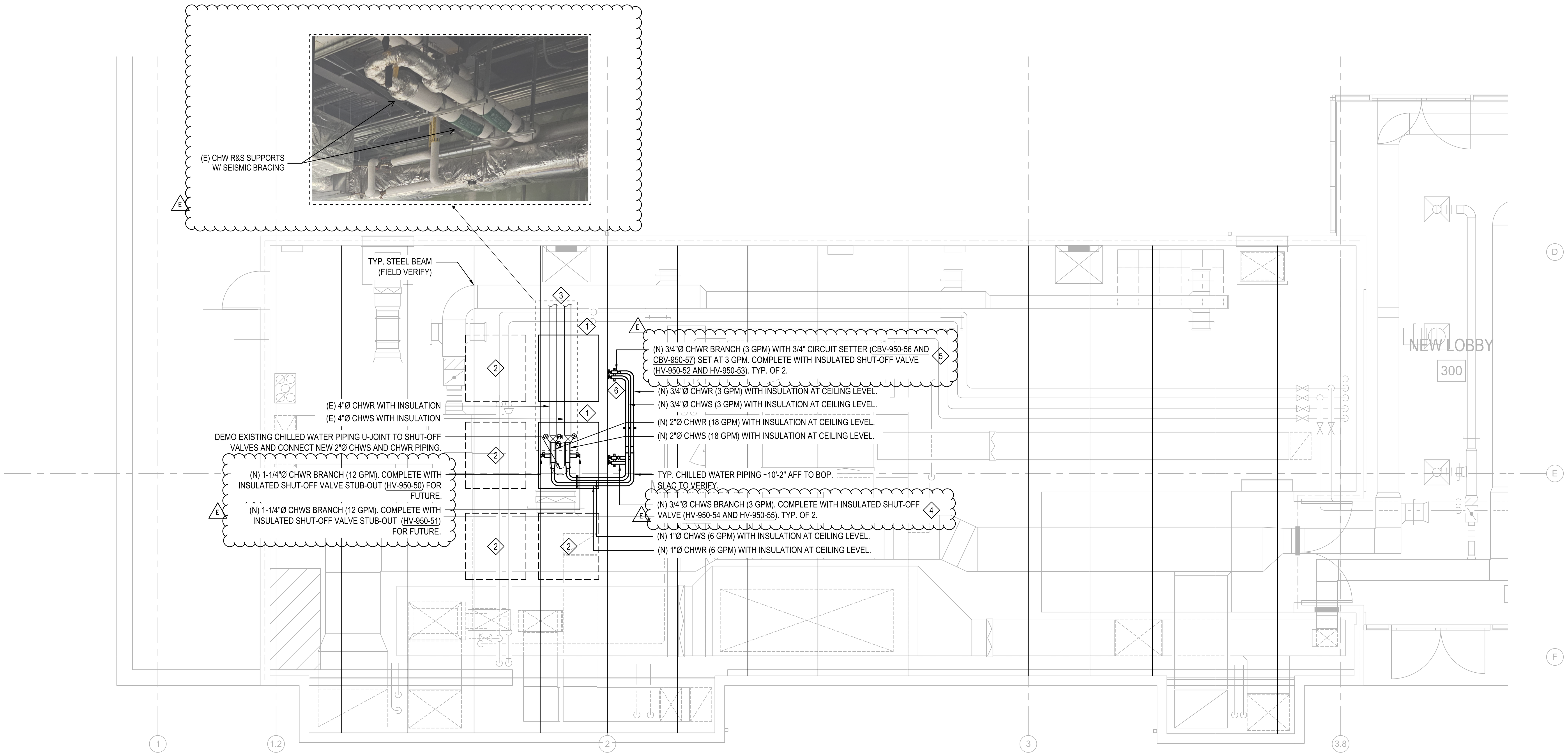
B950
MECH ROOM CO2 CHILLERS
MECHANICAL
COVER SHEET

		DATE	APPROVALS	DATE
ENGR	CHEN, MENG-HSIU	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TRAN, KEVIN	01-26-2026		
CHKR	LI, JIMMY	01-26-2026		

DRAWING NUMBER: B0950-M00-0416 REVISION: E E

B0950- M00 - 0416 E E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	KT	JL	MHC	07-23-2025
B	90% CD	KT	JL	MHC	08-20-2025
C	100% CD	KT	JL	MHC	11-05-2025
D	FAO REVIEW	KT	JL	MHC	11-21-2025
E	BIO REVIEW	KT	JL	MHC	01-26-2026



1 M2.00 **3RD FLOOR MECHANICAL PLAN - CHILLED WATER PIPING**
SCALE: 1/4" = 1'-0"

PLAN GENERAL NOTES

1. EXISTING MECHANICAL TO REMAIN WORK IS INDICATED BY BEING SHOWN LIGHT AND SHOWN ONLY DIAGRAMMATICALLY.
2. NEW MECHANICAL ITEMS ARE INDICATED BY BEING SHOWN BOLD.
3. CONTRACTOR SHALL VERIFY ALL EXISTING CONDITIONS IN THE FIELD.
4. CONTRACTOR SHALL COORDINATE WITH FINAL INSPECTION OF COMPLETED WORK WITH DESIGN ENGINEER AND ONWER OR OWNER'S REPRESENTATIVE, FOR FINAL APPROVAL AND ACCEPTANCE.
5. SWAY BRACING SHALL BE PROVIDED TO PREVENT/LIMIT POTENTIAL IMPACT AND DAMAGE TO ADJACENT COMPONENTS, AND COORDINATE AS NECESSARY WITH FIELD INSPECTION STAFF. THE EOR MAY EVALUATE, BY STRUCTURAL OBSERVATION, AND PROVIDE A WRITTEN DETERMINATION OF THE EXISTING CONDITION AT THE REQUEST OF THE CONTRACTOR AND/OR OWNER.

MECHANICAL SHEET NOTES

1. NEW CO2 CHILLERS TO BE PROVIDED BY SLAC. COORDINATE WITH SLAC FOR EXACT LOCATION AND MORE INFORMATION.
2. PLACEHOLDER FOR FUTURE CO2 CHILLERS TO BE PROVIDED BY SLAC. COORDINATE WITH SLAC FOR EXACT LOCATION AND MORE INFORMATION.
3. EXISTING 4"Ø CHILLED WATER SUPPLY AND RETURN WITH INSULATION. COORDINATE WITH MODULUS SCAN AND FIELD VERIFY FOR EXACT LOCATION AND CONDITIONS. FOR EXISTING CHILLER AND ASSOCIATED EQUIPMENT SCHEDULES REFER TO SLAC AS-BUILT DRAWINGS ID38390547 SHEETS 4 AND 5.
4. SLAC (MARCO ORIUNNO) TO BE RESPONSIBLE FOR CHILLED WATER SUPPLY PIPING CONNECTION TO CO2 CHILLERS AFTER SHUT-OFF VALVE.
5. SLAC (MARCO ORIUNNO) TO BE RESPONSIBLE FOR CHILLED WATER RETURN PIPING CONNECTION TO CO2 CHILLERS AFTER SHUT-OFF VALVE.
6. PROVIDE NEW 1-5/8"x1-5/8"x12" UNISTRUT SUPPORT. PROVIDE INSULATED PIPE SHIELDS CALCIUM SILICATE PIPE INSULATION FOR EACH PIPE. SEE DETAIL 2/M0.01 FOR MORE INFORMATION. (TYP. OF 4)

B0950- M01 - 0417 E E

SLAC BUILDING INSPECTION OFFICE

Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	**PRS#:** 26-009
Revised Authorization ☑ No ☐ Yes	
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MHC ENGINEERS
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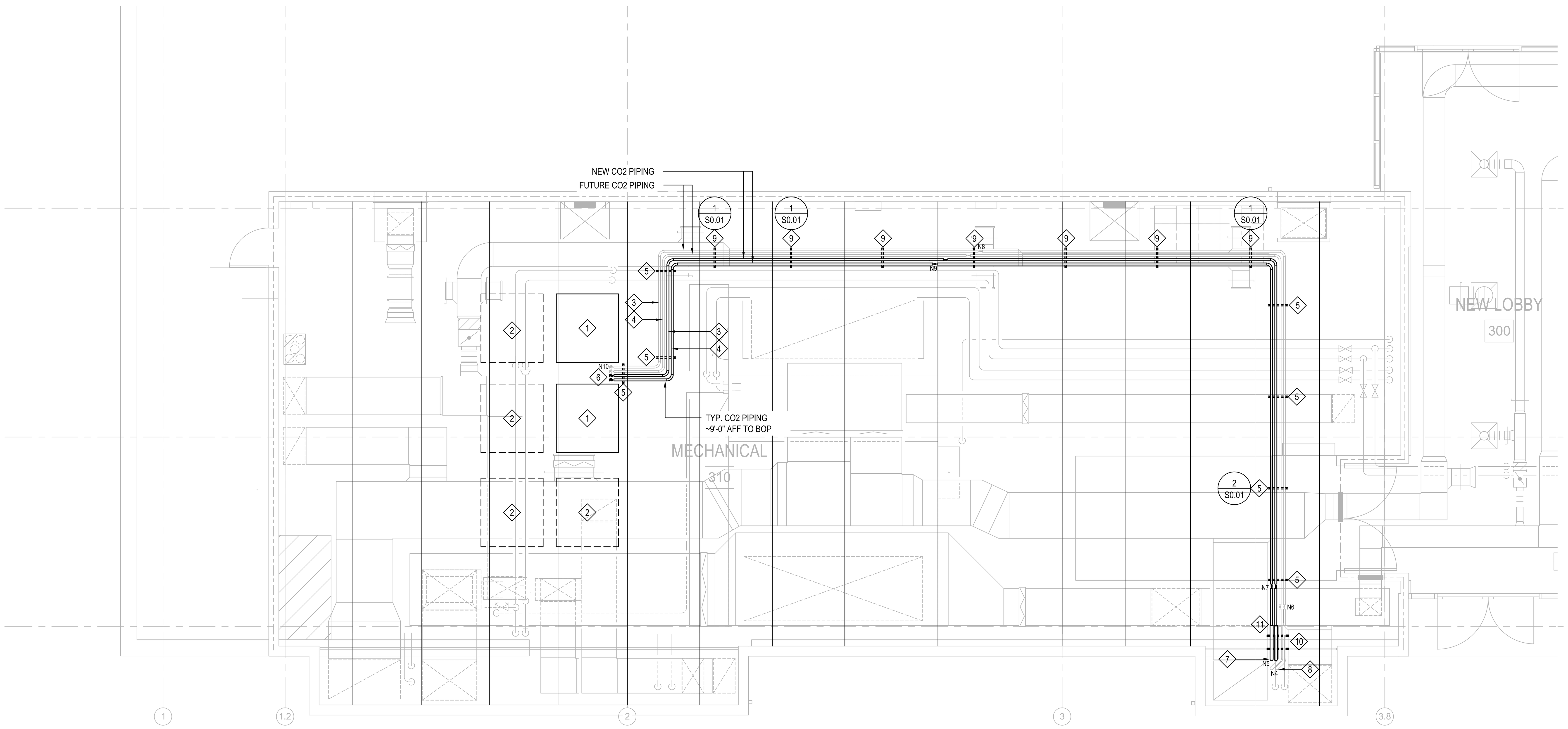
REGISTERED PROFESSIONAL ENGINEER MENG-HSIU CHEN M19582 EXP. 9/30/27 MECHANICAL STATE OF CALIFORNIA

SCALE: AS NOTED	DO NOT SCALE DRAWING	B0950.2025.05		M2.00
SLAC NATIONAL ACCELERATOR LABORATORY / U.S. DEPARTMENT OF ENERGY		B950 MECH ROOM CO2 CHILLERS MECHANICAL 3RD FL CHILLED WATER PIPING PLAN		
THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE COPIED, PUBLISHED OR OTHERWISE FURTHER DISSEMINATED WITHOUT PRIOR WRITTEN PERMISSION OF STANFORD UNIVERSITY / SLAC.				
ENGR CHEN, MENG-HSIU 01-26-2026 DES TRAN, KEVIN 01-26-2026 CHKR LI, JIMMY 01-26-2026	APPROVALS CHEUNG, CANON DATE 01-26-2026	DRAWING NUMBER B0950-M01-0417	REVISION E	E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	KT	JL	MHC	07-23-2025
B	90% CD	KT	JL	MHC	08-20-2025
C	100% CD	KT	JL	MHC	11-05-2025
D	FAO REVIEW	KT	JL	MHC	11-21-2025
E	BIO REVIEW	KT	JL	MHC	01-26-2026



1 M2.01 **3RD FLOOR MECHANICAL PLAN - CO2 PIPING**

SCALE: 1/4" = 1'-0"

PLAN GENERAL NOTES

1. EXISTING MECHANICAL TO REMAIN WORK IS INDICATED BY BEING SHOWN LIGHT.
2. NEW MECHANICAL ITEMS ARE INDICATED BY BEING SHOWN BOLD.
3. COORDINATE LOCATION OF EXISTING DUCTWORK AND PIPING WITH OTHER TRADES.
4. CONTRACTOR SHALL VERIFY ALL EXISTING CONDITIONS IN THE FIELD.
5. CONTRACTOR SHALL COORDINATE WITH FINAL INSPECTION OF COMPLETED WORK WITH DESIGN ENGINEER AND ONWER OR OWNER'S REPRESENTATIVE, FOR FINAL APPROVAL AND ACCEPTANCE.
6. SWAY BRACING SHALL BE PROVIDED TO PREVENT/LIMIT POTENTIAL IMPACT AND DAMAGE TO ADJACENT COMPONENTS, AND COORDINATE AS NECESSARY WITH FIELD INSPECTION STAFF. THE EOR MAY EVALUATE, BY STRUCTURAL OBSERVATION, AND PROVIDE A WRITTEN DETERMINATION OF THE EXISTING CONDITION AT THE REQUEST OF THE CONTRACTOR AND/OR OWNER.

E

MECHANICAL SHEET NOTES

1. NEW CO2 CHILLER TO BE PROVIDED BY SLAC AND INSTALLED PER LCLS-II-HE NEH B950 ROOM 302 RDS. SLAC TO CONFIRM WHETHER CMC CH.11/CFC 608.9/ASHRAE 15-2022 APPLIES TO EXISTING MECHANICAL ROOM. COORDINATE WITH SLAC FOR EXACT SIZE AND LOCATION.
2. POTENTIAL FUTURE CO2 CHILLERS TO BE PROVIDED BY SLAC. COORDINATE WITH SLAC FOR MORE INFORMATION, MAINTENANCE ACCESS, AND REQUIRED ELECTRICAL CLEARANCES.
3. SLAC (MARCO ORIUNNO) TO PROVIDE NEW 1/2"Ø ID CRYOWORKS TRANSFER FLEX HOSE CO2 SUPPLY PIPING. COORDINATE WITH SLAC.
4. SLAC (MARCO ORIUNNO) TO PROVIDE NEW 1/2"Ø ID CRYOWORKS TRANSFER FLEX HOSE CO2 RETURN PIPING. COORDINATE WITH SLAC.
5. NEW 1-5/8"x1-5/8"x18" UNISTRUT SUPPORT. SEE DETAILS 1/M0.01.
6. SLAC (MARCO ORIUNNO) TO BE RESPONSIBLE FOR THE DIRECT CONNECTION OF THE CO2 PIPING TO THE CO2 CHILLERS AFTER 1/2" INSULATED SHUTOFF VALVE STUB-OUTS AT CEILING.
7. SLAC (MARCO ORIUNNO) TO PROVIDE NEW 1/2"Ø ID CRYOWORKS TRANSFER FLEX HOSE CO2 SUPPLY AND RETURN RISERS UP FROM SHAFT TO CEILING OF MECHANICAL ROOM.
8. SLAC (MARCO ORIUNNO) TO BE RESPONSIBLE FOR FUTURE 1/2"Ø ID CRYOWORKS TRANSFER FLEX HOSE CO2 SUPPLY AND RETURN UP FROM SHAFT TO CEILING OF MECHANICAL ROOM. SHOWN FOR REFERENCE ONLY.
9. REFER TO STRUCTURAL DRAWINGS FOR NEW UNISTRUT SUPPORT ANCHORED OFF OF EXISTING DUCT TRAPEZE. SEE DETAIL 1/M0.01.
10. NEW 1-5/8"x1-5/8"x20" UNISTRUT SUPPORT FOR PIPE SUPPORT AT TRANSITION. SEE DETAIL 3/M0.01.
11. PROVIDE SCHEDULE 40 316 SS 2-1/2"Ø PIPE SLEEVE ELBOW FOR CO2 TRANSFER HOSE. REFER TO DETAIL 3/M0.01.

CO2 PIPE LENGTH ESTIMATIONS (REFER TO M2.03 FOR BEGINNING ROUTE)

ASSUMPTIONS:
- STARTING 12'-2" ABOVE SUB-BASEMENT FLOOR
- 3RD FLOOR MECHANICAL ROOM PIPING AT 2' BELOW CEILING
- 6.283" LENGTH PER 90 DEGREE ELBOW (4" RADIUS LENGTH)
- SLAC TO CONFIRM ASSUMPTIONS

LOCATION	NEW SUPPLY DISTANCE	NEW RETURN DISTANCE	FUTURE SUPPLY DISTANCE	FUTURE RETURN DISTANCE
N1 -> N3 (FROM SUB-BASEMENT)	-	-	4.14 FT	4.14 FT
N3 -> N4 (DISTANCE FROM 12'-2" ABOVE SUB-BASEMENT FLOOR TO 1' BELOW 3RD FLOOR MECHANICAL CEILING)	-	-	41.38 FT	41.38 FT
N4 -> N6 (RISER TO NEW SEGMENT CONNECTION)	-	-	4.48 FT	4.48 FT
N6-> N8 (NEW SEGMENT CONNECTION TO NEW SEGMENT CONNECTION)	-	-	50 FT	50 FT
N8-> N10 (NEW SEGMENT CONNECTION TO NEW SHUTOFF VALVE)	-	-	36.36 FT	35.44 FT
N1 -> N2 (FROM SUB-BASEMENT)	3.21 FT	3.21 FT	-	-
N2 -> N5 (DISTANCE FROM 12'-2" ABOVE SUB-BASEMENT FLOOR TO 2' BELOW 3RD FLOOR MECHANICAL CEILING)	41.38 FT	41.38 FT	-	-
N5-> N7 (RISER TO NEW SEGMENT CONNECTION)	5.41 FT	5.41 FT	-	-
N7-> N9 (NEW SEGMENT CONNECTION TO NEW SEGMENT CONNECTION)	50 FT	50 FT	-	-
N9-> N10 (NEW SEGMENT CONNECTION TO NEW SHUTOFF VALVE)	33.70 FT	32.89 FT	-	-
TOTAL PIPE LENGTH	133.7 FT	132.89 FT	138.53 FT	137.79 FT

SLAC BUILDING INSPECTION OFFICE

Construction Authorization (IFC)

BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization	☑ No ☐ Yes

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SAN FRANCISCO, CA 94103
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FAX (415) 512-7120

M19582
EXP. 9/30/27

SCALE: AS NOTED | DO NOT SCALE DRAWING | B0950.2025.05 | M2.01

SLAC NATIONAL ACCELERATOR LABORATORY
U.S. DEPARTMENT OF ENERGY

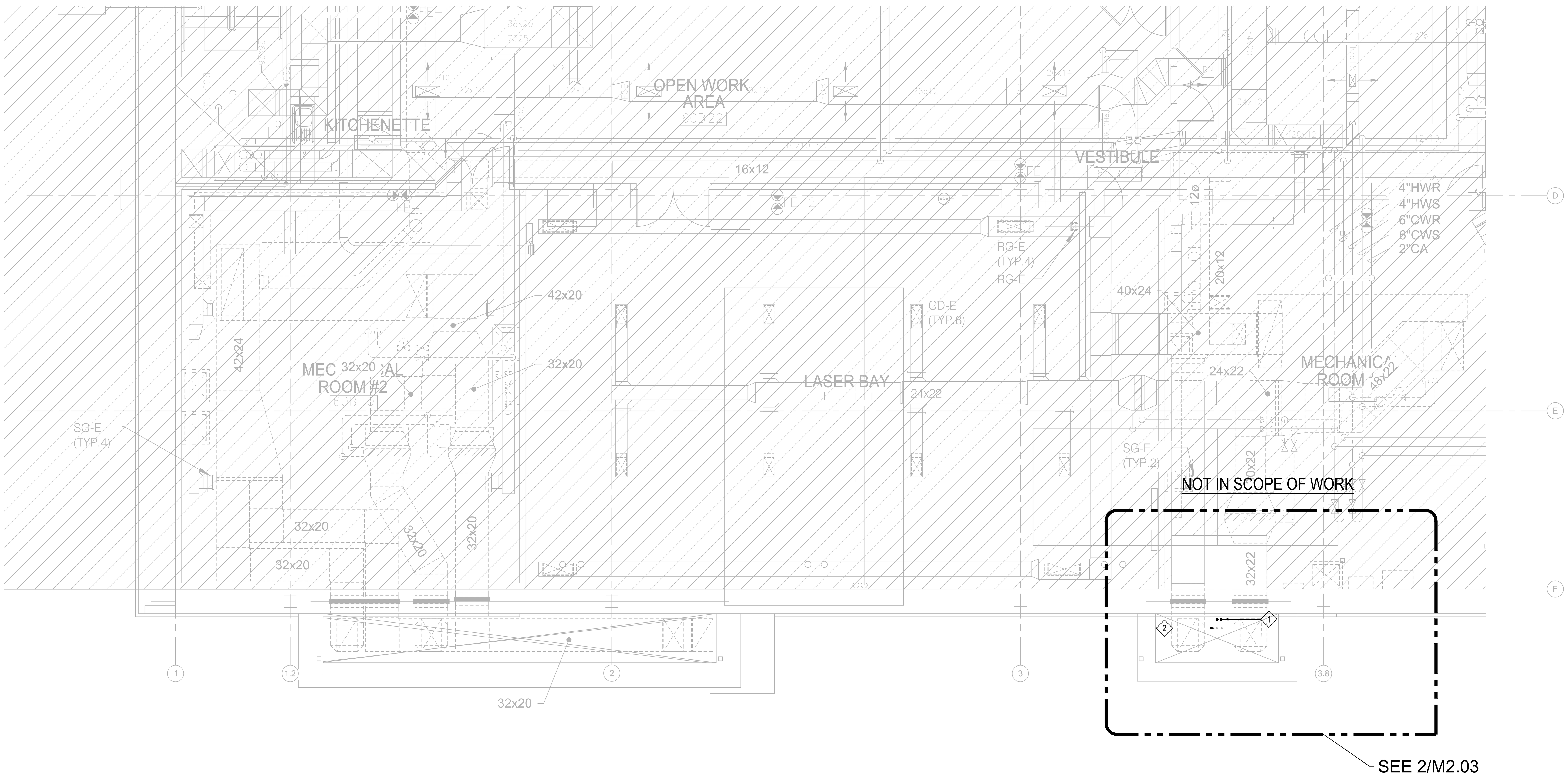
B950
MECH ROOM CO2 CHILLERS
MECHANICAL
3RD FL CO2 PIPING PLAN

		DATE	APPROVALS	DATE
ENGR	CHEN, MENG-HSIU	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TRAN, KEVIN	01-26-2026		
CHKR	LI, JIMMY	01-26-2026		

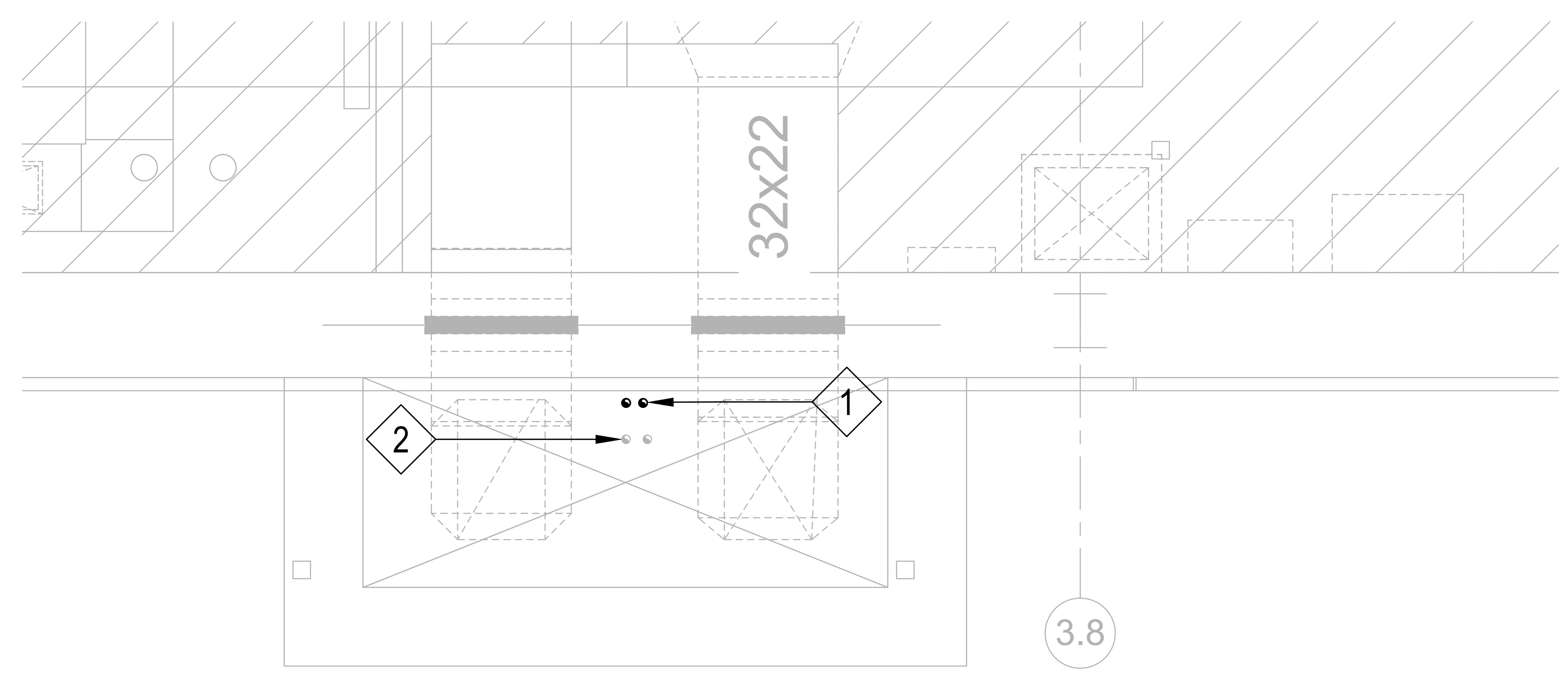
DRAWING NUMBER: B0950-M01-0418 | REVISION: E | E

B0950-M01-0418 E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	KT	JL	MHC	07-23-2025
B	90% CD	KT	JL	MHC	08-20-2025
C	100% CD	KT	JL	MHC	11-05-2025
D	FAO REVIEW	KT	JL	MHC	11-21-2025
E	BIO REVIEW	KT	JL	MHC	01-26-2026



1 M2.02 **BASEMENT MECHANICAL PLAN - CO2 PIPING**
SCALE: 1/4" = 1'-0"



2 M2.02 **BASEMENT ENLARGED MECHANICAL PLAN - CO2 PIPING**
SCALE: 1/2" = 1'-0"

PLAN GENERAL NOTES

1. EXISTING MECHANICAL TO REMAIN WORK IS INDICATED BY BEING SHOWN LIGHT.
2. NEW MECHANICAL ITEMS ARE INDICATED BY BEING SHOWN BOLD.
3. COORDINATE LOCATION OF EXISTING DUCTWORK AND NEW DUCTWORK WITH OTHER TRADES.
4. CONTRACTOR SHALL COORDINATE WITH FINAL INSPECTION OF COMPLETED WORK WITH DESIGN ENGINEER AND ONWER OR OWNER'S REPRESENTATIVE, FOR FINAL APPROVAL AND ACCEPTANCE.

MECHANICAL SHEET NOTES

1. SLAC (MARCO ORIUNNO) TO PROVIDE NEW 1/2"Ø ID CRYOWORKS TRANSFER FLEX HOSE CO2 SUPPLY AND RETURN RISERS WITH INSULATION DOWN. SLAC (MARCO ORIUNNO) TO BE RESPONSIBLE REGARDING SHAFT MOUNTING. PIPING SHOWN FOR REFERENCE ONLY.
2. SLAC (MARCO ORIUNNO) TO BE RESPONSIBLE FOR FUTURE 1/2"Ø ID CRYOWORKS TRANSFER FLEX HOSE CO2 SUPPLY AND RETURN RISERS WITH INSULATION UP. SLAC (MARCO ORIUNNO) TO BE RESPONSIBLE REGARDING SHAFT MOUNTING. PIPING SHOWN FOR REFERENCE ONLY.

SLAC BUILDING INSPECTION OFFICE
Construction Authorization (IFC)
BY: cnadler
DATE: 2/18/2026 PRS#: 26-009
Revised Authorization ☑ No ☐ Yes
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REGISTERED PROFESSIONAL ENGINEER MENG-HSIU CHEN M19582 EXP. 9/30/27 MECHANICAL STATE OF CALIFORNIA

SCALE: AS NOTED	DO NOT SCALE DRAWING	B0950.2025.05		M2.02

SLAC NATIONAL ACCELERATOR LABORATORY
U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
MECHANICAL
BASEMENT CO2 PIPING PLAN

		DATE	APPROVALS	DATE
ENGR	CHEN, MENG-HSIU	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TRAN, KEVIN	01-26-2026		
CHKR	LI, JIMMY	01-26-2026		

DRAWING NUMBER: B0950-M01-0419
REVISION: E
E

B0950- M01 - 0419 E E

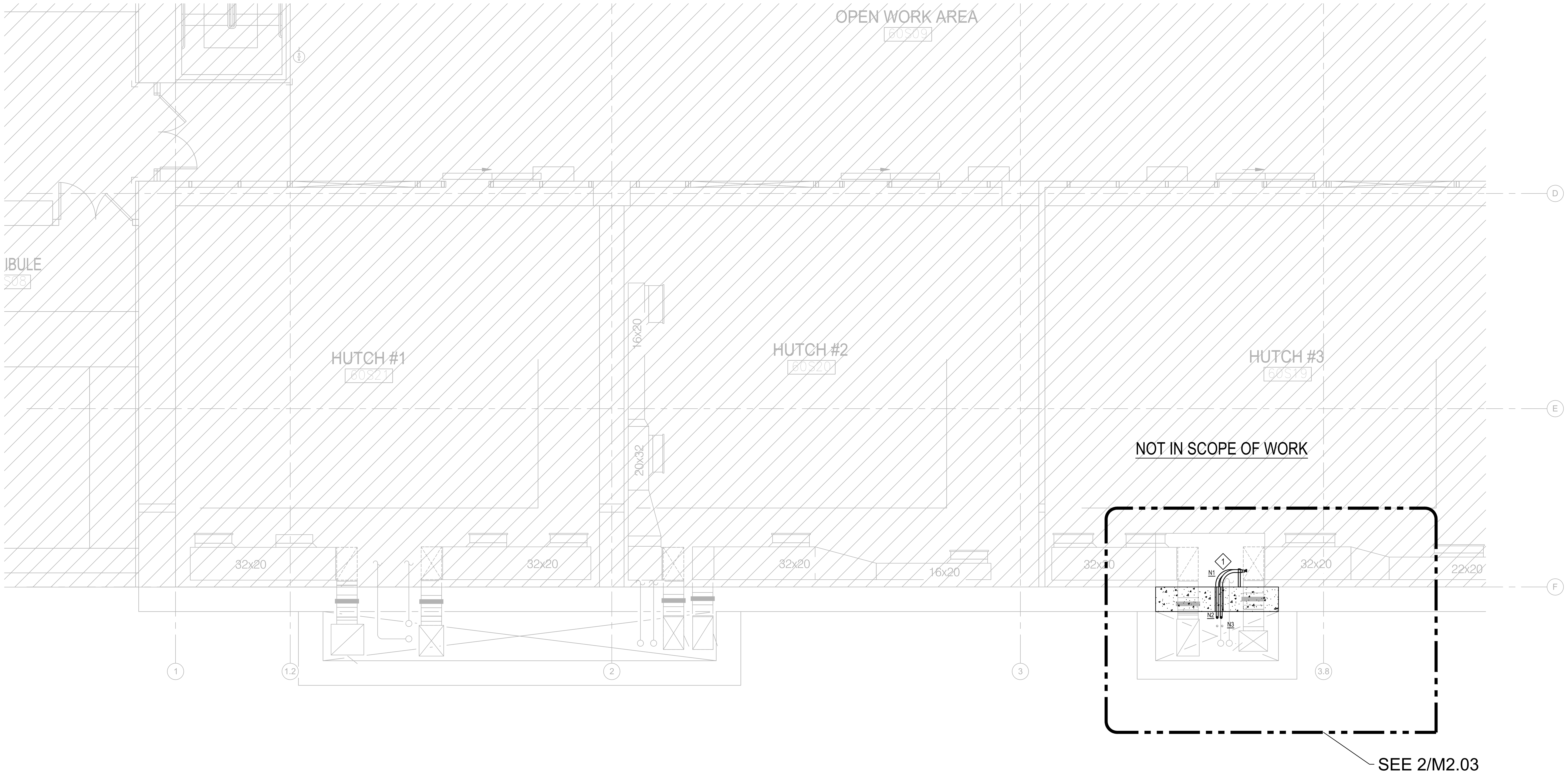
REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	KT	JL	MHC	07-23-2025
B	90% CD	KT	JL	MHC	08-20-2025
C	100% CD	KT	JL	MHC	11-05-2025
D	FAO REVIEW	KT	JL	MHC	11-21-2025
E	BIO REVIEW	KT	JL	MHC	01-26-2026

PLAN GENERAL NOTES

1. EXISTING MECHANICAL TO REMAIN WORK IS INDICATED BY BEING SHOWN LIGHT.
2. NEW MECHANICAL ITEMS ARE INDICATED BY BEING SHOWN BOLD.
3. COORDINATE LOCATION OF EXISTING DUCTWORK WITH OTHER TRADES.
4. CONTRACTOR SHALL COORDINATE WITH FINAL INSPECTION OF COMPLETED WORK WITH DESIGN ENGINEER AND ONWER OR OWNER'S REPRESENTATIVE, FOR FINAL APPROVAL AND ACCEPTANCE.

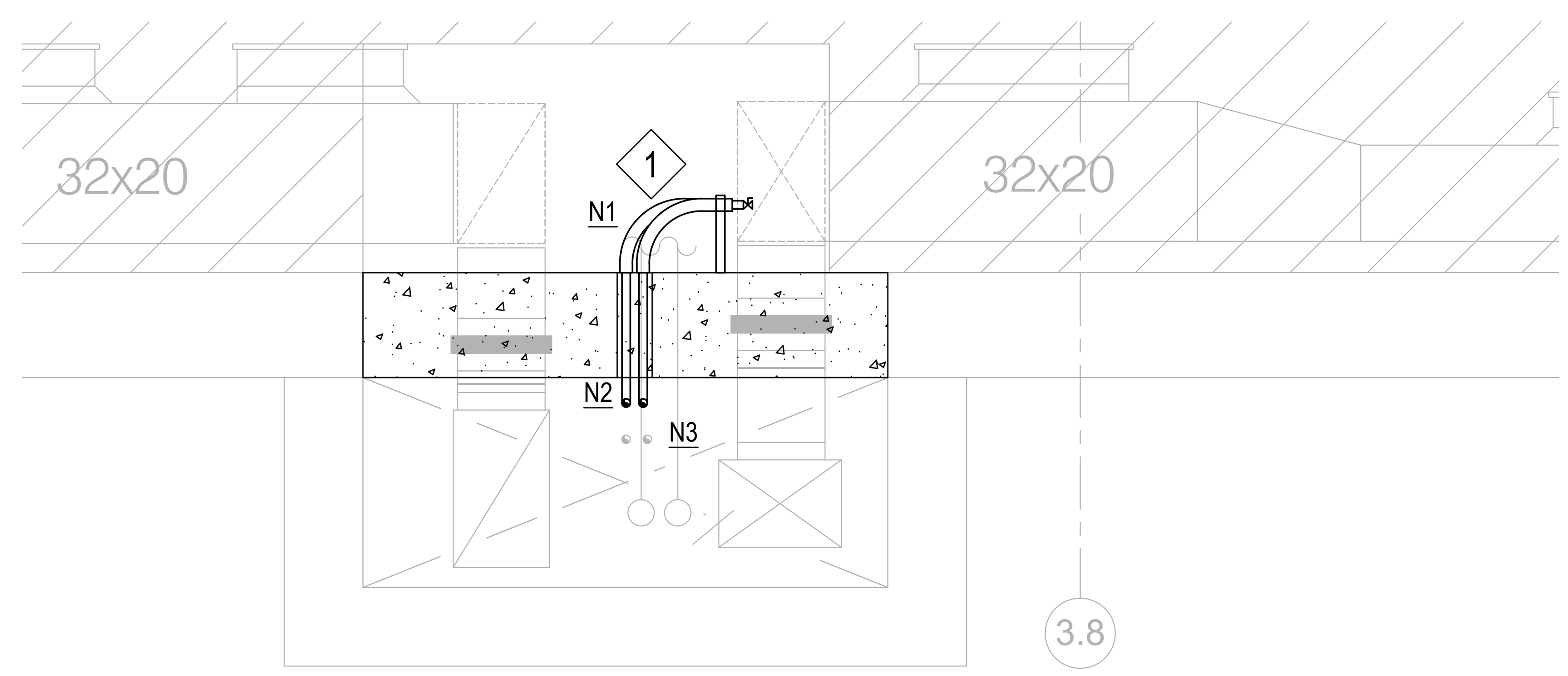
MECHANICAL SHEET NOTES

1 SLAC (MARCO ORIUNNO) TO EXTEND NEW AND FUTURE 1/2"Ø ID CRYOWORKS TRANSFER FLEX HOSE CO2 SUPPLY AND RETURN PIPING 1'-0" WITH CAPPED VALVES FROM SHAFT OPENING. PROVIDE SCHEDULE 40 316 SS 2-1/2"Ø PIPE SLEEVE ELBOW FOR CO2 TRANSFER HOSE. SEE DETAIL 4/M0.01 FOR MORE INFORMATION.



1 M2.03 **SUB-BASEMENT MECHANICAL PLAN - CO2 PIPING**
SCALE: 1/4" = 1'-0"

CO2 PIPE LENGTH ESTIMATIONS				
LOCATION	NEW SUPPLY DISTANCE	NEW RETURN DISTANCE	FUTURE SUPPLY DISTANCE	FUTURE RETURN DISTANCE
N1 -> N2	3.21 FT	3.21 FT	-	-
N1 -> N3	-	-	4.14 FT	4.14 FT



2 M2.03 **SUB-BASEMENT ENLARGED MECHANICAL PLAN - CO2 PIPING**
SCALE: 1/2" = 1'-0"

SLAC BUILDING INSPECTION OFFICE

Construction Authorization (IFC)

BY: cnadler

DATE: 2/18/2026 PRS#: 26-009

Revised Authorization ☑ No ☐ Yes

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M19582 EXP. 9/30/27

SCALE: AS NOTED	DO NOT SCALE DRAWING	B0950.2025.05	M2.03

SLAC NATIONAL ACCELERATOR LABORATORY — U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
MECHANICAL
SUB-BASEMENT CO2 PIPING PLAN

		DATE	APPROVALS	DATE
ENGR	CHEN, MENG-HSIU	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TRAN, KEVIN	01-26-2026		
CHKR	LI, JIMMY	01-26-2026		

DRAWING NUMBER: B0950-M01-0420 REVISION: E E

B0950- M01- 0420 E E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	OT	JO	SK	07-25-2025
B	90% CD	OT	JO	SK	08-20-2025
C	100% CD	OT	JO	SK	11-05-2025
D	FAO REVIEW	OT	JO	SK	01-05-2026
E	BIO REVIEW	OT	JO	SK	01-26-2026

CONTROLS

SYMBOL	DESCRIPTION
	SOLENOID OPEN VALVE
	INDUCTOR
	RESISTOR
	DIODE
	BATTERY
	THERMAL OVERLOAD
STOP	PUSH BUTTON SWITCH – MAINTAINED CONTACT, MUSHROOM HEAD.
	CONTACT – SINGLE POLE DOUBLE THROW.
H OFF A	HAND-OFF-AUTO SWITCH
	PUSH BOTTOM
P	PILOT LIGHT
PF	POWER FACTOR METER
R	RELAY
	CAPACITOR – ADJUSTABLE OR FIXED
	CONTACT – NORMALLY OPEN OR NORMALLY CLOSED.
	LEVEL SWITCH – NORMALLY OPEN OR NORMALLY CLOSED.
	PRESSURE SWITCH – NORMALLY OPEN OR NORMALLY CLOSED.
	FLOW SWITCH – NORMALLY OPEN OR NORMALLY CLOSED.
	TEMPERATURE SWITCH – NORMALLY OPEN OR NORMALLY CLOSED.
	PUSH BUTTON SWITCH – MOMENTARY CONTACT, NORMALLY OPEN OR NORMALLY CLOSED.
	RELAY CONTACT – TIME DELAY OPEN, INSTANTANEOUS CLOSED OR TIME DELAY CLOSED, INSTANTANEOUSLY OPEN.
	RELAY CONTACT – INSTANTANEOUS OPEN, TIME DELAY CLOSED OR INSTANTANEOUS CLOSED. TIME DELAY OPEN.
	LIMIT SWITCH – NORMALLY OPEN OR NORMALLY CLOSED.
	SWITCH – NORMALLY OPEN OR NORMALLY CLOSED.

SINGLE-LINE DIAGRAM

SYMBOL	DESCRIPTION
	CIRCUIT BREAKER
	ENCLOSED CIRCUIT BREAKER
ST	CIRCUIT BREAKER WITH SHUNT TRIP
	DRAW-OUT CIRCUIT BREAKER
	DRAW-OUT FUSED SWITCH
XA	VACUUM CIRCUIT BREAKER
M	UTILITY METER
	MAGNETIC STARTER – NEMA SIZE AS INDICATED FVNR UNLESS 2S1W, 2S2W OR FVNR INDICATED
	CONTROL TRANSFORMER WITH FUSING
	COMBINATION MOTOR STARTER/DISCONNECT
LA	LIGHTNING ARRESTER
	DISCONNECT
	FUSE DISCONNECT
	DELTA GROUNDED – "Y" CONNECTED
	DELTA – DELTA CONNECTED
	CURRENT TRANSFORMER
(3)	POTENTIAL TRANSFORMER (NUMBER INDICATES QUANTITY)
	NEUTRAL LINK
	MEDIUM VOLTAGE CABLE TERMINATION
	DRAW OUT
	GROUND
	CHASSIS
	FUSE
M	AC MOTOR WITH THERMAL MANUAL STARTER
K	KIRK KEY INTERLOCK
V	VOLTMETER
A	AMMETER
KW	KILOWATT-METER
GFI	GROUND FAULT INTERRUPTER
VFD	VARIABLE FREQUENCY DRIVE
VFD	VARIABLE FREQUENCY DRIVE W/BYPASS
	CONTINUOUS GROUND BUS
	TRANSFORMER
N E	AUTOMATIC TRANSFER SWITCH
G	ENGINE GENERATOR
	PANELBOARD

LIGHTING PLAN

SYMBOL	DESCRIPTION
	2x4 SURFACE MOUNTED LIGHT FIXTURE
	2x2 SURFACE MOUNTED LIGHT FIXTURE
	1x4 SURFACE MOUNTED LIGHT FIXTURE
	2x4 RECESSED MOUNTED LIGHT FIXTURE
	2x2 RECESSED MOUNTED LIGHT FIXTURE
	1x4 RECESSED MOUNTED LIGHT FIXTURE
	2x4 RECESSED MOUNTED EMERGENCY/NIGHT LIGHT FIXTURE
	2x2 RECESSED MOUNTED EMERGENCY/NIGHT LIGHT FIXTURE
	1x4 RECESSED MOUNTED EMERGENCY/NIGHT NIGHT FIXTURE
	8 FT. LONG PENDANT MOUNTED STRIP LIGHT FIXTURE
	4 FT. LONG PENDANT MOUNTED STRIP LIGHT FIXTURE
	8 FT. LONG SURFACE MOUNTED STRIP LIGHT FIXTURE
	4 FT. LONG SURFACE MOUNTED STRIP LIGHT FIXTURE
	8 FT. LONG PENDANT MOUNTED EMERGENCY/NIGHT LIGHT STRIP
	4 FT. LONG PENDANT MOUNTED EMERGENCY/NIGHT LIGHT STRIP
	TRACK LIGHTS
	RECESSED DOWNLIGHT
	RECESSED DOWNLIGHT ON EMERGENCY CIRCUIT
	SURFACE MOUNTED DOWNLIGHT
	SURFACE MOUNTED DOWNLIGHT ON EMERGENCY CIRCUIT
	RECESSED WALL-WASH DOWNLIGHT
	SURFACE WALL-WASH DOWNLIGHT
	WALL MOUNTED LIGHT FIXTURE
	WALL MOUNTED EMERGENCY LIGHT FIXTURE
	CEILING AND WALL MOUNTED EXIT LIGHT WITH DOUBLE FACE ARROW INDICATES DIRECTION
	CEILING AND WALL MOUNTED EXIT LIGHT WITH SINGLE FACE ARROW INDICATES DIRECTION
	EMERGENCY EXIT LIGHT WITH BATTERY BACK UP
	EMERGENCY EXIT LIGHT WALLPACK COMBO
OS	CEILING MOUNTED OCCUPANCY SENSOR
	CEILING MOUNTED OCCUPANCY SENSOR W/POWER PACK
OS	WALL MOUNTED OCCUPANCY SENSOR
OS	WALL MOUNTED OCCUPANCY SENSOR W/POWER PACK
OS	OCCUPANCY SENSOR, WALL MOUNTED, SMALL LETTER INDICATES SWITCH LEG FOR CONTROL
OS	COMBINE OCCUPANCY SENSOR AND DUAL RELAY SWITCH, SMALL LETTERS INDICATE SWITCH LEG FOR CONTROL
PC	PHOTO ELECTRIC CELL
$	LINE VOLTAGE MANUAL SWITCH
	DUAL POLE SWITCH, 277V LINE
$	TIMER SWITCH WITH 1-HR RANGE
$	THREE-WAY SWITCH, SMALL LETTER INDICATES SWITCH LEG FOR CONTROL
	LOW VOLTAGE MANUAL SWITCH
S	SENTRY SWITCH
D	DIMMER SWITCH
	WALL SCONCE
	WALL MOUNTED FLOOD LIGHT
	PENDANT MOUNTED HIGH BAY LIGHT FIXTURE
	WALL MOUNTED EXTERIOR LIGHT
	POLE MOUNTED STREET/PARKING SINGLE HEAD ROUND FIXTURE
	POLE MOUNTED STREET/PARKING DOUBLE HEAD ROUND FIXTURE
	POLE MOUNTED STREET/PARKING SINGLE HEAD SQUARED FIXTURE
	POLE MOUNTED STREET/PARKING DOUBLE HEAD SQUARED FIXTURE

POWER PLAN

SYMBOL	DESCRIPTION
	DUPLEX RECEPTACLE
	DUPLEX RECEPTACLE – GFCI ABOVE BACKSPLASH
	DOUBLE DUPLEX RECEPTACLE
	SPECIAL PURPOSE OUTLET
	DUPLEX RECEPTACLE, FLUSH IN FLOOR
	DOUBLE DUPLEX RECEPTACLE, FLUSH IN FLOOR
	COMBINATION POWER AND SIGNAL, FLUSH IN FLOOR
	PUSH BUTTON
S	SOLENOID VALVE
T	WALL MOUNTED THERMOSTAT
J	WALL MOUNTED JUNCTION BOX
J	CEILING MOUNTED JUNCTION BOX
J	FLOOR MOUNTED JUNCTION BOX
PP J	POWER POLE
$M	MANUAL MOTOR STARTER SWITCH
	DISCONNECT SWITCH – NON FUSED
	DISCONNECT SWITCH – FUSED
	DISCONNECT SWITCH (ENCLOSED CIRCUIT BREAKER)
	ENCLOSED STARTER
	COMBINATION MAGNETIC MOTOR STARTER WITH FUSES
M	CONNECTION POINT TO MOTOR SPECIFIED, SUPPLIED, AND INSTALLED UNDER OTHER DIVISIONS, RACEWAY, CONDUCTORS, AND CONNECTIONS UNDER THIS DIVISION. REFER TO ONE-LINE OR SCHEDULE
AVT	ABSENCE OF VOLTAGE TESTER
MH	MANHOLE
PB	PULLBOX
	GROUND ROD
	AIR TERMINAL
	TEST WELL
GND	GROUND BUS
	208/120V PANELBOARD, SURFACE MOUNTED
	208/120V PANELBOARD, RECESSED MOUNTED
	480/277V PANELBOARD, SURFACE MOUNTED
	480/277V PANELBOARD, RECESSED MOUNTED
	CABINET ENCLOSURE, SURFACE MOUNTED
	CABINET ENCLOSURE, RECESSED MOUNTED
T	TRANSFORMER
	CEILING FAN
	SURFACE MOUNTED WIREWAY (WIREMOLD)
	EQUIPMENT CONNECTION
	CONDUIT CONNECTION
	FLEX CONDUIT
	GUY WIRE POLE
	UTILITY POLE
G	CADWELD GROUND CONNECTION
G	GROUND CABLE
	CONDUIT SEAL, "EYS" SERIES, FILLED WITH COMPOUND.
	CONDUIT TURNING – UP
	CONDUIT TURNING – DOWN
	CONDUIT STUBBED AND CAPPED, LOCATION AS INDICATED
C	DUCTBANK, SITE PLAN C=CONTROL, E=EMERGENCY POWER, P=POWER, T=TELEPHONE/DATA, U=UTILITY COMPANY
	CONDUIT WITH CONDUCTORS – CONCEALED OR EXPOSED
	CONDUIT WITH CONDUCTORS – BELOW GRADE OR SLAB
E	CONDUIT AND CONDUCTORS ON EMERGENCY CIRCUIT
CO	CONDUIT ONLY
DB	DUCT BANK
BD	BUS DUCT
	PLUG-IN CIRCUIT BREAKER ON THE BUS DUCT, SIZE AS NOTED ON THE PLAN
ISOLATION GND, GND, CIRCUIT HOMERUN, HOT WIRES, NEUTRAL, IG	HOMERUN TO SWITCHBOARD, PANELBOARD, PANEL & CABINET, ETC.

TELECOMMUNICATION AND LOW VOLTAGE

SYMBOL	DESCRIPTION
	TELEPHONE OUTLET – WALL MOUNTED
	TELEPHONE OUTLET – FLUSH IN FLOOR
	DATA OUTLET – WALL MOUNTED
	DATA OUTLET – FLUSH IN FLOOR
	TEL/DATA – WALL MOUNTED
	TEL/DATA – FLUSH IN FLOOR
CP J	COMMUNICATION POLE
T	TELEPHONE BACKBOARD

ABBREVIATIONS

ABB	DESCRIPTION	ABB	DESCRIPTION
A	AMMETER, AMPERE	M	MAGNETIC. COIL OR CONTACT
AC	ALTERNATING CURRENT	MCA	MINIMUM CIRCUIT AMPS
AF	AMPERE FRAME	MCB	MAIN CIRCUIT BREAKER
AFF	ABOVE FINISHED FLOOR	MCC	MOTOR CONTROL CENTER
AFG	ABOVE FINISHED GRADE	MCCB	MOLDED CASE CIRCUIT BREAKER
ANN	ANNUNCIATOR	MCM	THOUSAND CIRCULAR MILS
AL	ALUMINUM	MCP	MOTOR CIRCUIT PROTECTOR
ARF	ABOVE RAISED FLOOR	MFR	MANUFACTURER
AS	AMMETER SWITCH	MH	MANHOLE
AT	AMPERE TRIP	MLO	MAIN LUGS ONLY
ATS	AUTOMATIC TRANSFER SWITCH	MO	MANUAL OPERATOR
AVT	ABSENCE OF VOLTAGE TESTER	MT, MTD	MOUNT, MOUNTED
		MTR	MOTOR
BRKR	BREAKER		
BLDG	BUILDING	(N)	NEW
B.O.C	BOTTOM OF CONCRETE	(NL)	NEW LOCATION
B.O.D	BOTTOM OF DUCT	N	NEUTRAL
B.O.P	BOTTOM OF PIPE	NC	NORMALLY CLOSED
		NEC	NATIONAL ELECTRICAL CODE
C	CONDUIT	NEMA	NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
CB	CIRCUIT BREAKER		
CCTV	CLOSED-CIRCUIT TELEVISION	NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
CEC	CALIFORNIA ELECTRICAL CODE		
CKT	CIRCUIT	NIC	NOT IN CONTRACT
CM	CONSTRUCTION MANAGER	NL	NIGHT LIGHT
CO	CONDUIT ONLY	NO	NORMALLY OPEN
CPT	CONTROL POWER TRANSFER	NTS	NOT TO SCALE
CR	CONTROL RELAY		
CRS	COATED RIGID STEEL	OFCI	OWNER FURNISHED, CONTRACTOR INSTALLED
CSRM	CONST. SAFETY REQMT. MANUAL	OFOI	OWNER FURNISHED, OWNER INSTALLED
CT	CURRENT TRANSFORMER	OL	OVERLOAD RELAY
CU	COPPER		
		PB	PUSH BUTTON
DB	DUCT BANK	PC	PHOTOCELL
DC	DIRECT CURRENT	PH	PHASE
DIV	DIVISION	PIV	POST INDICATOR VALVE
DPDT	DOUBLE POLE, DOUBLE THROW	PL	PILOT LIGHT
		PM	PROJECT MANAGER
E	EMPTY, EMERGENCY	PNL	PANEL
(E)	EXISTING	PVC	POLYVINYL CHLORIDE
EC	ELECTRICAL CONTRACTOR		
EG	ENGINE GENERATOR	QTY	QUANTITY
EMO	EMERGENCY MANUAL OFF		
EMT	ELECTRICAL METALLIC TUBING	RCPT	RECEPTACLE
ENCL	ENCLOSURE	(RE)	RELOCATED EXISTING
ENT	ELECTRICAL NONMETALLIC TUBING	RM	ROOM
EOL	END OF LINE	RNC	RIGID NONMETALLIC CONDUIT
EWC	ELECTRIC WATER COOLER	RQD	REQUIRED
EWH	ELECTRIC WATER HEATER	(RR)	REMOVE AND REPLACE
		RS	RIGID STEEL OR RAPID START
F	FUSE	RVNR	REDUCED VOLTAGE NONREVERSING
FA	FIRE ALARM	RVR	REDUCED VOLTAGE REVERSING
FC	FOOT CANDLE		
(F)	FUTURE	SCP	SECURITY CONTROL PANEL
FDR	FEEDER	SMD	SEE MECHANICAL DRAWINGS
FLA	FULL LOAD AMPS	SPKR	SPEAKER
FLR	FLOOR	SPST	SINGLE POLE, SINGLE THROW
FSD	FIRE SMOKE DETECTOR	SS	STAINLESS STEEL
FT	FEET, FOOT	SSD	SEE STRUCTURAL DRAWINGS
FU	FUSE	SURF	SURFACE
FVNR	FULL VOLTAGE NONREVERSING	SW	SWITCH
FVR	FULL VOLTAGE REVERSING	SWBD	SWITCHBOARD
		SWGR	SWITCHGEAR
(G)	GROUND	T	TRANSFORMER
G, GND	GROUND	TB	TERMINAL BLOCK
GFI	GROUND FAULT INTERRUPTER	TBD	TO BE DETERMINED
GFCI	GROUND FAULT CIRCUIT INTERRUPTER	TEL	TELEPHONE
GFR	GROUND FAULT RELAY	TSP	TWISTED SHIELDED PAIR
GRC	GALVANIZED RIGID CONDUIT	TVSS	TRANSIENT VOLTAGE SURGE SUPPRESSOR
		TYP	TYPICAL
H	HORN		
HH	HANDHOLE	UON	UNLESS OTHERWISE NOTED
HID	HIGH-INTENSITY DISCHARGE	UPS	UNINTERRUPTIBLE POWER SUPPLY
HOA	HAND-OFF-AUTO		
HTR	HEATER	V	VOLTMETER, VOLT
HT	HEIGHT	VA	VOLT-AMPERE
		VAR	VOLT-AMPERE REACTIVE
ID	INSIDE DIAMETER	VFD	VARIABLE FREQUENCY DRIVE
IMC	INTERMEDIATE METALLIC CONDUIT		
IG	ISOLATED GROUND	W	WATT, WIRE
I/O	INPUT-OUTPUT	WH	WATT-HOUR METER
		WP	WEATHERPROOF
JB	JUNCTION BOX		
		XP	EXPLOSIONPROOF-CLASS. DIVISION AND GROUP AS NOTED
K	KIRK KEY INTERLOCK OR KEY OPERATED	XFMR	TRANSFORMER
KCMIL	THOUSAND CIRCULAR MILS		
KA	KILOAMPERE	Y	WYE CONNECTED
KVA	KILOVOLT-AMPERE	Z	IMPEDANCE
KVAR	KILOVOLT-AMPERE REACTIVE		
KW	KILOWATT	1ϕ	SINGLE PHASE
KWH	KILOWATT HOUR	3ϕ	THREE PHASE

MOUNTING HEIGHTS

MOUNTING HEIGHT SHALL BE TO CENTER LINE (C/L) OF BOX, UNLESS OTHERWISE NOTED. WHERE BACK BOXES OF DIFFERENT SIZES ARE USED AND MOUNTED NEXT TO EACH OTHER OR ON THE SAME WALL, USE A COMMON CENTER LINE.

RECEPTACLES	18" TO C/L	VOLUME CONTROLS	48" TO C/L
RECEPTACLE WIREWAYS	43" TO C/L	THERMOSTATS	48" TO C/L
SWITCHES	48" TO C/L	TELE/COMMUNICATION OUTLET	18" TO C/L
MOTOR STARTERS	72" TO TOP	PANELBOARDS	72" TO TOP
DISCONNECTS	72" TO TOP	CABINETS	72" TO TOP
JUNCTION BOXES	18" TO C/L	FIRE ALARM PULL STATIONS	48" TO C/L

REFERENCE SYMBOLS

SYMBOL	DESCRIPTION
	DETAIL # / SHEET #
	SECTION # / SHEET #
	EQUIPMENT TAG # / SCHEDULE
	LIGHT FIXTURE # / WINDOW TAG #
	FEEDER TAG #
	REVISION #
	KEYED NOTE #
	POINT OF CONNECTION TO EXISTING (POC)
	ELEVATION DATUM
	EXISTING TO BE REMOVED (DEMOLISHED)
	EXISTING TO REMAIN
	NEW

GENERAL NOTES

1. THE COMPLETE ELECTRICAL INSTALLATION SHALL BE IN ACCORDANCE WITH THE LATEST EDITION OF NATIONAL ELECTRICAL CODE, THE LATEST RULES AND REGULATIONS OF THE SAFETY ORDERS ISSUED BY THE DIVISION OF INDUSTRIAL SAFETY, THE NATIONAL FIRE PROTECTION ASSOCIATION (NFPA), AND ALL APPLICABLE STATE AND LOCAL CODES ISSUED BY AUTHORITIES HAVING JURISDICTION.
2. LOCATION(S) OF ELECTRICAL EQUIPMENT AS SHOWN ON THE PLAN IS APPROXIMATE AND MAY BE CHANGED TO SUIT FIELD CONDITIONS AS DIRECTED BY THE CM.
3. UNLESS OTHERWISE NOTED, ALL GROUNDING/BONDING CONDUCTORS WIRE SHALL BE 4/0 AWG BARE COPPER.
4. UPON COMPLETION OF ELECTRICAL WORK, THE CONTRACTOR SHALL SCHEDULE AND PERFORM A COMPLETE FIELD AND FUNCTIONAL TESTING TO DEMONSTRATE TO THE CONSTRUCTION MANAGER/PM THAT THE NEW INSTALLATION IS OPERATING AS INTENDED.
5. ALL ITEMS SHOWN ON NEW WORK DRAWINGS ARE NEW, U.O.N. ALL ITEMS SHOWN ON REMOVAL WORK DRAWINGS ARE EXISTING, U.O.N.
6. ONLY IRREVERSIBLE TYPES OF GROUNDING CONNECTORS SHALL BE USED. MECHANICAL CONNECTORS SHALL NOT BE USED FOR GROUNDING.
7. THE ELECTRICAL DRAWINGS ARE DIAGRAMMATIC AND DO NOT SHOW ALL OFFSETS, BENDS, FITTINGS, JUNCTION BOXES, PULL BOXES, AND EXPANSION FITTINGS REQUIRED TO MEET FIELD CONDITIONS. CONTRACTOR SHALL DETERMINE ACTUAL MATERIAL AND HARDWARE REQUIREMENTS AND VERIFY ALL DIMENSIONS.
8. ELECTRICAL EQUIPMENT MUST BE LISTED AND LABELED BY A NATIONALLY RECOGNIZED TESTING LABORATORY, SUCH AS UL, AND BE INSTALLED AND USED IN ACCORDANCE WITH ANY INSTRUCTIONS INCLUDED IN THE LISTING AND LABELING. EQUIPMENT THAT IS NOT LISTED AND LABELED, OR HAS BEEN MODIFIED, REQUIRES 3RD PARTY CERTIFICATION FROM A FIELD EVALUATION BODY APPROVED BY SLAC BIO.
9. INSPECTOR WILL VERIFY TORQUE TO SPECS ON CONDUCTOR TERMINATIONS, TRANSFORMERS, BREAKERS, PANELS, BUSWAY TAP CANS, WIRE CONNECTORS, BOLTED BUS CONNECTIONS, ETC. IF NO MARKINGS ARE PRESENT, USE THE VALUES IN NEC ANNEX I. TORQUE VALUES SHALL BE DEMONSTRATED IN THE PRESENCE OF THE INSPECTOR USING A CALIBRATED TOOL TO ACHIEVE THE INDICATED TORQUE VALUE OF THE EQUIPMENT. TYPICAL TORQUE WRENCH CALIBRATIONS ARE VALID FOR A PERIOD OF SIX MONTHS OR 5,000 USES, WHICHEVER OCCURS FIRST. THE INSPECTION MUST BE DONE WITH THE EQUIPMENT IN A DE-ENERGIZED CONDITION.

DEMOLITION NOTES (CONTRACTORS)

1. REFER TO THE CONSTRUCTION SAFETY MANUAL.

SLAC ELECTRICAL SAFETY PROCEDURES

1. NO ENERGIZED ELECTRICAL WORK IS ALLOWED. FOR ALL WORK REQUIRING LOTO, SLAC AUTHORIZED WORKERS SHALL LOCK FIRST THEN CONTRACTORS WILL LOCK OVER THE SLAC LOCK(S).
2. ALL BREAKER AND DISCONNECT OPERATION/SWITCHING SHALL BE PERFORMED BY SLAC QUALIFIED ELECTRICAL WORKERS. CONTACT THE FIELD CONSTRUCTION MANAGER FOR ASSISTANCE WITH SWITCHING AND BREAKER OPERATION.
3. COORDINATE INSTALLATION OF ARC-FLASH LABELS WITH ELECTRICAL EQUIPMENT BEFORE THE EQUIPMENT IS TIED INTO THE ELECTRICAL SUPPLY. AFTER EQUIPMENT IS TIED INTO THE ELECTRICAL SUPPLY, ACCESS TO THE EQUIPMENT SHALL BE CONTROLLED UNDER THE LOTO PROGRAM. ARC-FLASH LABELS SUPPLIED BY SLAC.
4. SEE CONSTRUCTION SAFETY REQUIREMENTS MANUAL (CSRM) FOR ELECTRICAL DEMOLITION REQUIREMENTS.
5. PRIOR TO ENERGIZATION OF ELECTRICAL EQUIPMENT OR JUNCTION BOXES WHERE SPLICES HAVE BEEN MADE, ALLOW SLAC ELECTRICIANS TO INSPECT INSTALLATION INCLUDING WIRING TERMINATION, WORKMENSHIP, TORQUE SEALS, ETC. OPEN COVERS & DEAD FRONTS FOR INSPECTION. COORDINATE WORK WITH SLAC CM.

SCOPE SUMMARY

1. REMOVE (E) ATS AND TRIM BACK (E) FEEDERS AS REQUIRED TO SUPPORT THE (N) SCOPE OF WORK. GRIND DOWN (E) ATS CONC. PAD TO FLOOR LEVEL.
2. AT (E) 4MCC04-950 RELOCATE (E) BUCKET D3 TO BELOW LOCATION E4. PROVIDE (N) 400A BUCKET W/ 400A LSIG BREAKER AND ALL REQ. HARDWARE INTO LOCATION D3.
3. RELABEL (E) AND (N) EQUIPMENT TO SLAC EQUIPMENT NAMING CONVENTION.
4. PROVIDE (N) FEEDER FROM (N) 400A BUCKET AT (E) 4MCC04-950 TO (E) 480VAC PANEL B950-4PB-1001 (4SDBP1A-950)*.
5. PROVIDE (N) FEEDER FROM (E) 480VAC PANEL B950-4PB-1001 TO (N) 480VAC PANEL B950-4PB-1002.
6. PROVIDE (N) 480VAC PANEL B950-4PB-1002, W/ MIN. (6) 3P 40A LSIG BREAKERS FOR (2) (N) AND (4) FUTURE CO2 CHILLERS.
7. PROVIDE (6) (N) DROP CORDS W/ MELTRICS RECEPTACLES FOR (2) (N) AND (4) FUTURE CO2 CHILLERS, RESULTING PHASE ROTATION ORDER SHALL BE THE SAME FOR ALL RECEPTACLES, WIRE RECEPTACLES ACCORDINGLY.
8. FURNISH (2) (N) NON-FUSED DISCONNECTS W/ MELTRICS INLETS FOR (2) (N) CO2 CHILLERS, HAND-OFF TO SLAC. INSTALLATION BY SLAC.
9. PROVIDE (N) 1P 20A BREAKER AT (E) PANEL B950-2PB-1002 (2SDBP1A-950)* FOR (N) CONVENIENCE RECEPTACLE.
10. PROVIDE (N) CONVENIENCE RECEPTACLE W/ (N) BRANCH CIRCUIT FROM (E) PANEL B950-2PB-1002 (2SDBP1A-950)*.
11. PERFORM PHASE ROTATION TEST FOR (N) MCC BUCKET, (N) AND (E) PANELS, (E) TRANSFORMER, AND (N) MELTRICS RECEPTACLES IN THE SCOPE OF WORK. SUBMIT RESULTS TO EEOR AND SLAC FOR REVIEW AND COORDINATION, MARK EACH PIECE OF EQUIPMENT W/ PHASE ROTATION ORDER.

* NOTE: PANELS W/ NAME IN PARENTHESIS ARE (E) AND REQUIRE RELABELING, OLD NAME SHOWN IN PARENTHESIS.

SLAC BUILDING INSPECTION OFFICE

Construction Authorization (IFC)

BY: cnadler

DATE: 2/18/2026 PRS#: 26-009

Revised Authorization ☑ No ☐ Yes

These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.

The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.

THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

SYMBOLS, NOTES AND ABBREVIATIONS LISTED ARE FOR GENERAL USE. DISREGARD THOSE WHICH ARE NOT USED ON THE DRAWINGS.

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REGISTERED PROFESSIONAL ENGINEER SERGEY V. KOROLEV E20491 EXP: 09/30/27 ELECTRICAL STATE OF CALIFORNIA

SCALE: AS NOTED | DO NOT SCALE DRAWING | B0950.2025.05 | E0.01

SLAC NATIONAL ACCELERATOR LABORATORY — U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
ELECTRICAL
COVER SHEET

		DATE	APPROVALS	DATE
ENGR	KOROLEV, SERGEY	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TURCHINA, OLENA	01-26-2026		
CHKR	ORTIZ, JUSTIN	01-26-2026		

DRAWING NUMBER: B0950-E00-0421 REVISION: E E

B0950-E00-0421 - E E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	OT	JO	SK	07-25-2025
B	90% CD	OT	JO	SK	08-20-2025
C	100% CD	OT	JO	SK	11-05-2025
D	FAO REVIEW	OT	JO	SK	01-05-2026
E	BIO REVIEW	OT	JO	SK	01-26-2026

(N) ELECTRICAL EQUIPMENT SCHEDULE

TAG	(APPROXIMATE) QUANTITY	DESCRIPTION	MANUFACTURER CATALOG NUMBER	VOLTAGE (L-L/L-N)	PHASE	FRAME RATING (AMPS)	LUG CONDUCTOR RANGE, PER PHASE (AWG)	DIMENSIONS (EA. UNIT) H" x W" x D"	MOUNTING TYPE	APPROXIMATE WEIGHT (LBS.)	REMARKS & AREA OF USE
PNL 1	1	SIEMENS PANELBOARD P1 TYPE, SYSTEM VOLTAGE 480VAC, 3-PH, 3 WIRE, #6 -350 KCMIL LUGS, 250A RATED BUS, TOP FEED. COMPLETE ORDER W/ LUG KIT, GROUND BAR AND ALL REQUIRED HARDWARE. UL LISTED, WARRANTY (12) MONTHS FROM INITIAL OPERATION AND (18) MONTHS FROM SHIPMENT.	SIEMENS: P1F30WC250CTS W/ MLKA3A (LUG KIT) W/ EGK (GND BUS)	480/-	3	250	#6AWG-350 KCMIL	44" X 20" X 5.75"	SURFACE	140	(N) PANEL B950-4PB-1002
BCK 1	1	EATON FREEDOM FLASHGARD MCC BUCKET, 5X, 480VAC, 400A, LSIG BREAKER, 3-PH, UL LISTED, SEE SUBMITTAL FOR ADDITIONAL INFORMATION.	LSF0017016-006	480/-	3	400	-	SEE SUBMITTAL	-	SEE SUBMITTAL	(N) BUCKET D3
RCP 1	6	MELTRIC DSN60 RECEPTACLE. POLY BLUE, SIZE 3, TYPE 4X, IP69, 3P+G, 60A, 480VAC, 60HZ, NO AUX, W/MELTRIC HANDLE. UL LISTED, WARRANTY 1 YEAR FROM THE DATE OF SHIPMENT.	MELTRIC: 63-64043 (RECEPTACLE) W/713P0EP4 (HANDLE)	480/-	3	60	#14AWG-4AWG	3.07" X 3.07" X 6.54"	-	-	(N) RECEPTACLE

NOTES:

1. INSTALL (N) EQUIPMENT LISTED IN THIS SCHEDULE COMPLETE W/ALL REQUIRED HARDWARE.
2. INSTALL (N) EQUIPMENT PER MANUFACTURER'S INSTRUCTION UNLESS OTHERWISE NOTED.
3. STRUCTURAL CALCULATION ARE NOT REQUIRED PER ASCE 7-16 CHAPTER 13, AND PER SLAC SEISMIC DESIGN SPECIFICATION FOR BUILDINGS, STRUCTURES, EQUIPMENT, AND SYSTEMS – 2023 EDITION – SECTION 1.3.

(N) ELECTRICAL EQUIPMENT SCHEDULE (FURNISHED BY CONTRACTOR, INSTALLED BY SLAC)

TAG	(APPROXIMATE) QUANTITY	DESCRIPTION	MANUFACTURER CATALOG NUMBER	VOLTAGE (L-L/L-N)	PHASE	FRAME RATING (AMPS)	LUG CONDUCTOR RANGE, PER PHASE (AWG)	DIMENSIONS (EA. UNIT) H" x W" x D"	MOUNTING TYPE	APPROXIMATE WEIGHT (LBS.)	REMARKS & AREA OF USE
DS 1	2	MELTRIC DISCONNECT SWITCH, METAL, TYPE 4X, 3P+G, 60A, 480VAC, NON-FUSED W/MELTRIC DSN60 INLET POLY BLUE, SIZE 3, TYPE 4X, IP69 3P+G, 60A, 480VAC 60 HZ NO AUX. UL LISTED WARRANTY 1 YEAR FROM THE DATE OF SHIPMENT.	MELTRIC: 67-67043-329-NF (DIS. SW.) W/63-68043 (INLET)	480/-	3	60	#14AWG-3AWG	16.93" X 8.92" X 7.26"	-	-	(N) DISCONNECT SWITCH W/INLET

NOTES:

1. INSTALL (N) EQUIPMENT LISTED IN THIS SCHEDULE COMPLETE W/ALL REQUIRED HARDWARE.
2. INSTALL (N) EQUIPMENT PER MANUFACTURER'S INSTRUCTION UNLESS OTHERWISE NOTED.
3. STRUCTURAL CALCULATION ARE NOT REQUIRED PER ASCE 7-16 CHAPTER 13, AND PER SLAC SEISMIC DESIGN SPECIFICATION FOR BUILDINGS, STRUCTURES, EQUIPMENT, AND SYSTEMS – 2023 EDITION – SECTION 1.3.

(N) FEEDER SCHEDULE:

TAG	RATED AMPS	UNGROUNDED CONDUCTOR (AWG)	UNGROUNDED CONDUCTOR QTY. IN EA. CONDUIT	GROUNDED CONDUCTOR (AWG)	GROUNDED CONDUCTOR QTY. IN EA. CONDUIT	INSULATION TYPE	INSULATION RATING (DEG. C)	VOLTAGE (L-L/L-N)	PHASE	CONDUIT TRADE SIZE (DIA. INCH.)	CONDUIT TYPE	CONDUIT QTY. PER FEEDER	VERTICAL RISER/DISTANCE (LF)	HORIZONTAL DISTANCE (LF)	TOTAL DISTANCE W/ 15% ADDED (LF)	VOLTAGE DROP AT EQUIPMENT (% OF NOM.)
F001	400	500KCMIL	3	#3	1	THHN/THWN	75	480/-	3	3"	GALV. IMC	1	20	10	35	0.25%
F002	250	250KCMIL	3	#4	1	THHN/THWN	75	480/-	3	2 1/2"	GALV. IMC	1	10	5	18	0.12%
F005	45	#6	3	#6	1	SOOW	90	480/-	3	1"	GALV. IMC	1	15	25	45	0.33%

NOTES:

1. ALL CONDUCTORS SHALL BE STRANDED COPPER UNLESS OTHERWISE NOTED.
2. VOLTAGE DROP CALCULATED BASED ON HORIZONTAL AND VERTICAL DISTANCES OF ROUTES SHOWN ON DRAWINGS, CHANGES IN THESE ROUTES MAY CHANGE VOLTAGE DROP, 15% SAFETY FACTOR APPLIED TO ALL DISTANCES, 85% PF IS ASSUMED.

(E) FEEDER SCHEDULE:

TAG	RATED AMPS	UNGROUNDED CONDUCTOR (AWG)	UNGROUNDED CONDUCTOR QTY. IN EA. CONDUIT	GROUNDED CONDUCTOR (AWG)	GROUNDED CONDUCTOR QTY. IN EA. CONDUIT	INSULATION TYPE	INSULATION RATING (DEG. C)	VOLTAGE (L-L/L-N)	PHASE	CONDUIT TRADE SIZE (DIA. INCH.)	CONDUIT TYPE	CONDUIT QTY. PER FEEDER	VERTICAL RISER/DISTANCE (LF)	HORIZONTAL DISTANCE (LF)	TOTAL DISTANCE W/ 15% ADDED (LF)	VOLTAGE DROP AT EQUIPMENT (% OF NOM.)
F003	125	#1	3	#6	1	THHN/THWN	75	480/-	3	2"	GALV. IMC	1	5	5	12	0.09%
F004	250	4/0	4	#4	1	THHN/THWN	75	208/120	3	2 1/2"	GALV. IMC	1	5	10	18	0.30%

NOTES:

1. FEEDER F003 AND F004 ONLY REQUIRED IF FEEDERS DO NOT ALREADY EXIST BETWEEN (E) EQUIPMENT. SEE SINGLE-LINE DIAGRAM DETAIL 1/E5.03.

SLAC BUILDING INSPECTION OFFICE

Construction Authorization (IFC)

BY: cnadler

DATE: 2/18/2026 **PRS#:** 26-009

Revised Authorization ☑ No ☐ Yes

These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.

The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.

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San Ramon, CA 94583
+1 (866) VEKTOR1 (835-8671)

REGISTERED PROFESSIONAL ENGINEER
SERGEY V. KOROLEV
E20491
EXP: 09/30/27
ELECTRICAL
STATE OF CALIFORNIA

SCALE: AS NOTED | DO NOT SCALE DRAWING | B0950.2025.05 | E0.02

SLAC NATIONAL ACCELERATOR LABORATORY
U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
ELECTRICAL
EQUIPMENT SCHEDULES

		DATE	APPROVALS	DATE
ENGR	KOROLEV, SERGEY	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TURCHINA, OLENA	01-26-2026		
CHKR	ORTIZ, JUSTIN	01-26-2026		

DRAWING NUMBER: B0950-E06-0422 REVISION: E E

B0950-E06-0422 - E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	OT	JO	SK	07-25-2025
B	90% CD	OT	JO	SK	08-20-2025
C	100% CD	OT	JO	SK	11-05-2025
D	FAO REVIEW	OT	JO	SK	01-05-2026
E	BIO REVIEW	OT	JO	SK	01-26-2026

DEMOLITION GENERAL NOTES

1. THE QUANTITY AND LOCATION OF (E) ELECTRICAL DEVICES, PATHWAYS AND EQUIPMENT THAT SHALL BE REMOVED AS PART OF THIS PROJECT ARE SHOWN ON RESPECTIVE ELECTRICAL DEMOLITION DRAWINGS BASED ON OLDER AND PARTIAL AS-BUILT DRAWINGS, AND SITE OBSERVATIONS, NOT EVERYTHING COULD BE VERIFIED WITHOUT REMOVING DEAD FRONTS AND DISTURBING EQUIPMENT FUNCTION. THEREFORE, THE CONTRACTOR SHALL USE THIS INFORMATION FOR REFERENCE ONLY, AND PRIOR TO BID, SHALL VISIT THE SITE AND FURTHER OBSERVE (E) CONDITIONS AS NECESSARY TO VERIFY THE RESPECTIVE DEMOLITION DRAWINGS TO PERFORM THE DEMOLITION OF THE (E) ELECTRICAL SYSTEMS ACCORDING TO THE RESPECTIVE DEMOLITION DRAWINGS, NOTIFY E.E.O.R. OF ANY DISCREPANCIES FOUND.
2. CONTRACTOR SHALL COORDINATE AND PLAN ALL DEMOLITION WORK WITH THE SLAC OWNER REPRESENTATIVE, ALL OTHER DISCIPLINES ASSIGNED TO THIS PROJECT, AND UTILITY OR SERVICE PROVIDERS, INCLUDING, BUT NOT LIMITED TO THE ELECTRICAL PROVIDER.
3. CONTRACTOR SHALL COORDINATE ALL DEMOLITION WORK WITH REQUIREMENTS OF TECHNICAL SPECIFICATIONS FOR HAZARDOUS MATERIALS ABATEMENT REPORT PREPARED BY INDUSTRIAL HYGIENE CONSULTANT (IF REQUIRED BY SLAC OWNER REPRESENTATIVE).
4. CONTRACTOR SHALL MAINTAIN AND VERIFY CONTINUITY OF UPSTREAM AND DOWNSTREAM ELECTRICAL CIRCUITS DURING AND AFTER WORK IS COMPLETE.
5. DEMOLITION WORK SHALL INCLUDE, BUT NOT BE LIMITED TO, SPECIFIC DEMOLITION WORK SHOWN ON THESE DEMOLITION DRAWINGS AND THE FOLLOWING:

   A. AIR GAP, DISCONNECT AND REMOVE (E) ELECTRICAL EQUIPMENT.

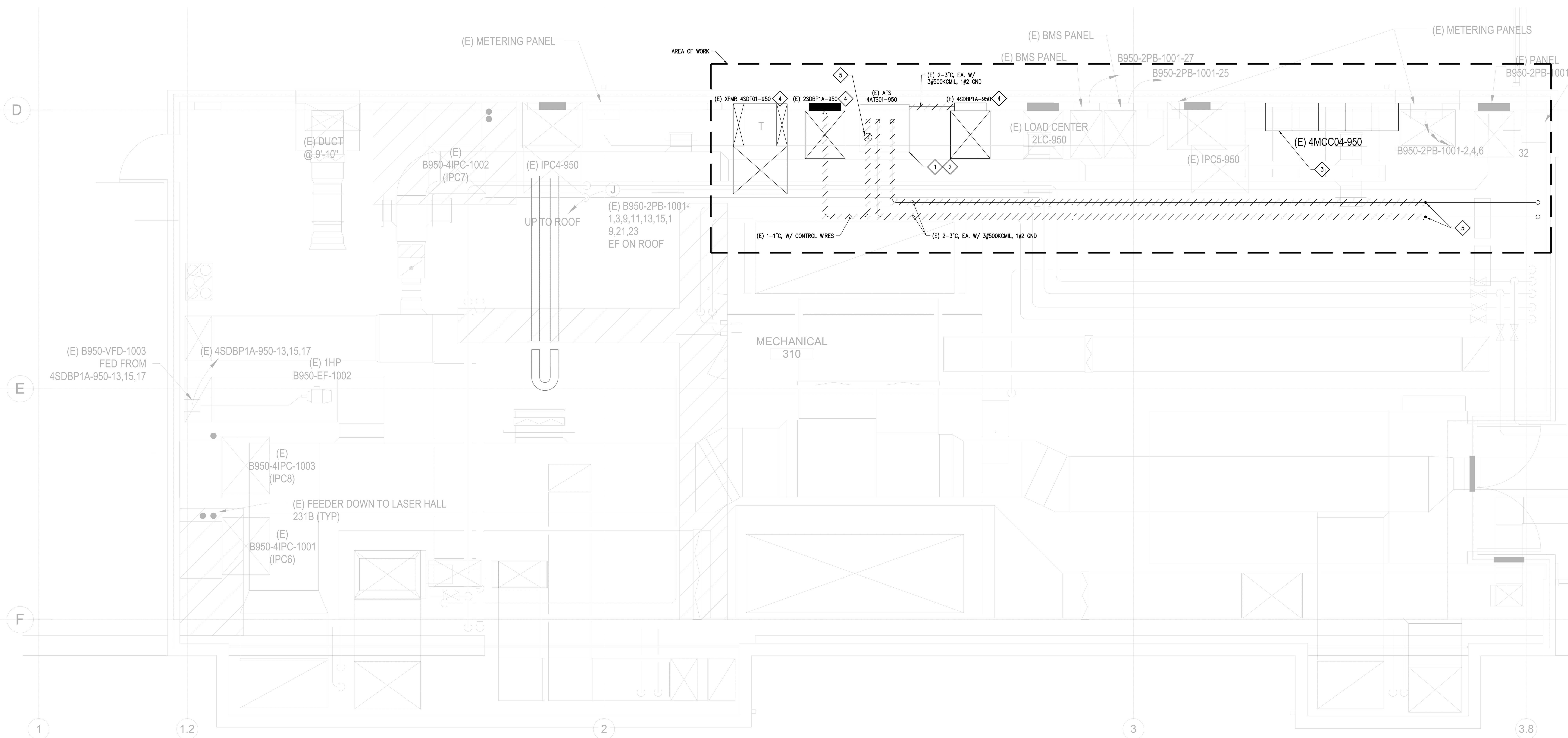
   B. AIR GAP, DISCONNECT AND REMOVE (E) CONDUCTORS AND RACEWAYS WHERE SPECIFIED TO DO SO.

   C. REMOVE AND DISPOSE OF ALL REMOVED CONDUITS, RACEWAYS AND WIRING FROM THE PREMISES ACCORDING TO FEDERAL AND STATE REGULATIONS.

   D. PATCH, REPAIR AND PAINT ALL SURFACES ON WALLS AND CEILINGS THAT ARE TO REMAIN, WHERE WORK HAS BEEN PERFORMED TO MATCH (E) ADJACENT CONDITIONS, AND MAINTAIN (E) FIRE RATED CONSTRUCTION.
6. ALL (E) EQUIPMENT BEING REMOVED SHALL REMAIN AS THE PROPERTY OF THE SLAC OWNER REPRESENTATIVE AND SHALL BE PREPARED FOR STORAGE BY CONTRACTOR AND RETURNED TO THE SLAC OWNER REPRESENTATIVE. CONTRACTOR SHALL COORDINATE WITH THE SLAC OWNER REPRESENTATIVE FOR STORAGE LOCATION AND METHOD, AND STORE THE EQUIPMENT AS DIRECTED.
7. ELECTRICAL CONTRACTOR SHALL PROVIDE TEMPORARY LIGHTING AND POWER CONNECTIONS FOR THEMSELVES AND OTHER DISCIPLINES AS REQUIRED DURING THE CONSTRUCTION EFFORT.
8. ALL (E) SURVEILLANCE, FIRE LIFE SAFETY, AND SECURITY DEVICES SHALL REMAIN AND BE PROTECTED DURING CONSTRUCTION.
9. ALL (E) ELECTRICAL, WIRING, AND CONDUIT TO REMAIN, SIGNAL AND FIRE ALARM EQUIPMENT, SERVICES, AND RACEWAYS SHOWN ON THIS DEMOLITION PLAN AND ANY OTHER ADDITIONAL EQUIPMENT IDENTIFIED BY GPR, POTHOLING, OR X-RAY SCANS OR OTHERWISE DISCOVERED IN FIELD SHALL REMAIN AND BE PROTECTED DURING THE DEMOLITION EFFORT UNLESS OTHERWISE INSTRUCTED BY THE E.E.O.R. DURING CONSTRUCTION.
10. ELECTRICAL DEMOLITION SHALL COMPLY WITH THE CONSTRUCTION SAFETY REQUIREMENTS MANUAL.
11. CAP AND SEAL ALL (E) CONDUITS TO REMAIN. USE COPPER ANTI-SEIZE ON THREADED CAPS. USE GALV. SPRAY ON (E) METAL CONDUITS WHERE GALV. COATING HAS BEEN DAMAGED FROM CUTTING OR HANDLING. APPLY EPOXY TO PLASTIC OR PVC CONDUIT CAPS.
12. SEE TABLE ON APPLICABLE SCHEDULE SHEET FOR SCHEDULES OF (E) ELECTRICAL EQUIPMENT, WIRING, AND CONDUITS TO BE DEMOLISHED.
13. GENERALLY, (E) ELECTRICAL EQUIPMENT, WIRING, AND CONDUIT TO BE DEMOLISHED ARE SHOWN IN DARKER AND HEAVIER LINE TYPE AND LISTED IN SCHEDULES, SPECIFIC AREAS WHERE EQUIPMENT SHALL BE DEMOLISHED AND (E) LARGER EQUIPMENT TO BE DEMOLISHED ARE EXPLICITLY CALLED OUT WITH KEYNOTES.
14. PROVIDE CIRCUIT TRACE, AND COORDINATE WITH AND ASSIST SLAC OWNER REPRESENTATIVE AND SLAC ENGINEERS TO LOCATE (E) CONDUCTORS AND (E) SIGNAL WIRING END TO END AS REQ. FOR DEMOLITION

DEMOLITION KEY NOTES

1. REMOVE (E) ATS 4ATS01-950. SEE DETAIL 1/E5.01D FOR REMOVAL OF ASSOCIATED CONDUIT AND CONDUCTORS. PREPARE FOR STORAGE, AND COORDINATE W/ SLAC OWNER REPRESENTATIVE FOR STORAGE METHODS AND LOCATION.
2. GRIND DOWN (E) ATS CONC. PAD TO FLOOR LEVEL.
3. RELOCATE (E) BUCKET D3 TO BELOW BUCKET E4. RENAME TO "E5". SEE DETAIL 2/E5.03 FOR (N) CONFIGURATION.
4. REMOVE (E) EQUIPMENT IDENTIFICATION LABELS. USE TEMPORARY LABELS OF (E) IDENTIFICATION UNTIL APPLICATION OF (N) LABELS.
5. DISCONNECT (E) CONDUITS ALONG ROUTE, PULL BACK (E) CONDUCTORS TO NEAREST UNION.



1 LCLS B950 R310 (R302 GPS) DEMO POWER PLAN SCALE: 3/8" = 1'-0"

NORTH

SLAC BUILDING INSPECTION OFFICE

Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization ☑ No ☐ Yes	
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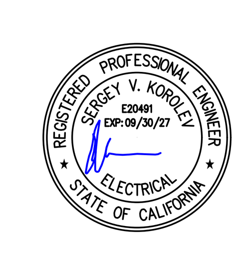


SCALE: AS NOTED	DO NOT SCALE DRAWING	B0950.2025.05		E3.01D

SLAC NATIONAL ACCELERATOR LABORATORY — U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
ELECTRICAL
LCLS B950 R310 DEMO POWER PLAN

		DATE	APPROVALS	DATE
ENGR	KOROLEV, SERGEY	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TURCHINA, OLENA	01-26-2026		
CHKR	ORTIZ, JUSTIN	01-26-2026		

DRAWING NUMBER: B0950-ED1-0423 REVISION: E E

B0950-ED1-0423 - E

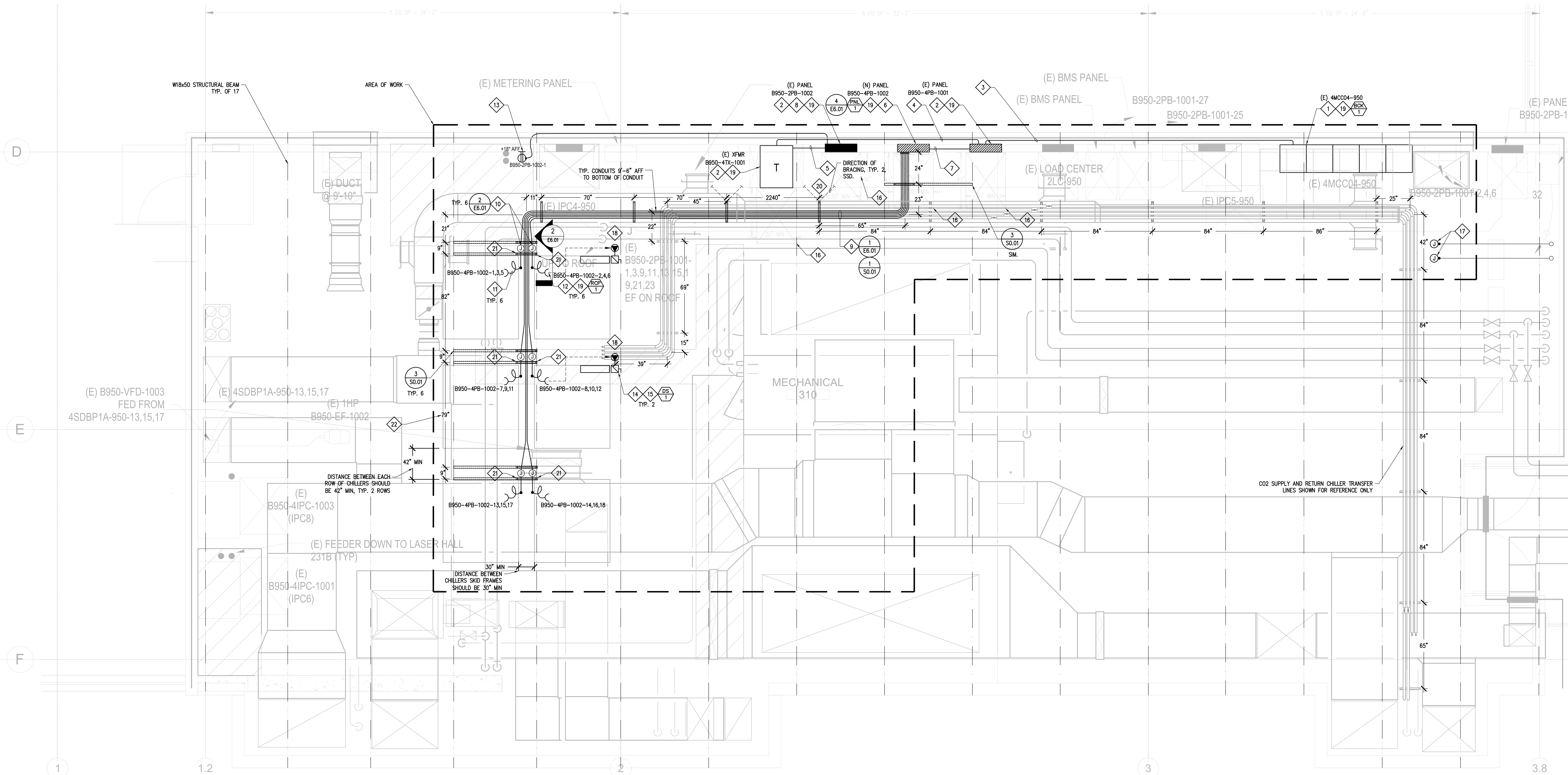
REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	OT	JO	SK	07-25-2025
B	90% CD	OT	JO	SK	08-20-2025
C	100% CD	OT	JO	SK	11-05-2025
D	FAO REVIEW	OT	JO	SK	01-05-2026
E	BIO REVIEW	OT	JO	SK	01-26-2026

PLAN GENERAL NOTES

1. INFORMATION PROVIDED ON THESE CONSTRUCTION DOCUMENTS WAS GATHERED FROM FIELD OBSERVATION AND PREVIOUS RECORD DRAWINGS AND SHALL BE USED WITH CAUTION; CONTRACTOR SHALL FIELD VERIFY ALL QUANTITIES, ESPECIALLY THE EXACT COUNT OF DEVICES, ROUTING, SIZES, LENGTHS AND COUNTS OF CONDUIT AND CONDUCTORS TO BE INSTALLED AND OR REMOVED PRIOR TO STARTING THE RESPECTIVE WORK.
2. FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS SEE ELECTRICAL COVER SHEET.
3. WORK SHALL NOT INTERFERE WITH ADJACENT FACILITY USE AND OR FUNCTION. CONTRACTOR SHALL COORDINATE WITH SLAC OWNER REPRESENTATIVE AT LEAST 2-WEEKS PRIOR TO STARTING WORK TO ESTABLISH A MUTUALLY AGREED UPON SCHEDULE.
4. CONTRACTOR SHALL COORDINATE ALL WORK WITH ALL OTHER TRADES.
5. CONTRACTOR SHALL MAINTAIN AND VERIFY CONTINUITY OF UPSTREAM AND DOWNSTREAM ELECTRICAL CIRCUITS DURING AND AFTER WORK IS COMPLETE.
6. CONTRACTOR SHALL FOLLOW MANUFACTURER'S MOUNTING, WIRING AND INSTALLATION INSTRUCTIONS FOR ALL ASSEMBLIES AND DEVICES UNLESS OTHERWISE NOTED.
7. CONTRACTOR SHALL COORDINATE ALL ROUTING OF CONDUIT AND WIRING WITH (E) CONDITIONS.
8. CONTRACTOR SHALL COORDINATE FINAL LOCATIONS OF RECEPT. W/ SLAC OWNER REPRESENTATIVE.
9. IF REQUIRED BY SLAC OWNER REPRESENTATIVE CONTRACTOR SHALL PAINT (N) CONDUIT IN ACCORDANCE W/APPLICABLE PROCEDURES.
10. ELECTRICAL CIRCUITS AND EQUIPMENT THAT ARE SHOWN IN LIGHTER LINE WEIGHT OR IN GRAY COLOR ARE EXISTING.
11. SEE EQUIPMENT SCHEDULE ON APPLICABLE SCHEDULE SHEET FOR RESPECTIVE EQUIPMENT TAGS.
12. SEE FEEDER SCHEDULE ON APPLICABLE SCHEDULE SHEET FOR RESPECTIVE FEEDER TAGS.

PLAN KEY NOTES

1. PROVIDE (N) 400A BUCKET INTO 5X LOCATION D3 AND LABEL "D3". SEE SINGLE-LINE DIAGRAM DETAIL 2/E5.03. CIRCUIT BREAKER SHALL BE FURNISHED W/ FULL TESTING REPORT FROM MFR. CONTRACTOR SHALL PROVIDE ADDITIONAL 3RD PARTY CIRCUIT BREAKER TESTING AND CERTIFICATION AS REQUIRED BY SLAC OWNER REPRESENTATIVE.
2. RELABEL (E) EQUIPMENT AS FOLLOWS:
   4SDBP1A-950 TO B950-4PB-1001
   2SDBP1A-950 TO B950-2PB-1002
   4SDT01-950 TO B950-4TX-1001
3. PROVIDE (N) FEEDER FROM (E) 4MCC04-950 (N) BRK. D3 TO (E) PANEL B950-4PB-1001. COORDINATE ALL ROUTING W/ (E) FIELD CONDITION AND SLAC OWNER REPRESENTATIVE. SEE SINGLE-LINE DIAGRAM 1/E5.03 AND FEEDER SCHEDULE SH. E0.02
4. VERIFY IF CONDUCTORS PRESENT BETWEEN (E) PANEL B950-4PB-1001 AND (E) XFMR B950-4TX-1001. IF NOT, PROVIDE (N) FEEDER. COORDINATE ALL ROUTING W/ (E) FIELD CONDITION AND SLAC OWNER REPRESENTATIVE. SEE SINGLE-LINE DIAGRAM 1/E5.03 AND FEEDER SCHEDULE SH. E0.02.
5. VERIFY IF CONDUCTORS PRESENT BETWEEN (E) XFMR B950-4TX-1001 AND (E) B950-2PB-1002. IF NOT, PROVIDE (N) FEEDER. COORDINATE ALL ROUTING W/ (E) FIELD CONDITION AND SLAC OWNER REPRESENTATIVE. SEE SINGLE-LINE DIAGRAM 1/E5.03 AND FEEDER SCHEDULE SH. E0.02
6. PROVIDE (N) 480VAC 250A PANEL B950-4PB-1002, W/ 250A MAIN CB AND MIN. (6) 3P 40A LSIG BREAKERS FOR (2) (N) AND (4) FUTURE CO2 CHILLERS.
7. PROVIDE (N) FEEDER FROM (E) PANEL B950-4PB-1001 (E) 250A BREAKER TO (N) PANEL B950-4PB-1002. COORDINATE ALL ROUTING W/ (E) FIELD CONDITION AND SLAC OWNER REPRESENTATIVE. SEE SINGLE-LINE DIAGRAM 1/E5.03 AND FEEDER SCHEDULE SH. E0.02
8. PROVIDE CKT. TO SUPPORT CONVENIENCE RECEPTACLE USING (E) 1P 15A BREAKER AT (E) PANEL B950-2PB-1002.
9. PROVIDE (6) 1" IMC EACH W/3#8AWG + #10G TO THE CHILLER AREA. UTILIZE CEILING SUSPENDED MECHANICAL PIPE RACK SUPPORT FOR CONDUIT ROUTING. SEE DETAEL E6.01. REFER TO DRAWING M0.01 DET. 1, 2. COORDINATE W/ SLAC OWNER REPRESENTATIVE AND MECHANICAL FOR EXACT ROUTING.
10. PROVIDE (N) J-BOX FOR CONDUIT TERMINATION ON UNISTRUT SUPPORT FROM STRUCTURAL BEAM ABOVE. COORDINATE W/ SLAC OWNER REPRESENTATIVE FOR EXACT LOCATION.
11. E PROVIDE 3#6AWG + #6G SOOW CORD W/ WIRE MESH CORD GRIP AT THE J-BOX. CONNECT TO FEEDER WITHIN J-BOX USING POLARIS 4-PORT CONNECTOR P/N IPLD4-4 OR APPROVED EQUAL. PROVIDE (N) FERRULES FOR STRIPPED SOFT STRANDED ENDS OF SOOW CORD PRIOR TO TERMINATION INTO CONNECTOR. COORDINATE FINAL LENGTH OF CORD W/ SLAC OWNER REPRESENTATIVE.
12. PROVIDE (N) MELTRIC RECEPTACLE W/ INSTALLATION ACCESSORIES. SEE EQUIPMENT SCHEDULE SH. E0.02. RESULTING PHASE ROTATION ORDER SHALL BE THE SAME FOR ALL RECEPTACLES.
13. PROVIDE (N) 120V DUPLEX RECEPTACLE. PROVIDE 3/4" EMT W/ 2#12AWG + #12G. PROVIDE (N) 20A CB IN PNL. "B950-2PB-1002". COORDINATE FINAL LOCATION OF RECEPTACLE W/ SLAC OWNER REPRESENTATIVE.
14. TERMINATE 480VAC SOURCE CONNECTION EGC INSIDE OF THE DISCONNECT SWITCH AT COMMON GROUND BAR.
15. FURNISH (2) (N) 60A NON-FUSIBLE DISCONNECT SWITCHES, W/ 480V 60A MELTRIC INLET. INSTALLATION ON CHILLER SKID BY SLAC.
16. PROVIDE CLEARANCE MARKING FOR EQUIPMENT ONTO FLOOR.
17. PROVIDE (2) (N) 30"W X 30"H X 10"D MIN. PULL BOXES TO TERMINATE (E) CONDUITS AND CONDUCTORS OF NORMAL AND STAND-BY POWER FEEDERS, SEE SLD E5.01D.
18. MAINTAIN 42" CLEARANCE IN FRONT OF DISCONNECT TO NEAREST GROUNDED CONDUCTIVE STRUCTURE.
19. PROVIDE PHASE ROTATION TEST SERVICES. SUBMIT RESULTS TO EEOR AND SLAC OWNER REPRESENTATIVE FOR REVIEW AND COORDINATION. MARK EQUIPMENT W/ PHASE ROTATION ORDER. PRIOR TO RE-ENERGIZING, COORDINATE W/ SLAC OWNER REPRESENTATIVE FOR EQUIPMENT READINESS REQUIREMENTS AND PROCEDURES, REFER TO SLD ON E5.03.
20. LOWER (E) LIGHT BY PROVIDING (N) LONGER V-HOOK CHAINS TO AVOID (N) CONDUIT ALONG ROUTE.
21. PROVIDE (N) PADDED J-HOOK BELOW CHANNEL STRUT, COIL UNUSED SOOW CORD FOR FUTURE CHILLER, BAG AND TAPE OFF RECEPT.
22. DIMENSIONS HAVE BEEN TAKEN FROM AS-BUILTS AND FIELD MEASUREMENTS. THE CONTRACTOR SHALL VERIFY DIMENSIONS IN FIELD DURING CONSTRUCTION. ADJUST AS REQUIRED TO ADDRESS FIELD CONDITIONS AND MEANS AND METHODS OF CONSTRUCTION.



1 LCLS B950 R310 (R302 GPS) POWER PLAN SCALE : 3/8" = 1'-0"

NORTH

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SCALE: AS NOTED	DO NOT SCALE DRAWING	B0950.2025.05	E3.01

SLAC NATIONAL ACCELERATOR LABORATORY — U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
ELECTRICAL
LCLS B950 R310 POWER PLAN

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CHKR	ORTIZ, JUSTIN	01-26-2026		

DRAWING NUMBER: B0950-EP1-0424 REVISION: E E

B0950-EP1-0424 - E E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	OT	JO	SK	07-25-2025
B	90% CD	OT	JO	SK	08-20-2025
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D	FAO REVIEW	OT	JO	SK	01-05-2026
E	BIO REVIEW	OT	JO	SK	01-26-2026

1 (E) SINGLE LINE DIAGRAM SCALE : N.T.S.

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2. CONTRACTOR SHALL FOLLOW SLAC SAFETY PROCEDURES FOR ANY ELECTRICAL WORK. CONTRACTOR SHALL ENSURE TO FOLLOW PROCEDURES FOR LOCKOUT/TAG-OUT, VERIFYING NO VOLTAGE ON LINE /LOAD TERMINALS AND CONTROL WIRING, AND TREAT THE WORK ON MCC AS AN ARC FLASH CATEGORY 4 OR HIGHER.
3. PRIOR TO RE-ENERGIZING ANY EQUIPMENT, SLAC OWNER REPRESENTATIVE SHALL PROVIDE READINESS REVIEW OF ALL DISTRIBUTION EQUIPMENT IN THE SCOPE OF WORK, CONTRACTOR SHALL ASSIST AND COORDINATE W/ SLAC STAFF AS REQUIRED TO COMPLETE THE READINESS REVIEW.

SLD DEMOLITION KEY NOTES

1. SINGLE LINE DIAGRAM COPIED FROM DCR STAMPED DRAWING "ID-383-904-25" (REV. "06", DATED 06/21/2024). SINGLE LINE FIELD VERIFIED BY CIBIL ZACHARIAH ON 07/02/2025.
2. SEE SH. E3.01D FOR DEMOLITION NOTES.
3. ROOM NUMBER 302 CHANGED TO 310 PER LCLS-II-HE NEH B950 ROOM 310 RDS R2, DATED 7/29/2025.

B0950-ED7-0425 - E | E

SLAC BUILDING INSPECTION OFFICE

Construction Authorization (IFC)

BY: cnadler

DATE: 2/18/2026 | PRS#: 26-009

Revised Authorization ☑ No ☐ Yes

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SLAC NATIONAL ACCELERATOR LABORATORY — U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
ELECTRICAL
SINGLE LINE DIAGRAM DEMO

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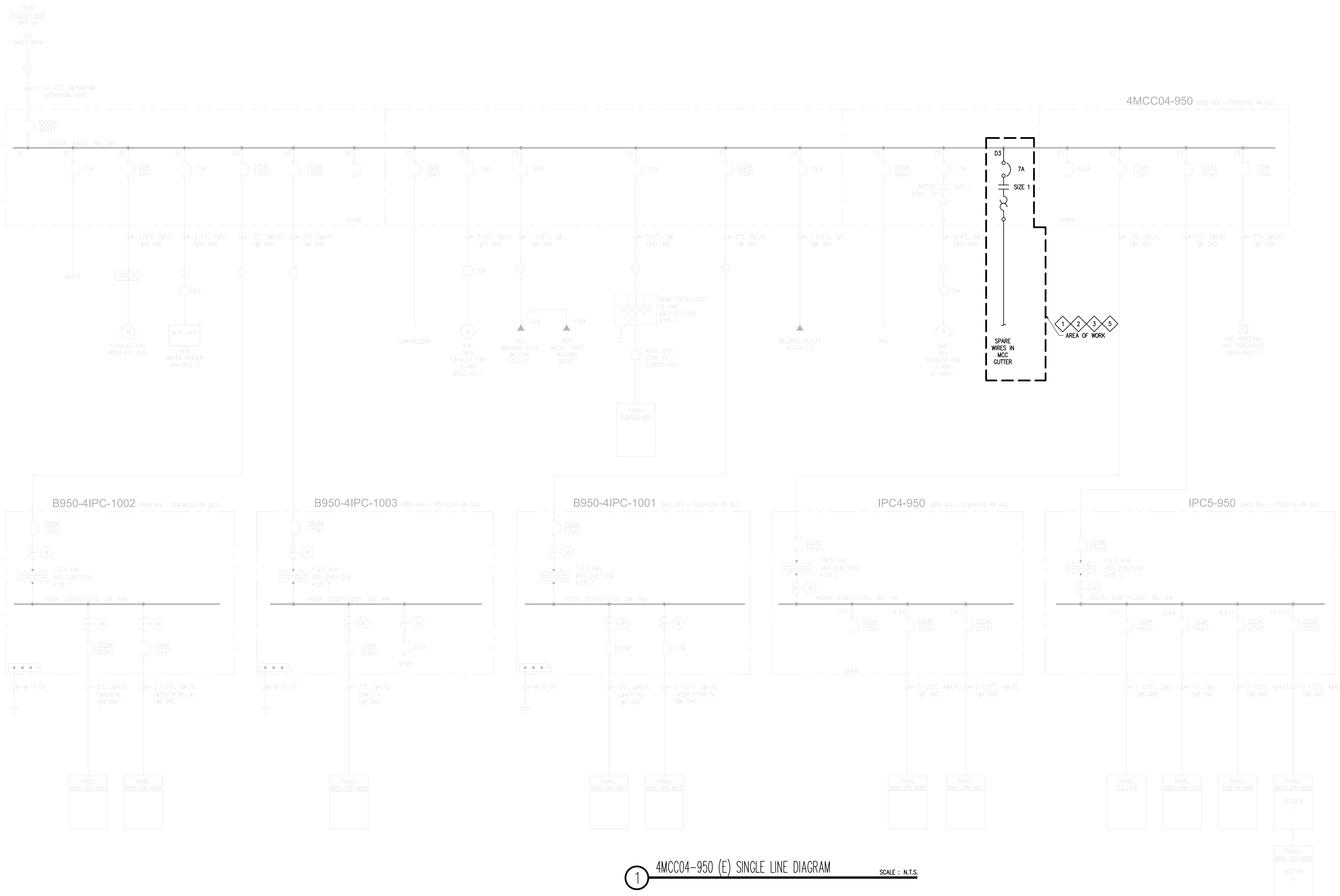
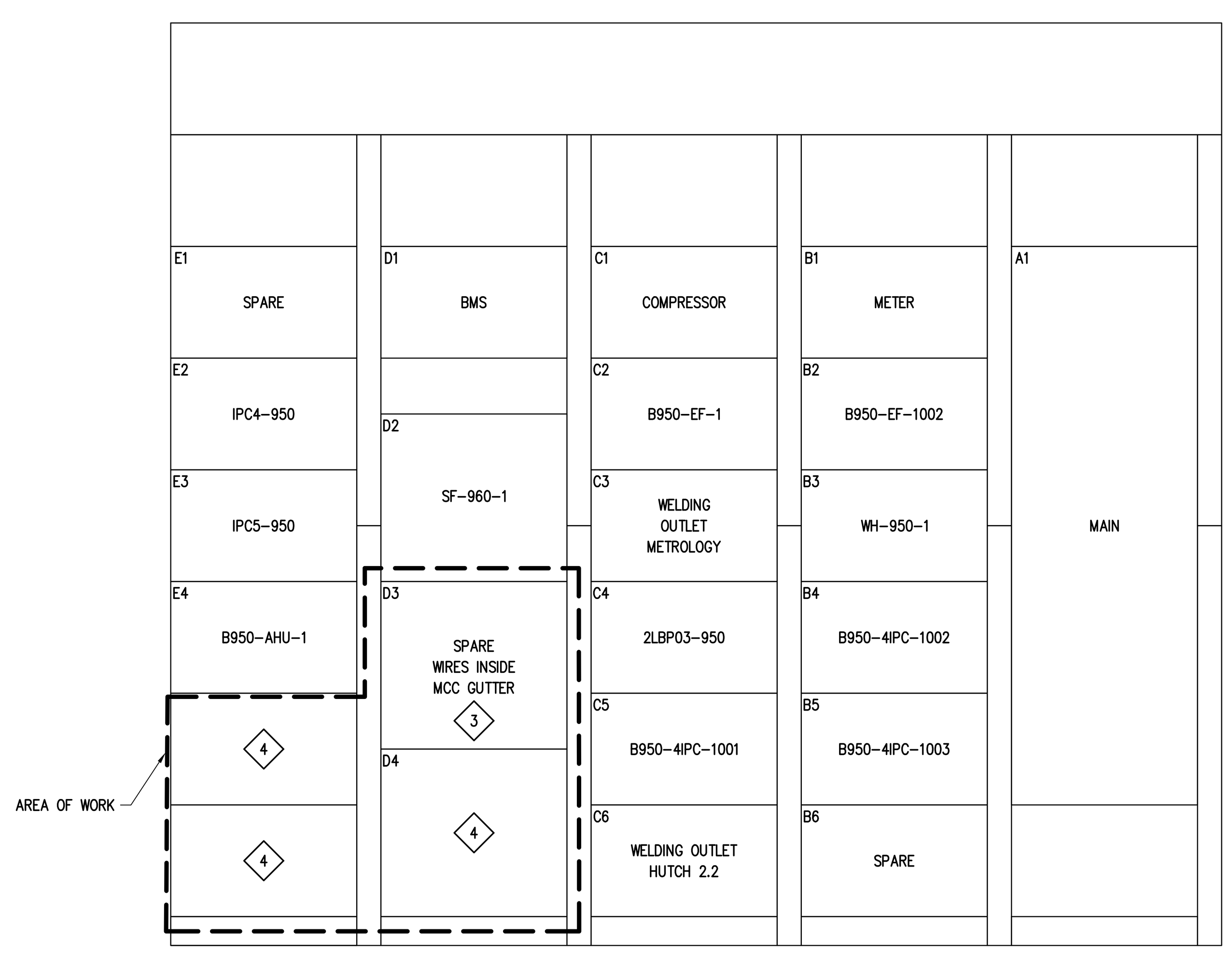
E	BIO REVIEW	OT	JO	SK	01-26-2026



1 4MCC04-950 (E) SINGLE LINE DIAGRAM SCALE : N.T.S.



2 4MCC04-950 (E) FRONT ELEVATION SCALE : N.T.S.

SLD GENERAL NOTES

1. INFORMATION PROVIDED ON THESE CONSTRUCTION DOCUMENTS WAS GATHERED FROM FIELD OBSERVATION AND PREVIOUS RECORD DRAWINGS AND SHALL BE USED WITH CAUTION; CONTRACTOR SHALL FIELD VERIFY ALL QUANTITIES, ESPECIALLY THE EXACT COUNT OF DEVICES, ROUTING, SIZES, LENGTHS AND COUNTS OF CONDUIT AND CONDUCTORS TO BE INSTALLED AND OR REMOVED PRIOR TO STARTING THE RESPECTIVE WORK.
2. CONTRACTOR SHALL FOLLOW SLAC SAFETY PROCEDURES FOR ANY ELECTRICAL WORK. CONTRACTOR SHALL ENSURE TO FOLLOW PROCEDURES FOR LOCKOUT/TAG-OUT, VERIFYING NO VOLTAGE ON LINE /LOAD TERMINALS AND CONTROL WIRING, AND TREAT THE WORK ON MCC AS AN ARC FLASH CATEGORY 4 OR HIGHER.
3. PRIOR TO RE-ENERGIZING ANY EQUIPMENT, SLAC OWNER REPRESENTATIVE SHALL PROVIDE READINESS REVIEW OF ALL DISTRIBUTION EQUIPMENT IN THE SCOPE OF WORK, CONTRACTOR SHALL ASSIST AND COORDINATE W/ SLAC STAFF AS REQUIRED TO COMPLETE THE READINESS REVIEW.

SLD KEY NOTES

1 SINGLE LINE DIAGRAM COPIED FROM DCR STAMPED DRAWING "ID-383-904-25" (REV. "06", DATED 06/21/2024). SINGLE LINE FIELD VERIFIED BY CIBIL ZACHARIAH ON 04/01/2025.

2 CONTRACTOR MAY NOT PERFORM WORK ON LEGACY SINGLE-PHASE CIRCUITS OR IN LEGACY (PRIOR TO 2019) ELECTRICAL PANELBOARDS THAT CONTAIN SINGLE PHASE CIRCUITS UNLESS NEUTRAL WIRING VERIFICATION OF THE SINGLE-PHASE CIRCUIT IN THE PANEL HAS BEEN PERFORMED AND DOCUMENTED IN ACCORDANCE WITH SLAC-I-708-404-096 "OPERATIONS ON SYSTEMS WITH SUSPECTED SHARED NEUTRALS". SEE ESH MANUAL CHAPTER 8 ELECTRICAL SAFETY MANUAL SECTION 7.2.13. IF THE VERIFICATION HAS NOT BEEN PERFORMED, THEN IT MUST BE PERFORMED PRIOR TO START OF CONSTRUCTION.

3 REMOVE, AND RETAIN (E) DISCONNECTED BUCKET AT MCC LOCATION D3 IN ACCORDANCE W/ MFR. INSTRUCTIONS. REMOVE TEMPORARY LABEL "D3".

4 REMOVE (2) DEAD FRONTS BELOW E4 AND DEAD FRONT BELOW D3.

5 ROOM NUMBER 302 CHANGED TO 310 PER LCLS-II-HE NEH B950 ROOM 310 RDS R2, DATED 7/29/2025.

SLAC BUILDING INSPECTION OFFICE

Construction Authorization (IFC)

BY: cnadler

DATE: 2/18/2026 **PRS#:** 26-009

Revised Authorization ☑ No ☐ Yes

These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.

The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.

THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

Vektor Engineering & Consulting Services, Inc.
"Where engineering and technology drive innovation"
2603 Camino Ramon, Suite 417
San Ramon, CA 94583
+1 (866) VEKTOR1 (835-8671)

SCALE: AS NOTED | DO NOT SCALE DRAWING | B0950.2025.05 | E5.02

SLAC NATIONAL ACCELERATOR LABORATORY U.S. DEPARTMENT OF ENERGY

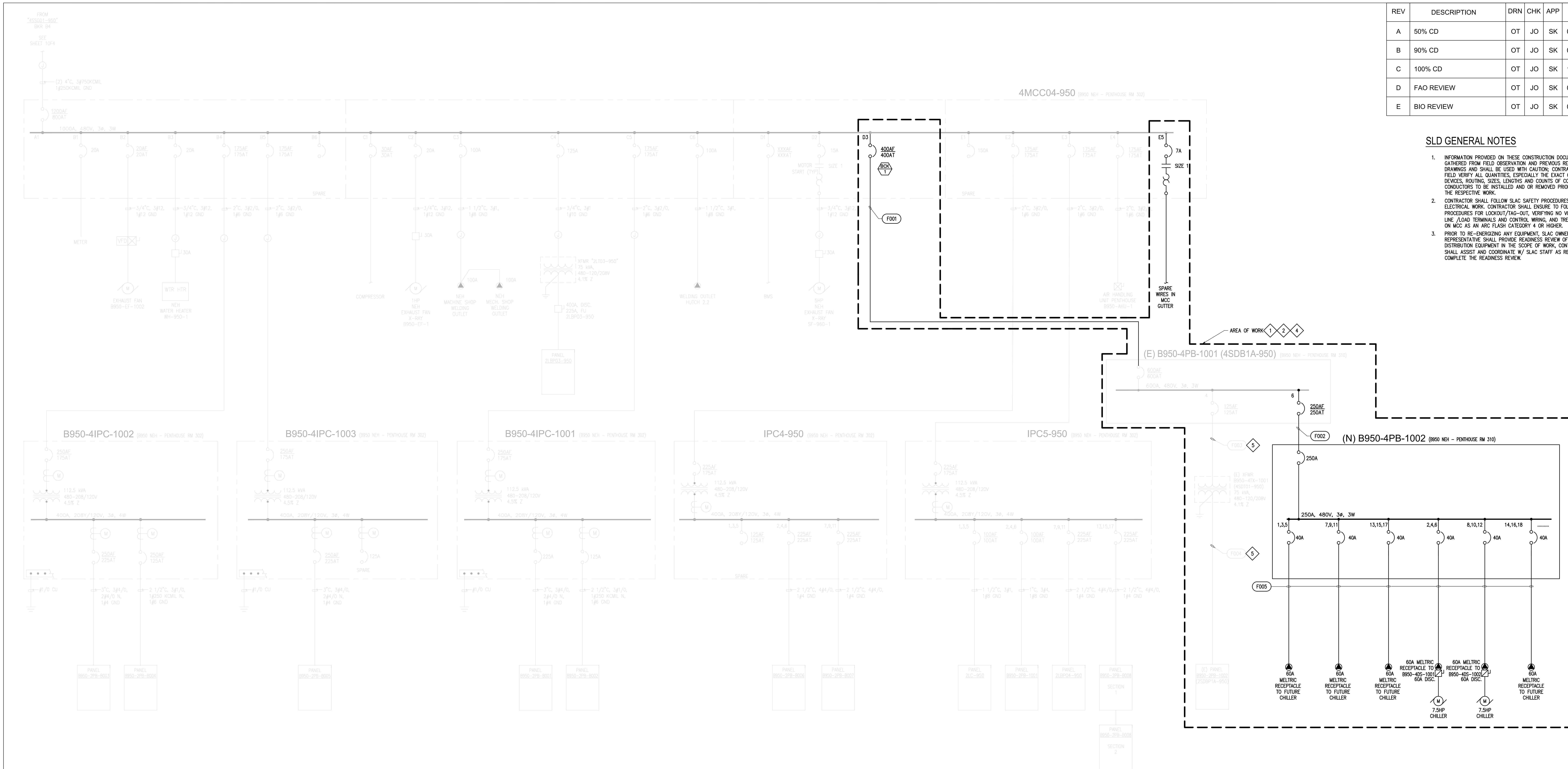
B950
MECH ROOM CO2 CHILLERS
ELECTRICAL
SINGLE LINE DIAGRAM

		DATE	APPROVALS	DATE
ENGR	KOROLEV, SERGEY	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TURCHINA, OLENA	01-26-2026		
CHKR	ORTIZ, JUSTIN	01-26-2026		

DRAWING NUMBER B0950-E07-0426 REVISION E E

B0950-E07-0426 - E E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	OT	JO	SK	07-25-2025
B	90% CD	OT	JO	SK	08-20-2025
C	100% CD	OT	JO	SK	11-05-2025
D	FAO REVIEW	OT	JO	SK	01-05-2026
E	BIO REVIEW	OT	JO	SK	01-26-2026



1 4MCC04-950 (N) SINGLE LINE DIAGRAM SCALE : N.T.S.

2 4MCC04-950 (N) FRONT ELEVATION NEW SCALE : N.T.S.

SLD GENERAL NOTES

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2. CONTRACTOR SHALL FOLLOW SLAC SAFETY PROCEDURES FOR ANY ELECTRICAL WORK. CONTRACTOR SHALL ENSURE TO FOLLOW PROCEDURES FOR LOCKOUT/TAG-OUT, VERIFYING NO VOLTAGE ON LINE /LOAD TERMINALS AND CONTROL WIRING, AND TREAT THE WORK ON MCC AS AN ARC FLASH CATEGORY 4 OR HIGHER.
3. PRIOR TO RE-ENERGIZING ANY EQUIPMENT, SLAC OWNER REPRESENTATIVE SHALL PROVIDE READINESS REVIEW OF ALL DISTRIBUTION EQUIPMENT IN THE SCOPE OF WORK, CONTRACTOR SHALL ASSIST AND COORDINATE W/ SLAC STAFF AS REQUIRED TO COMPLETE THE READINESS REVIEW.

SLD KEY NOTES

1. SINGLE LINE DIAGRAM COPIED FROM DCR STAMPED DRAWING "ID-383-904-25" (REV. "06", DATED 06/21/2024). SINGLE LINE FIELD VERIFIED BY CIBIL ZACHARIAH ON 07/02/2025.
2. CONTRACTOR MAY NOT PERFORM WORK ON LEGACY SINGLE-PHASE CIRCUITS OR IN LEGACY (PRIOR TO 2019) ELECTRICAL PANELBOARDS THAT CONTAIN SINGLE PHASE CIRCUITS UNLESS NEUTRAL WIRING VERIFICATION OF THE SINGLE-PHASE CIRCUIT IN THE PANEL HAS BEEN PERFORMED AND DOCUMENTED IN ACCORDANCE WITH SLAC-I-708-404-096 "OPERATIONS ON SYSTEMS WITH SUSPECTED SHARED NEUTRALS". SEE ESH MANUAL CHAPTER 8 ELECTRICAL SAFETY MANUAL SECTION 7.2.13. IF THE VERIFICATION HAS NOT BEEN PERFORMED, THEN IT MUST BE PERFORMED PRIOR TO START OF CONSTRUCTION.
3. PROVIDE (N) BLANK UNIT DOOR.
4. ROOM NUMBER 302 CHANGED TO 310 PER LCLS-II-HE NEH B950 ROOM 310 RDS R2, DATED 7/29/2025.
5. VERIFY (E) FEEDER IN FIELD, COMPARE TO EXPECTED FEEDER SCHEDULE ON SHEET E0.02, REPORT FINDINGS TO SLAC OWNER REPRESENTATIVE AND EEOR.

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization	☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

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SCALE: AS NOTED	DO NOT SCALE DRAWING	B0950.2025.05		E5.03

SLAC NATIONAL ACCELERATOR LABORATORY — U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
ELECTRICAL
SINGLE LINE DIAGRAM NEW

		DATE	APPROVALS	DATE
ENGR	KOROLEV, SERGEY	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TURCHINA, OLENA	01-26-2026		
CHKR	ORTIZ, JUSTIN	01-26-2026		

DRAWING NUMBER B0950-E07-0427 REVISION E E

B0950-E07-0427 - E E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	OT	JO	SK	07-25-2025
B	90% CD	OT	JO	SK	08-20-2025
C	100% CD	OT	JO	SK	11-05-2025
D	FAO REVIEW	OT	JO	SK	01-05-2026
E	BIO REVIEW	OT	JO	SK	01-26-2026

PANELS GENERAL NOTES

1. CONTRACTOR SHALL COMPLY WITH ALL NOTES AS LISTED FOR EACH RESPECTIVE PNL.
2. CONTRACTOR SHALL PROVIDE (N) TYPED PANEL SCHEDULE FOR EACH ELECTRICAL PANEL, PLACE IN CLEAR PLASTIC SLEEVE AND ATTACH TO INNER SURFACE OF PANEL DOOR.
3. CONTRACTORS SHALL NOT PERFORM WORK ON LEGACY SINGLE–PHASE CIRCUITS OR IN LEGACY ELECTRICAL PANELBOARDS (LEGACY = INSTALLED PRIOR TO 2019) THAT CONTAIN SINGLE PHASE CIRCUITS UNLESS NEUTRAL WIRING VERIFICATION OF ALL SINGLE–PHASE CIRCUITS IN THE PANEL HAS BEEN PERFORMED AND DOCUMENTED IN ACCORDANCE WITH SLAC–I–708–404–096 OPERATIONS ON SYSTEMS WITH SUSPECTED SHARED NEUTRALS. SEE ESH MANUAL CHAPTER 8 ELECTRICAL SAFETY MANUAL SECTION 7.2.13. IF THE VERIFICATION HAS NOT BEEN PERFORMED, THEN IT MUST BE PERFORMED BY F&O EPD PRIOR TO START OF PROJECT CONSTRUCTION.

PANELS KEY NOTES

1. EXISTING PANEL SCHEDULE VERIFIED BY CIBIL ZACHARIAH ON 03/05/2024.
2. ALL ELECTRICAL PANELS SHALL BE INSPECTED FOR SHARED NEUTRALS BY ELECTRICAL CONTRACTOR PRIOR TO BEING RE–ENERGIZED, REPORT FINDINGS TO EEOR AND SLAC OWNER REPRESENTATIVE PRIOR TO PROCEEDING.

2 1

PANEL SCHEDULE 4SDBP1A-950

VOLTS: 480V | PHASE: 3 | WIRE: 3 | BUSSING: 600A | POLES: 10
EXIST ☑ NEW ☐
MAIN: 600A | FED FROM: 4ATS1A-950 | LOCATION: MECH.R. 310 B950 | MOUNTED: SURFACE | AIC RATING: 10kAIC

AREA OF WORK

LOAD DESCRIPTION	TYPE	A	B	C	BRKR.	CKT.	CKT.	BRKR.	A	B	C	TYPE	LOAD DESCRIPTION
-	*	0.00			-	1	2	-	0.00			*	-
SPARE	*		0.00		15/3	1	2	15/3		0.00		*	SPARE
-	*			0.00	-	1	2	-			0.00	*	-
-	*	0.00			-	3	4	-	25.00			R	75 KVA XFMR 4SDT01-950
SPARE	*		0.00		15/3	3	4	125/3		25.00		R	PNL (2SDBP1A)
-	*			0.00	-	3	4	-			25.00	R	-
-	*				*	5	6	-	0.00			*	-
SPACE	*				*	5	6	250/3		0.00		*	SPARE
-	*				*	5	6	-			0.00	*	-
-	*				*	7	8	*				*	-
SPACE	*				*	7	8	*				*	SPACE
-	*				*	7	8	*				*	-
-	*				*	9	10	*				*	-
SPACE	*				*	9	10	*				*	SPACE
-	*				*	9	10	*				*	-
		0.00	0.00	0.00					25.00	25.00	25.00		

DRAWING # EQUIP. TAG A00593	CONN. KVA	DEMAND FACTOR	DEMAND KVA
TYPE "M": NON-CONTINUOUS / MISC. LOADS	0.00	100%	0.00
TYPE "L": LIGHTING / CONTINUOUS LOADS	0.00	125%	0.00
TYPE "R": RECEPTACLES (FIRST 10KVA)	75.00	100%	10.00
TYPE "R": RECEPTACLES (OVER 10KVA)		50%	32.50
TYPE "H": LARGEST MOTOR LOAD	0.00	125%	0.00
TYPE "H": HVAC / MECHANICAL LOADS	0.00	100%	0.00
TOTALS:	75.00		42.50

BOLD = NEW LOAD, NEW BREAKER

PHASE A: 25.00 KVA
PHASE B: 25.00 KVA
PHASE C: 25.00 KVA
51.12 AMPS

3 (E) "B950-4PB-1001" (4SDP1A-950) W/ (E) SCHED.

2 1

PANEL SCHEDULE 4MCC04-950

VOLTS: 480 | PHASE: 3 | WIRE: 4 | BUSSING: 1200A | POLES: N/A
EXIST ☑ NEW ☐
MCB: 800A | FED FROM: 4SSG01-950 #B4 | LOCATION: RM. 310 B950 | MOUNTED: SURFACE | AIC RATING: 65kAIC

AREA OF WORK

LOAD DESCRIPTION	TYPE	A	B	C	BRKR.	CKT.	CKT.	BRKR.	A	B	C	TYPE	LOAD DESCRIPTION
-	*	0.00			-			-	0.58			M	-
METER	*		0.00		-	B1	B2	20/3		0.58		M	B950-EF-1002
-	*			0.00	-			-			0.58	M	-
-	M	1.20			-			-	16.00			M	-
WH-950-1	M		1.20		20/3	B3	B4	175/3		16.00		M	B950-4IPC-1002
-	M			1.20	-			-			16.00	M	-
-	M	8.50			-			-	0.00			*	-
B950-4IPC-1003	M		8.50		175/3	B5	B6	175/3		0.00		*	SPARE
-	M			8.50	-			-			0.00	*	-
-	H	2.80			-			-	0.46			M	-
COMPRESSOR	H		2.80		30/3	C1	C2	20/3		0.46		M	B950-EF-1
-	H			2.80	-			-			0.46	M	-
-	R	15.00			-			-	11.50			M	-
WELDING OUTLET	R		15.00		100/3	C3	C4	125/3		11.50		M	2LBP03-950
-	R			15.00	-			-			11.50	M	-
-	M	16.00			-			-	15.00			R	-
B950-4IPC-1001	M		16.00		175/3	C5	C6	100/3		15.00		R	WELDING OUTLET HUTCH 2.2
-	M			16.00	-			-			15.00	R	-
-	M	0.20			-			-	2.10			H	-
BMS	M		0.20		10/3	D1	D2	15/3		2.10		H	SF-960-1
-	M			0.20	-			-			2.10	H	-
-	*	0.00			-							*	-
SPARE WIRES INSIDE MCC GUTTER	*		0.00		7/3	D3	D4					*	SPACE
-	*			0.00	-							*	-
-	*	0.00			-			-	20.00			M	-
SPARE	*		0.00		150/3	E1	E2	175/3		20.00		M	IPC4-950
-	*			0.00	-			-			20.00	M	-
-	M	20.00			-			-	50.00			H	-
IPC5-950	M		20.00		175/3	E3	E4	175/3		50.00		H	B950-AHU-1
-	M			20.00	-			-			50.00	H	-
-	*												
SPACE	*					E5							
-	*												
		63.70	63.70	63.70					115.64	115.64	115.64		

DRAWING # EQUIP. TAG A12595	CONN. KVA	DEMAND FACTOR	DEMAND KVA
TYPE "M": NON-CONTINUOUS / MISC. LOADS	283.32	100%	283.32
TYPE "L": LIGHTING / CONTINUOUS LOADS	0.00	125%	0.00
TYPE "R": RECEPTACLES (FIRST 10KVA)	90.00	100%	10.00
TYPE "R": RECEPTACLES (OVER 10KVA)		50%	40.00
TYPE "H": LARGEST MOTOR LOAD	150.00	125%	187.50
TYPE "H": HVAC / MECHANICAL LOADS	14.70	100%	14.70
TOTALS:	538.02		535.52

BOLD = NEW LOAD, NEW BREAKER

PHASE A: 179.34 KVA
PHASE B: 179.34 KVA
PHASE C: 179.34 KVA
644.13 AMPS

1 (E) "4MCC04-950" W/ (E) SCHED.

PANEL SCHEDULE B950-4PB-1001

VOLTS: 480V | PHASE: 3 | WIRE: 3 | BUSSING: 600A | POLES: 10
EXIST ☑ NEW ☐
MAIN: 600A | FED FROM: 4MCC04-950 E4 | LOCATION: MECH.R. 310 B950 | MOUNTED: SURFACE | AIC RATING: 65kAIC

AREA OF WORK

LOAD DESCRIPTION	TYPE	A	B	C	BRKR.	CKT.	CKT.	BRKR.	A	B	C	TYPE	LOAD DESCRIPTION
-	*	0.00			-	1	2	-	0.00			*	-
SPARE	*		0.00		15/3	1	2	15/3		0.00		*	SPARE
-	*			0.00	-	1	2	-			0.00	*	-
-	*	0.00			-	3	4	-	**0.18**			**M**	75 KVA XFMR B950-4TX-1001
SPARE	*		0.00		15/3	3	4	125/3		**0.00**		*	PNL B950-2PB-1002
-	*			0.00	-	3	4	-			**0.00**	*	-
-	*				*	5	6	-	**13.80**			**M**	**-**
SPACE	*				*	5	6	250/3		**13.80**		**M**	**PNL B950-4PB-1002**
-	*				*	5	6	-			**13.80**	**M**	**-**
-	*				*	7	8	*				*	-
SPACE	*				*	7	8	*				*	SPACE
-	*				*	7	8	*				*	-
-	*				*	9	10	*				*	-
SPACE	*				*	9	10	*				*	SPACE
-	*				*	9	10	*				*	-
		0.00	0.00	0.00					13.98	13.80	13.80		

DRAWING # EQUIP. TAG A00593	CONN. KVA	DEMAND FACTOR	DEMAND KVA
TYPE "M": NON-CONTINUOUS / MISC. LOADS	41.58	100%	41.58
TYPE "L": LIGHTING / CONTINUOUS LOADS	0.00	125%	0.00
TYPE "R": RECEPTACLES (FIRST 10KVA)	0.00	100%	0.00
TYPE "R": RECEPTACLES (OVER 10KVA)		50%	0.00
TYPE "H": LARGEST MOTOR LOAD	0.00	125%	0.00
TYPE "H": HVAC / MECHANICAL LOADS	0.00	100%	0.00
TOTALS:	41.58		41.58

BOLD = NEW LOAD, NEW BREAKER

PHASE A: 13.98 KVA
PHASE B: 13.80 KVA
PHASE C: 13.80 KVA
50.01 AMPS

4 (E) "B950-4PB-1001" W/ (N) SCHED.

PANEL SCHEDULE 4MCC04-950

VOLTS: 480 | PHASE: 3 | WIRE: 4 | BUSSING: 1200A | POLES: N/A
EXIST ☑ NEW ☐
MCB: 800A | FED FROM: 4SSG01-950 #B4 | LOCATION: RM. 310 B950 | MOUNTED: SURFACE | AIC RATING: 65kAIC

AREA OF WORK
LSIG BREAKER
AREA OF WORK

LOAD DESCRIPTION	TYPE	A	B	C	BRKR.	CKT.	CKT.	BRKR.	A	B	C	TYPE	LOAD DESCRIPTION
-	*	0.00			-			-	0.58			M	-
METER	*		0.00		-	B1	B2	20/3		0.58		M	B950-EF-1002
-	*			0.00	-			-			0.58	M	-
-	M	1.20			-			-	16.00			M	-
WH-950-1	M		1.20		20/3	B3	B4	175/3		16.00		M	B950-4IPC-1002
-	M			1.20	-			-			16.00	M	-
-	M	8.50			-			-	0.00			*	-
B950-4IPC-1003	M		8.50		175/3	B5	B6	175/3		0.00		*	SPARE
-	M			8.50	-			-			0.00	*	-
-	H	2.80			-			-	0.46			M	-
COMPRESSOR	H		2.80		30/3	C1	C2	20/3		0.46		M	B950-EF-1
-	H			2.80	-			-			0.46	M	-
-	R	15.00			-			-	11.50			M	-
WELDING OUTLET	R		15.00		100/3	C3	C4	125/3		11.50		M	2LBP03-950
-	R			15.00	-			-			11.50	M	-
-	M	16.00			-			-	15.00			R	-
B950-4IPC-1001	M		16.00		175/3	C5	C6	100/3		15.00		R	WELDING OUTLET HUTCH 2.2
-	M			16.00	-			-			15.00	R	-
-	M	0.20			-			-	2.10			H	-
BMS	M		0.20		10/3	D1	D2	15/3		2.10		H	SF-960-1
-	M			0.20	-			-			2.10	H	-
-	**M**	**13.98**			**-**							*	-
B950-4PB-1001	**M**		**13.80**		**400/3**	D3	D4					*	SPACE
-	**M**			**13.80**	**-**							*	-
-	*	0.00			-			-	20.00			M	-
SPARE	*		0.00		150/3	E1	E2	175/3		20.00		M	IPC4-950
-	*			0.00	-			-			20.00	M	-
-	M	20.00			-			-	50.00			H	-
IPC5-950	M		20.00		175/3	E3	E4	175/3		50.00		H	B950-AHU-1
-	M			20.00	-			-			50.00	H	-
-	*	**0.00**			**-**								
SPARE WIRES INSIDE MCC GUTTER	*		**0.00**		**7/3**	E5							
-	*			**0.00**	**-**								
		77.68	77.50	77.50					115.64	115.64	115.64		

DRAWING # EQUIP. TAG	CONN. KVA	DEMAND FACTOR	DEMAND KVA
TYPE "M": NON-CONTINUOUS / MISC. LOADS	324.90	100%	324.90
TYPE "L": LIGHTING / CONTINUOUS LOADS	0.00	125%	0.00
TYPE "R": RECEPTACLES (FIRST 10KVA)	90.00	100%	10.00
TYPE "R": RECEPTACLES (OVER 10KVA)		50%	40.00
TYPE "H": LARGEST MOTOR LOAD	150.00	125%	187.50
TYPE "H": HVAC / MECHANICAL LOADS	14.70	100%	14.70
TOTALS:	579.60		577.10

BOLD = NEW LOAD, NEW BREAKER

PHASE A: 193.32 KVA
PHASE B: 193.14 KVA
PHASE C: 193.14 KVA
694.14 AMPS

2 (E) "4MCC04-950" W/ (N) SCHED.

SLAC BUILDING INSPECTION OFFICE
Construction Authorization (IFC)
BY: cnadler
DATE: 2/18/2026 | PRS#: 26-009
Revised Authorization ☑ No ☐ Yes
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+1 (866) VEKTOR1 (835-8671)

REGISTERED PROFESSIONAL ENGINEER SERGEY V. KOROLEV E20491 EXP: 09/30/27 ELECTRICAL STATE OF CALIFORNIA

SCALE: AS NOTED | DO NOT SCALE DRAWING | B0950.2025.05 | E5.04

SLAC NATIONAL ACCELERATOR LABORATORY | U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
ELECTRICAL
PANEL SCHEDULES

		DATE	APPROVALS	DATE
ENGR	KOROLEV, SERGEY	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TURCHINA, OLENA	01-26-2026		
CHKR	ORTIZ, JUSTIN	01-26-2026		

DRAWING NUMBER B0950-E06-0428 | REVISION E | E

B0950-E06-0428 - E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	OT	JO	SK	07-25-2025
B	90% CD	OT	JO	SK	08-20-2025
C	100% CD	OT	JO	SK	11-05-2025
D	FAO REVIEW	OT	JO	SK	01-05-2026
E	BIO REVIEW	OT	JO	SK	01-26-2026

2 1

AREA OF WORK

AREA OF WORK

VOLTS: 208 **PHASE:** 3 **WIRE:** 4 **BUSSING:** 250A **POLES:** 42

EXIST ☑ NEW ☐

PANEL SCHEDULE

2SDBP1A-950

MAIN 250A **FED FROM:** PNL "4SDBP1A-950" **LOCATION:** MECH.R.310 B 950 **MOUNTED:** SURFACE **AIC RATING:** 10kAIC

LOAD DESCRIPTION	TYPE	A	B	C	BRKR.	CKT.	CKT.	BRKR.	A	B	C	TYPE	LOAD DESCRIPTION
SPARE	*	0.00			20/1	1	2	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	3	4	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	5	6	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	7	8	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	9	10	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	11	12	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	13	14	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	15	16	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	17	18	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	19	20	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	21	22	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	23	24	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	25	26	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	27	28	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	29	30	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	31	32	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	33	34	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	35	36	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	37	38	20/1	0.00			*	SPARE
GENSET CONTROLS	M		8.30		100/2	39	40	20/1		0.00		*	SPARE
GENSET CONTROLS	M			8.30	-	41	42	20/1			0.00	*	SPARE
	*	0.00	8.30	8.30					0.00	0.00	0.00		

DRAWING # EQUIP. TAG A00595	CONN. KVA	DEMAND FACTOR	DEMAND KVA
TYPE "M": NON-CONTINUOUS / MISC. LOADS	16.60	100%	16.60
TYPE "L": LIGHTING / CONTINUOUS LOADS	0.00	125%	0.00
TYPE "R": RECEPTACLES (FIRST 10KVA)	0.00	100%	0.00
TYPE "R": RECEPTACLES (OVER 10KVA)		50%	0.00
TYPE "H": LARGEST MOTOR LOAD	0.00	125%	0.00
TYPE "H": HVAC / MECHANICAL LOADS	0.00	100%	0.00
TOTALS:	16.60		16.60

BOLD = NEW LOAD, NEW BREAKER

PHASE A: 0.00 **KVA**
PHASE B: 8.30 **KVA**
PHASE C: 8.30 **KVA**

46.08 **AMPS**

2 (E) PNL. "B950-2PB-1002" (2SDBPA1-950) W/ (E) SCHED.

AREA OF WORK

LSIG BREAKER (TYP. 6)

VOLTS: 480V **PHASE:** 3 **WIRE:** 3 **BUSSING:** 250A **POLES:** 30

EXIST ☐ NEW ☑

PANEL SCHEDULE

B950-4PB-1002

MAIN: 250A **FED FROM:** PNL B950-4PB-1001 **LOCATION:** RM. 302, B950 **MOUNTED:** SURFACE **AIC RATING:** 22kAIC

LOAD DESCRIPTION	TYPE	A	B	C	BRKR.	CKT.	CKT.	BRKR.	A	B	C	TYPE	LOAD DESCRIPTION
MELTRIC RCP	*	0.00			-	1	2	-	6.90			H	-
TO FUTURE CHILLER #4	*		0.00		40/3	3	4	40/3		6.90		H	CHILLER #1
-	*			0.00	-	5	6	-			6.90	H	-
MELTRIC RCP	*	0.00			-	7	8	-	6.90			H	-
TO FUTURE CHILLER #5	*		0.00		40/3	9	10	40/3		6.90		H	CHILLER #2
-	*			0.00	-	11	12	-			6.90	H	-
MELTRIC RCP	*	0.00			-	13	14	-	0.00			*	MELTRIC RCP
TO FUTURE CHILLER #6	*		0.00		40/3	15	16	40/3		0.00		*	TO FUTURE CHILLER #3
-	*			0.00	-	17	18	-			0.00	*	-
-	*	0.00			-	19	20	*				*	-
SPARE	*		0.00		25/3	21	22	*				*	SPACE
-	*			0.00	-	23	24	*				*	-
-	*				*	25	26	*				*	-
SPACE	*				*	27	28	*				*	SPACE
-	*				*	29	30	*				*	-
		0.00	0.00	0.00					13.80	13.80	13.80		

DRAWING # EQUIP. TAG	CONN. KVA	DEMAND FACTOR	DEMAND KVA
TYPE "M": NON-CONTINUOUS / MISC. LOADS	0.00	100%	0.00
TYPE "L": LIGHTING / CONTINUOUS LOADS	0.00	125%	0.00
TYPE "R": RECEPTACLES (FIRST 10KVA)	0.00	100%	0.00
TYPE "R": RECEPTACLES (OVER 10KVA)		50%	0.00
TYPE "H": LARGEST MOTOR LOAD	20.70	125%	25.88
TYPE "H": HVAC / MECHANICAL LOADS	20.70	100%	20.70
TOTALS:	41.40		46.57

BOLD = NEW LOAD, NEW BREAKER

PHASE A: 13.80 **KVA**
PHASE B: 13.80 **KVA**
PHASE C: 13.80 **KVA**

56.02 **AMPS**

1 (N) PNL. "B950-4PB-1002" W/ (N) SCHED.

PANELS GENERAL NOTES

1. CONTRACTOR SHALL COMPLY WITH ALL NOTES AS LISTED FOR EACH RESPECTIVE PNL.
2. CONTRACTOR SHALL PROVIDE (N) TYPED PANEL SCHEDULE FOR EACH ELECTRICAL PANEL, PLACE IN CLEAR PLASTIC SLEEVE AND ATTACH TO INNER SURFACE OF PANEL DOOR.
3. CONTRACTORS SHALL NOT PERFORM WORK ON LEGACY SINGLE-PHASE CIRCUITS OR IN LEGACY ELECTRICAL PANELBOARDS (LEGACY = INSTALLED PRIOR TO 2019) THAT CONTAIN SINGLE PHASE CIRCUITS UNLESS NEUTRAL WIRING VERIFICATION OF ALL SINGLE-PHASE CIRCUITS IN THE PANEL HAS BEEN PERFORMED AND DOCUMENTED IN ACCORDANCE WITH SLAC-I-708-404-096 OPERATIONS ON SYSTEMS WITH SUSPECTED SHARED NEUTRALS. SEE ESH MANUAL CHAPTER 8 ELECTRICAL SAFETY MANUAL SECTION 7.2.13. IF THE VERIFICATION HAS NOT BEEN PERFORMED, THEN IT MUST BE PERFORMED BY F&O EPD PRIOR TO START OF PROJECT CONSTRUCTION.

PANELS KEY NOTES

1. EXISTING PANEL SCHEDULE VERIFIED BY CIBIL ZACHARIAH ON 04/01/2025.
2. ALL ELECTRICAL PANELS SHALL BE INSPECTED FOR SHARED NEUTRALS BY ELECTRICAL CONTRACTOR PRIOR TO BEING RE-ENERGIZED, REPORT FINDINGS TO EEOR AND SLAC OWNER REPRESENTATIVE PRIOR TO PROCEEDING.

AREA OF WORK

AREA OF WORK

VOLTS: 208 **PHASE:** 3 **WIRE:** 4 **BUSSING:** 250A **POLES:** 42

EXIST ☑ NEW ☐

PANEL SCHEDULE

B950-2PB-1002

MAIN 250A **FED FROM:** PNL "B950-4PB-1001" **LOCATION:** MECH.R.310 B950 **MOUNTED:** SURFACE **AIC RATING:** 10kAIC

LOAD DESCRIPTION	TYPE	A	B	C	BRKR.	CKT.	CKT.	BRKR.	A	B	C	TYPE	LOAD DESCRIPTION
CONV. RECEPTACLE	**R**	**0.18**			**20/1**	**1**	2	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	3	4	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	5	6	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	7	8	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	9	10	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	11	12	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	13	14	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	15	16	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	17	18	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	19	20	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	21	22	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	23	24	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	25	26	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	27	28	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	29	30	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	31	32	20/1	0.00			*	SPARE
SPARE	*		0.00		20/1	33	34	20/1		0.00		*	SPARE
SPARE	*			0.00	20/1	35	36	20/1			0.00	*	SPARE
SPARE	*	0.00			20/1	37	38	20/1	0.00			*	SPARE
SPARE	*		**0.00**		100/2	39	40	20/1		0.00		*	SPARE
-	*			**0.00**	-	41	42	20/1			0.00	*	SPARE
	*	0.18	0.00	0.00					0.00	0.00	0.00		

DRAWING # EQUIP. TAG A00595	CONN. KVA	DEMAND FACTOR	DEMAND KVA
TYPE "M": NON-CONTINUOUS / MISC. LOADS	0.00	100%	0.00
TYPE "L": LIGHTING / CONTINUOUS LOADS	0.00	125%	0.00
TYPE "R": RECEPTACLES (FIRST 10KVA)	0.18	100%	0.18
TYPE "R": RECEPTACLES (OVER 10KVA)		50%	0.00
TYPE "H": LARGEST MOTOR LOAD	0.00	125%	0.00
TYPE "H": HVAC / MECHANICAL LOADS	0.00	100%	0.00
TOTALS:	0.18		0.18

BOLD = NEW LOAD, NEW BREAKER

PHASE A: 0.18 **KVA**
PHASE B: 0.00 **KVA**
PHASE C: 0.00 **KVA**

0.50 **AMPS**

3 (E) PNL. "B950-2PB-1002" W/ (N) SCHED.

SLAC BUILDING INSPECTION OFFICE

Construction Authorization (IFC)

BY: cnadler

DATE: 2/18/2026 PRS#: 26-009

Revised Authorization ☑ No ☐ Yes

These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.

The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.

THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

Vektor Engineering & Consulting Services, Inc.

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San Ramon, CA 94583
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REGISTERED PROFESSIONAL ENGINEER SERGEY V. KOROLEV E20491 EXP: 09/30/27 ELECTRICAL STATE OF CALIFORNIA

SCALE: AS NOTED | DO NOT SCALE DRAWING | B0950.2025.05 | E5.05

SLAC NATIONAL ACCELERATOR LABORATORY U.S. DEPARTMENT OF ENERGY

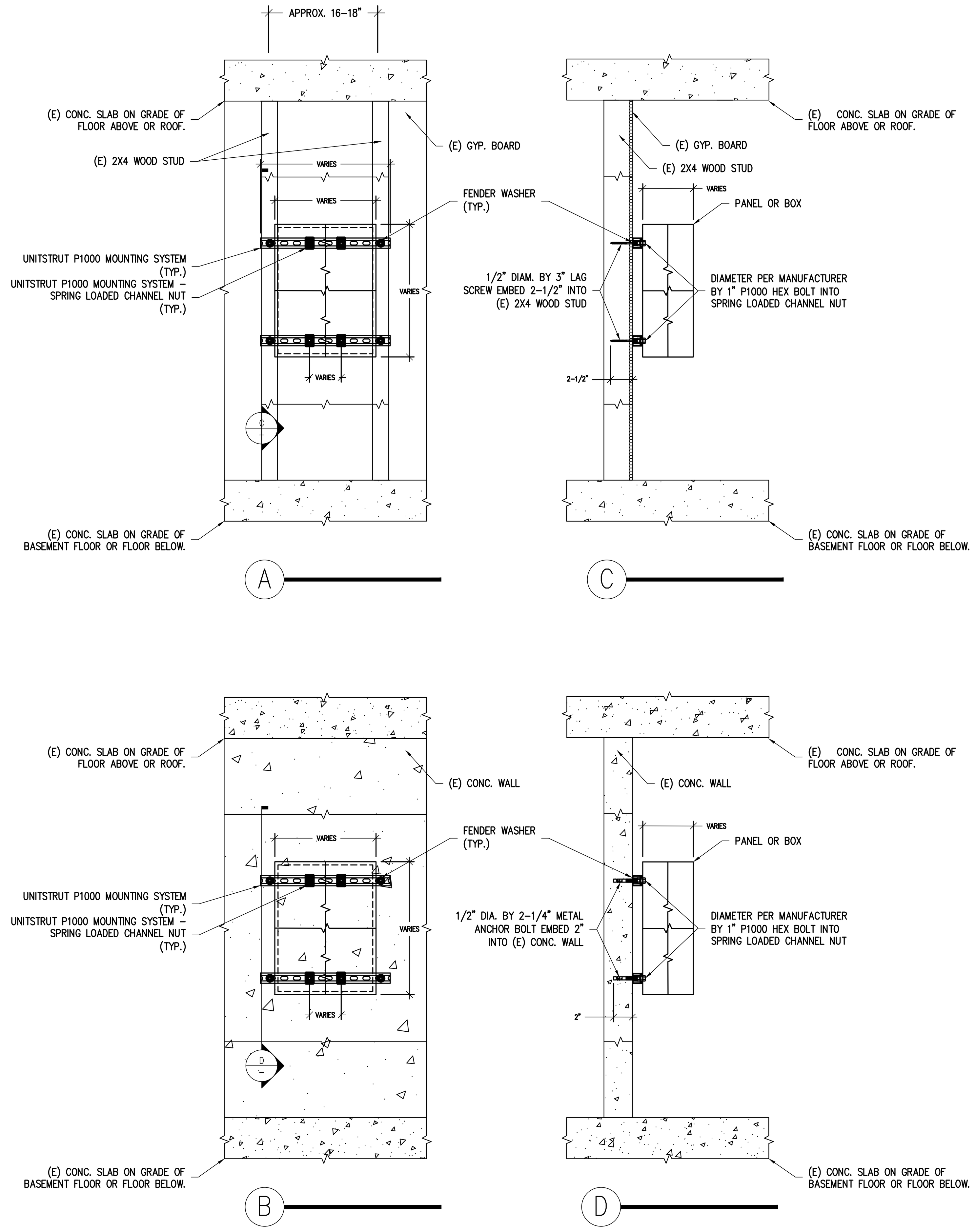
B950
MECH ROOM CO2 CHILLERS
ELECTRICAL
PANEL SCHEDULES

		DATE	APPROVALS	DATE
ENGR	KOROLEV, SERGEY	01-26-2026	CHEUNG, CANON	01-26-2026
DES	TURCHINA, OLENA	01-26-2026		
CHKR	ORTIZ, JUSTIN	01-26-2026		

DRAWING NUMBER B0950-E06-0429 REVISION E E

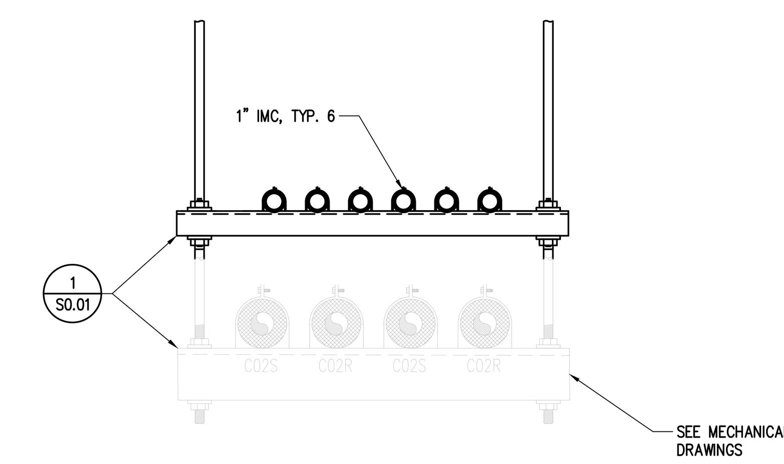
B0950-E06-0429 - E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	OT	JO	SK	07-25-2025
B	90% CD	OT	JO	SK	08-20-2025
C	100% CD	OT	JO	SK	11-05-2025
D	FAO REVIEW	OT	JO	SK	01-05-2026
E	BIO REVIEW	OT	JO	SK	01-26-2026

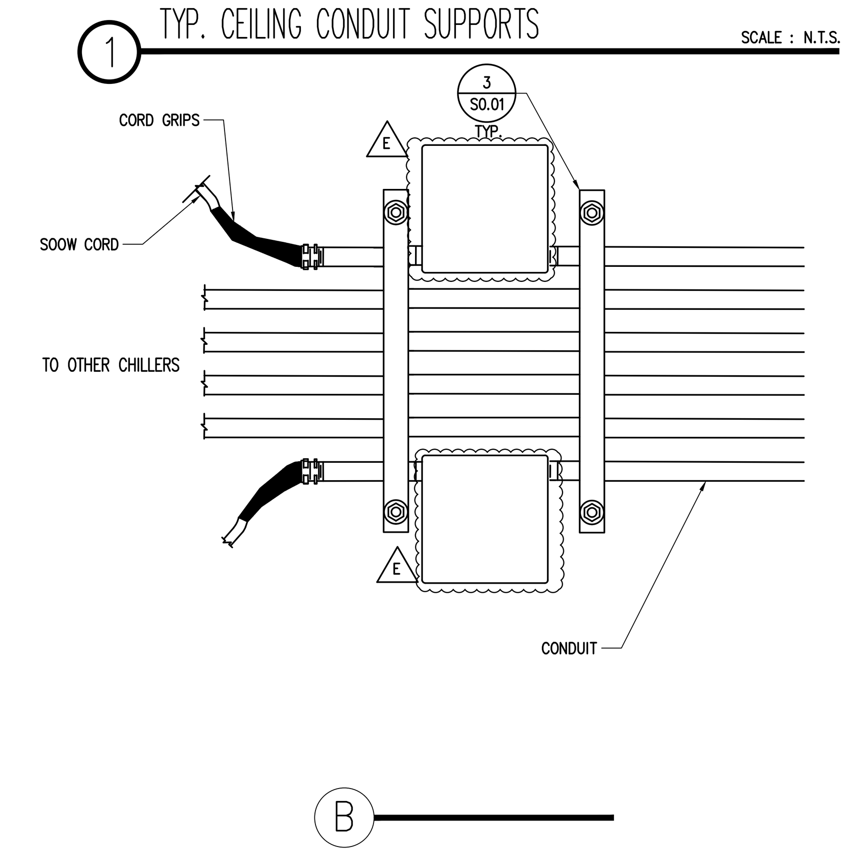


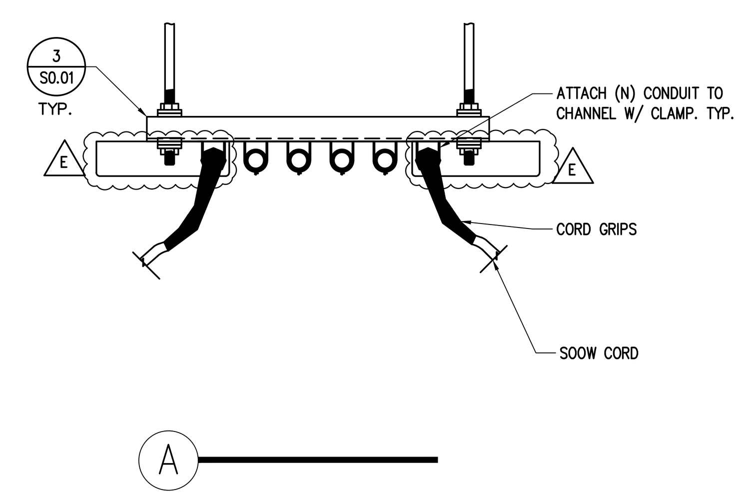
NOTES:
1. DIAMETER "D" IMPLIES FASTENER THAT MATCHES MANUFACTURER'S FASTENER REQUIREMENTS AND HARDWARE DIMENSIONS.
2. CONTRACTOR SHALL UTILIZE ALL MOUNTING HOLES AS DIRECTED BY MANUFACTURER'S INSTALLATION INSTRUCTIONS.
3. PROVIDE A SEPARATE UNISTRUT CHANNEL FOR EACH HORIZONTAL ROW OF MOUNTING HOLES AT PANEL OR BOX TO BE MOUNTED.
4. PROVIDE A FASTENER PER UNISTRUT CHANNEL INTO EACH VERTICAL (E) WOODEN STUD ALONG THE HORIZONTAL UNISTRUT CHANNEL PATH, DISTANCE BETWEEN FASTENERS SHALL MATCH THE O.C. SPACING OF THE (E) WOODEN STUDS IN WALL AND SHALL NOT EXCEED 18"; WHERE WOODEN STUDS ARE MISSING OR DAMAGED, AND THEREFORE 18" SPACING IS EXCEEDED, CONTRACTOR SHALL OPEN GYP. WALL AND PROVIDE (N) FULL HEIGHT WOODEN STUDS AS NECESSARY TO REDUCE SPACING DISTANCE TO LESS THAN THE REQUIRED 18".
5. PROVIDE A FASTENER PER UNISTRUT CHANNEL INTO (E) CONC. WALL ALONG THE HORIZONTAL UNISTRUT CHANNEL PATH, DISTANCE BETWEEN FASTENERS SHALL NOT EXCEED 24".
6. FOR METAL STUDS, USE 1/4" DIA. X 1-1/2" LONG SMS W/ FENDER WASHERS, SIM. TO WOOD STUD DETAIL.

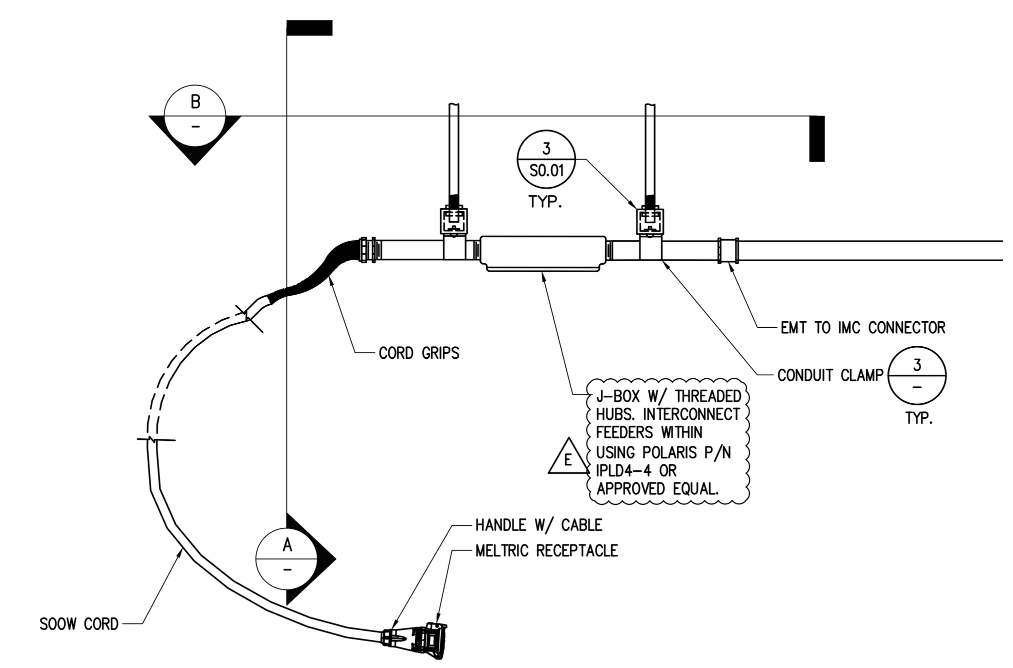
4 TYP. PANEL MOUNTING — SCALE: N.T.S.



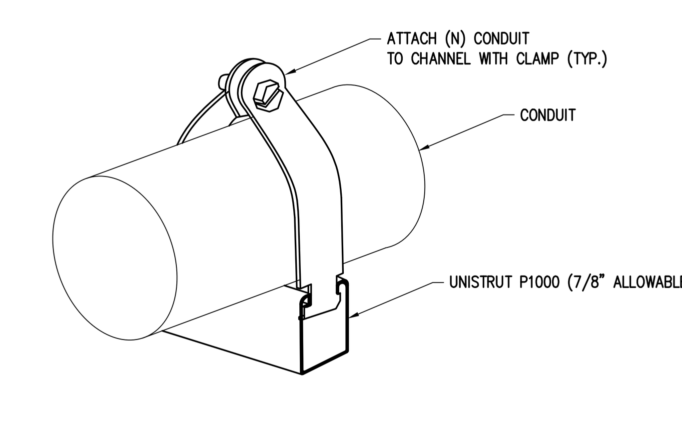
1 TYP. CEILING CONDUIT SUPPORTS — SCALE: N.T.S.







2 TYP. CONDUIT CABLE CONNECTION — SCALE: N.T.S.



NOTE: MAXIMUM SPACING DISTANCE BETWEEN CONDUIT SUPPORT SHALL BE 8'-0"

3 TYP. CONDUIT SUPPORT DETAIL — SCALE: N.T.S.

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization ☑ No ☐ Yes	
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REGISTERED PROFESSIONAL ENGINEER SERGEY V. KOROLEV ELECTRICAL STATE OF CALIFORNIA

SCALE: AS NOTED	DO NOT SCALE DRAWING	B0950.2025.05	E6.01

SLAC NATIONAL ACCELERATOR LABORATORY — U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
ELECTRICAL
DETAILS

		DATE	APPROVALS	DATE
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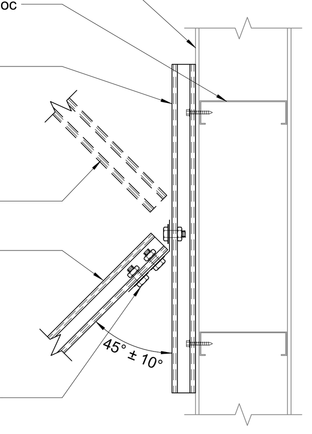
DES	TURCHINA, OLENA	01-26-2026		
CHKR	ORTIZ, JUSTIN	01-26-2026		

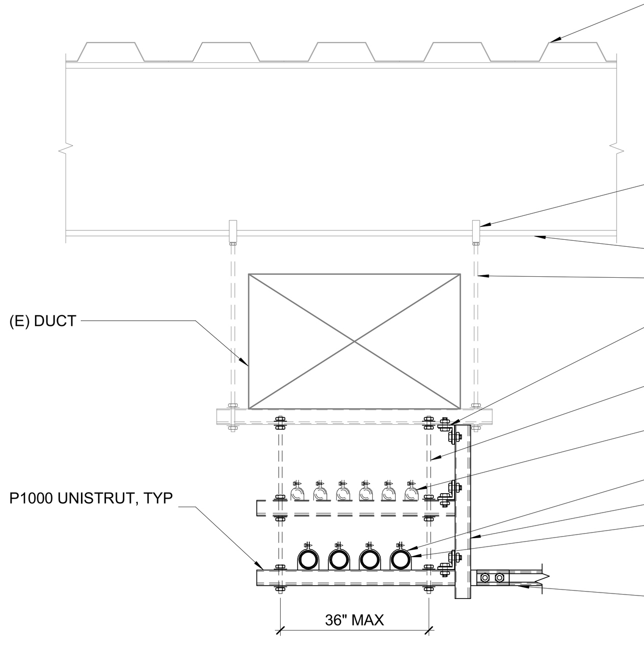
DRAWING NUMBER: B0950-E05-0438 — REVISION: E — E

B0950-E05-0438 - E

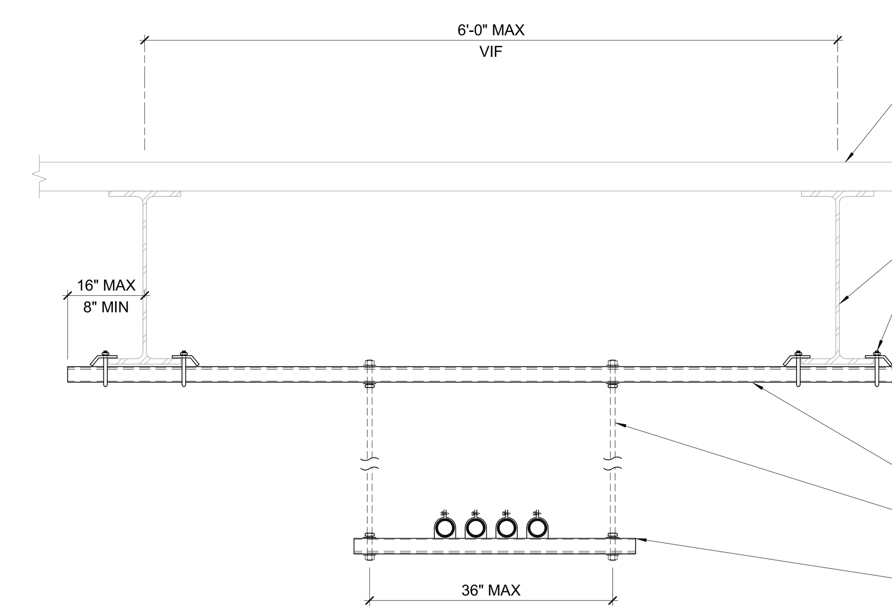
REV	DESCRIPTION	DRN	CHK	APP	DATE
C	100% CD	DJM	RSB	ZFA	11-3-2025
D	FAO REVIEW	CEF	RSB	RSB	01-05-2026



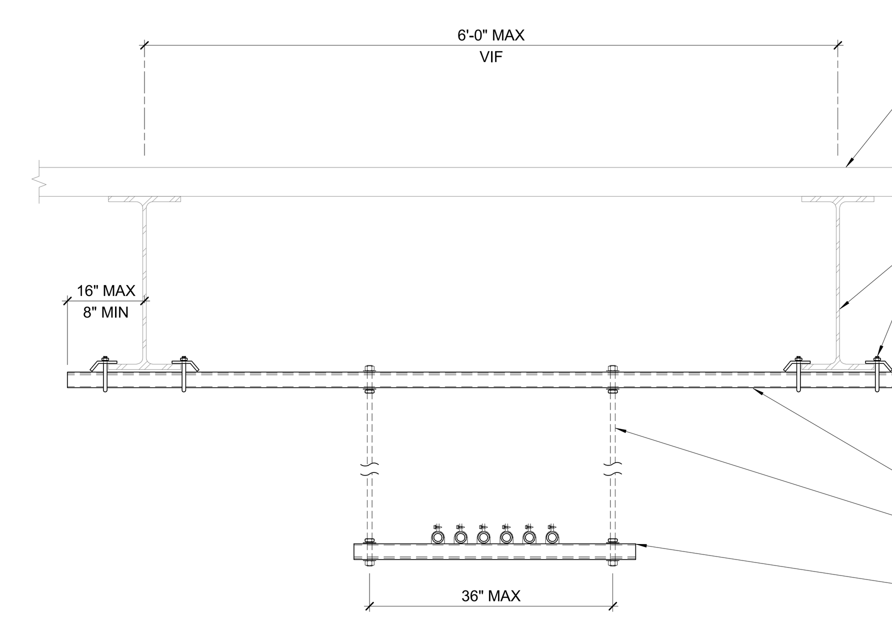
5 HORIZONTAL BRACE TO WALL CONNECTION
1 1/2" = 1'-0"



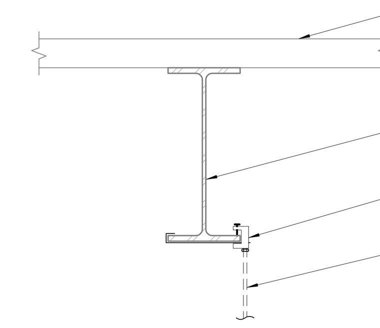
1 TYPICAL TRAPEZE @ (E) DUCT SUPPORT
1" = 1'-0"



2 TRAPEZE PIPE SUPPORT BETWEEN (E) ROOF FRAMING
1" = 1'-0"



3 TRAPEZE CONDUIT SUPPORT BETWEEN (E) ROOF FRAMING
1" = 1'-0"



4 TRAPEZE MEP SUPPORT PERPENDICULAR TO ROOF FRAMING
1" = 1'-0"

ABBREVIATIONS

AB	ANCHOR BOLT
ABV	ABOVE
AC	AIR CONDITIONING
ADJ	ADJACENT
ADDL	ADDITIONAL
ALT	ALTERNATE
ALUM	ALUMINUM
ARCH	ARCHITECT
AYC	ALASKAN YELLOW CEDAR
@	AT
BF	BRACED FRAME
BLDG	BUILDING
BLK/BLKG	BLOCK/BLOCKING
BLW	BELOW
BM	BEAM
BN	BOUNDARY NAIL
BOT	BOTTOM
BRG	BEARING
BTWN	BETWEEN
BU	BUILT-UP
BYND	BEYOND
C	AMERICAN STANDARD CHANNEL
CA	CALIFORNIA
CANT	CANTILEVER
CB	CARRIAGE BOLT
CFS	COLD FORMED STEEL
CIP	CAST IN PLACE
CGL	CERTIFIED GLUED LUMBER
CJ	CONTROL JOINT
℄	CENTERLINE
CJP	COMPLETE JOINT PENETRATION
CLG	CEILING
CLR	CLEAR
COL	COLUMN
CONC	CONCRETE
CONN	CONNECTION
CONT	CONTINUOUS
COORD	COORDINATE/ COORDINATION
CMU	CONCRETE MASONRY UNIT
CSK	COUNTERSINK
CW	CUT WASHER
DBA	DEFORMED BAR ANCHOR
DBL	DOUBLE
DCW	DEMAND CRITICAL WELD
DF	DOUGLAS FIR
DIA or Ø	DIAMETER
DIAG	DIAGONAL
DIM	DIMENSION
DIST	DISTANCE
DJ	DOWEL JOINT
DL	DEAD LOAD
DN	DOWN
DO	DITTO
DWG	DRAWING
DWL	DOWEL
EA	EACH
EE	EACH END
EF	EACH FACE
ELEC	ELECTRICAL
ELEV	ELEVATOR/ELEVATION
EMBED	EMBEDMENT
EQ	EQUAL
EQUIP	EQUIPMENT
ES	EACH SIDE
EW	EACH WAY
(E)	EXISTING
EXP	EXPANSION
EXT	EXTERIOR
FDN	FOUNDATION
FIN	FINISH
FG	FINISH GRADE
FLR	FLOOR
FN	FACE NAIL
FOC	FACE OF CONCRETE
FOM	FACE OF MASONRY
FOS	FACE OF STUD
FRMG	FRAMING
FS	FAR SIDE

FTG	FOOTING
GA	GAGE or GAUGE
GALV	GALVANIZED
GB	GRADE BEAM
GL	GRIDLINE
GLB	GLUE LAMINATED BEAM
GR	GRADE
HD	HOLDOWN
HDG	HOT-DIP GALVANIZED
HDR	HEADER
HGR	HANGER
HK	HOOK
HORIZ	HORIZONTAL
HSB	HIGH STRENGTH BOLT
HSG	HIGH STRENGTH GROUT
HSH	HORIZONTAL SLOTTED HOLE
HSS	HOLLOW STRUCTURAL SECTION
HT	HEIGHT
ID	INSIDE DIAMETER
IJ	I SHAPED WOOD BUILT UP TRUSS
INT	INTERIOR
JST	JOIST
JT	JOINT
KP	KING POST
L	STEEL ANGLE
Lb or #	POUND(s)
LGMF	LIGHT GAGE METAL FRAMING
LGMFC	LIGHT GAGE METAL FRAMING CONTRACTOR
LL	LIVE LOAD
LLH	LONG LEG HORIZONTAL
LLV	LONG LEG VERTICAL
LOC	LOCATION
LS	LAG SCREW
LSL	LAMINATED STRAND LUMBER
LVL	LAMINATED VENEER LUMBER
LWC	LIGHTWEIGHT CONCRETE
MAX	MAXIMUM
MB	MACHINE BOLT
MBM	METAL BUILDING MANUFACTURER
MC	MISCELLANEOUS CHANNEL
MECH	MECHANICAL
MEZZ	MEZZANINE
MF	MOMENT FRAME
MFR	MANUFACTURER
MIN	MINIMUM
MISC	MISCELLANEOUS
MIW	MALLEABLE IRON WASHER
MTL	METAL
MU	MECH UNIT
(N)	NEW
N/A	NOT APPLICABLE
NO or #	NUMBER
NS	NEAR SIDE
NSG	NON-SHRINK GROUT
NTS	NOT TO SCALE
NWC	NORMAL-WEIGHT CONCRETE
O/	OVER
oc	ON CENTER
OD	OUTSIDE DIAMETER
OH	OPPOSITE HAND
OPNG	OPENING
OPP	OPPOSITE
OVS	OVERSIZED
OW	OTHERWISE
OWT	OPEN WEB TRUSS
⅊	PLATE or PROPERTY LINE
PA	POST ABOVE
PAF	POWER ACTUATED FASTENERS
PEN	PANEL EDGE NAIL
PERP	PERPENDICULAR
PES	PANEL EDGE SCREWS
PJP	PARTIAL JOINT PENETRATION
PLF	POUNDS PER LINEAR FOOT

PNL	PANEL
PSF	POUNDS PER SQUARE FOOT
PSI	POUNDS PER SQUARE INCH
PSL	PARALLEL STRAND LUMBER
PTDF	PRESSURE TREATED DOUGLAS FIR
PT	POINT
R	RADIUS
RBS	REDUCED BEAM SECTION
RFTR	RAFTER
REF	REFERENCE
REINF	REINFORCING
REQD	REQUIRED
RET	RETAINING
REV	REVISION
RF	ROOF
RWD	REDWOOD
S	AMERICAN STANDARD BEAM
SAD	SEE ARCHITECTURAL DRAWINGS
SB	SOLID BLOCK
SC	SLIP CRITICAL
SCD	SEE CIVIL DRAWINGS
SCHED	SCHEDULE
SED	SEE ELECTRICAL DRAWINGS
SEOR	STRUCTURAL ENGINEER OF RECORD
SFRS	SEISMIC FORCE RESISTING SYSTEM
SHTG	SHEATHING
SIM	SIMILAR
SKYLT	SKYLIGHT
SLD	SEE LANDSCAPE DRAWINGS
SMS	SHEET METAL SCREW
SMD	SEE MECHANICAL DRAWINGS
SOG	SLAB ON GROUND
SPCG	SPACING
SPD	SEE PLUMBING DRAWINGS
SPEC	SPECIFICATION
SQ	SQUARE
SS	SELECT STRUCTURAL or STAINLESS STEEL
STGR	STAGGERED
STD	STANDARD
STIFF	STIFFENER
STL	STEEL
STRUCT	STRUCTURAL
SW	SHEAR WALL
SYM	SYMMETRICAL
T&B	TOP AND BOTTOM
T&G	TONGUE AND GROOVE
THK	THICK
THRD	THREADED
THRU	THROUGH
TL	TOTAL LOAD
TN	TOE NAIL
TO	TOP OF
TOC	TOP OF CONCRETE
TOF	TOP OF FRAMING
TOM	TOP OF MASONRY
TOP	TOP OF PLYWOOD
TOS	TOP OF STEEL
TOT	TOTAL
TU	TILT UP
TYP	TYPICAL
UNO	UNLESS NOTED OTHERWISE
VERT	VERTICAL
VIF	VERIFY IN FIELD
VSH	VERTICAL SLOTTED HOLE
W	WIDE FLANGE STEEL BEAM
W/	WITH
W/O	WITHOUT
WD	WOOD
WHS	WELDED HEADED STUD
WLD	WELDED
WP	WORK POINT/WATERPROOF
WS	WOOD SCREW
WT	WEIGHT
WTS	WELDED THREADED STUD
WWR	WELDED WIRE REINFORCEMENT

SLAC BUILDING INSPECTION OFFICE

Construction Authorization (IFC)

BY: cnadler

DATE: 2/18/2026 **PRS#:** 26-009

Revised Authorization ☑ No ☐ Yes

These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.

The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.

THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

ZFA STRUCTURAL ENGINEERS
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SCALE: AS NOTED	DO NOT SCALE DRAWING	B0950.2025.05		S0.01

SLAC NATIONAL ACCELERATOR LABORATORY | U.S. DEPARTMENT OF ENERGY

B950
MECH ROOM CO2 CHILLERS
STRUCTURAL
COVER SHEET AND DETAILS

		DATE	APPROVALS	DATE
ENGR	BOGART, RYAN	01-05-2026		
DES	MUCK, DALTON	01-05-2026		
CHKR	BOGART, RYAN	01-05-2026		

DRAWING NUMBER	REVISION	
B0950-S00-0439	D	E

B0950 - S00 - 0421 - D E